

A M B E R S O N G

Hardware Bible

Human Readable edition

What is in the machine and how it is joined: the boards, the power, the cables, the tube radio and the build.

Based on the Hardware Bible, LLM edition: the complete text, published with this book in the Ambersong repository (docs/llm-edition/), as of repository commit **4cbc6da** of 2026-10-02. Where the two differ, the LLM edition is right.

Revisions

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Contents

About this book	7
What this book covers	7
What it does not cover	7
How to use it	7
Notation	8
1. What Ambersong is	10
1.1 The machine at a glance	10
1.2 The sound path	11
1.3 The control side	12
1.4 The power at a glance	12
1.5 The front panel at a glance	12
2. Safety before you open it	13
2.1 Where the mains is, and when	13
2.2 The front power switch does not make it safe	13
2.3 The radio chassis is on mains neutral	14
2.4 Safety-rated parts	14
2.5 The low-voltage side	15
3. The physical build	16
3.1 The cabinet and how to open it	16
3.2 The front panel	16
3.3 Tuning, needle and index	17
3.4 Cooling	17
3.5 The Faraday cages	17
3.6 Connectors and labels	17
3.7 Printed parts	18
4. The two microcontroller boards	19
4.1 The boards	19
4.2 Main board pin map	20
4.3 Audio board pin map	22
4.4 Main board support parts	23
4.5 Audio board support parts	24
4.6 The back-panel USB-C ports	24
4.7 The real-time clock module	25
4.8 When something is wrong	25
5. Power, earth and grounds	27
5.1 The power at a glance	27
5.2 The mains side	28
5.3 The four grounds	28

5.4 Earth, cages and the antenna	30
5.5 The 5 V and 3.3 V buses	30
5.6 The amplifier supply and the box fan	31
5.7 When something is wrong	32
6. The audio chain	33
6.1 The path at a glance	33
6.2 The digital audio bus (I2S)	34
6.3 The ADC (PCM1802)	34
6.4 The tube radio into the ADC	36
6.5 The DAC (PCM5102A)	37
6.6 From the DAC to the amplifier	37
6.7 The AUX path and the amplifier's input panel	38
6.8 The three volume controls	38
6.9 When something is wrong	39
7. The clock display	40
7.1 The display at a glance	40
7.2 How it is driven	41
7.3 The segment map	42
7.4 The digit drivers	42
7.5 The segment drivers	43
7.6 The colon	43
7.7 Boards and cables	43
7.8 When something is wrong	44
8. Dial, needle and tuning sensors	45
8.1 The mechanics	45
8.2 The stepper motor and its driver	46
8.3 The index sensor (A3144) and its magnet	46
8.4 The angle sensor (AS5600)	47
8.5 The FM calibration receiver (RDA5807M)	47
8.6 Cables	48
8.7 When something is wrong	48
9. Front-panel controls and lamps	50
9.1 The FM panel lamps	50
9.2 The Bluetooth button and LED	51
9.3 The source switch	52
9.4 The front volume knob	53
9.5 The amplifier power sensor	54
9.6 When something is wrong	55
10. Cables and harnesses	56
10.1 How to read the cable pages	56
10.2 FM receiver power — main board to FM calibration receiver	59

10.3 FM receiver control — main board to FM calibration receiver	60
10.4 Main board 5 V — 5 V bus to main board	60
10.5 Segment drive — main board to display driver board	61
10.6 Needle motor control — main board to needle motor driver	62
10.7 Amplifier power sensor — main board to power sensor board	62
10.8 Digit select — main board to display driver board	63
10.9 Needle index sensor — main board and 5 V bus to the index sensor	64
10.10 Tuning angle sensor — main board to the angle sensor	65
10.11 Panel-lamp control — main board and 5 V bus to the panel-lamp board	66
10.12 Board link — main board to audio board	66
10.13 Front volume knob — audio board and 3.3 V to the volume knob	67
10.14 Bluetooth button and LED — audio board to front panel	68
10.15 The audio harness — audio board to ADC and DAC	69
10.16 Clock module power — 3.3 V jack to the real-time clock module	71
10.17 Clock module control — audio board to the real-time clock module	72
10.18 Audio board 5 V — 5 V bus to audio board	72
10.19 Audio board 3.3 V — audio board to the 3.3 V supply	73
10.20 Source switch — audio board and 3.3 V to the source-switch board	73
10.21 DAC to amplifier — DAC board to the amplifier's balanced input	75
10.22 Radio to ADC — tube radio's output transformer to the ADC	76
10.23 Colon 5 V — 5 V bus to the clock display's colon	76
10.24 Motor driver 5 V — 5 V bus to the needle motor driver	77
10.25 ADC 5 V — 5 V bus to the ADC board	78
10.26 Driver board 5 V — 5 V bus to the display driver board	78
10.27 Digit rails — display driver board to the display bus board	79
10.28 Segment buses — display driver board to the display bus board	80
10.29 Box-fan harness — amplifier supply to the box fan	80
10.30 Amplifier supply wires — amplifier supply to the amplifier	81
10.31 USB — back panel to each board	82
10.32 Needle motor lead — needle motor driver to the needle motor	83
10.33 Panel-lamp leads — panel-lamp board to the four FM lamps	83
10.34 Inside the clock display — colon and digits	84
10.35 FM receiver antenna and the antenna coax	85
10.36 Mains and earth wiring	86
10.37 When something is wrong	86
11. The FM antenna	87
11.1 The aerial	87
11.2 The feedline and its earthing	87
11.3 Build details from the notes	88
11.4 What is not measured	89
11.5 When something is wrong	89
12. The amplifier and speakers	90

12.1 What the amplifier and speakers are	90
12.2 The volume control	90
12.3 The input panel	91
12.4 What was measured at the panel and on the cable	91
12.5 When something is wrong	92
13. The tube radio (LLOYDS TM-838N)	93
13.1 Overview	93
13.2 FM RF amplifier and mixer (V1, 12DT8)	98
13.3 FM local oscillator and AFC (V2, 12DT8)	101
13.4 The FM IF path (V4, V5, 12BA6)	104
13.5 The FM ratio detector and AFC control	108
13.6 The 455 kHz AM IF and AM detector	111
13.7 AM and SW input, converter (V3, 12BE6) and the SW and AM local oscillator	113
13.8 The audio amplifier (V6, 12AV6, and V7, 50C5), tone switch and output transformer	117
13.9 The series heater string, the mode supply and the indicator lamps	121
13.10 The original transformerless AC supply and B+	125
13.11 Where the tube radio meets Ambersong	128
13.12 When something is wrong	129
Appendix A. Datasheet facts used in this machine	130
A.1 Main board: ESP32-S3	130
A.2 Audio board: ESP32	131
A.3 The ADC: PCM1802 (Texas Instruments)	131
A.4 The DAC: PCM5102A (Texas Instruments)	132
A.5 Amplifier power sensor and box-fan regulator	132
A.6 Display and motor drivers	133
A.7 Connectors	133
A.8 Needle, tuning and lamp parts	134
A.9 Tubes	135
Appendix B. Where the schematics differ from this book	136
B.1 Drawing errors still in the published sheets	136
B.2 Parts drawn where they read best	136
B.3 Reading notes	137
Appendix C. What this book does not cover, and open points	139
C.1 Outside the book's scope	139
C.2 Open points	139
Appendix D. Printed parts	143
D.1 Speakers	143
D.2 Back	143
D.3 Front panel	144
D.4 Board cases and holders, inside the cabinet	144
D.5 The tube radio's Faraday cage	146

D.6 Clock display	146
Glossary	147
Index	156

About this book

This book describes the hardware of Ambersong: a vintage tube radio, rebuilt in a wooden cabinet with two small computer boards, a digital clock, a motor-driven dial needle and a modern amplifier. It says what every part is, where it sits when that is known, and what it is joined to. With it you should be able to repair, debug, change or rebuild a machine like this one without any other document.

What this book covers

- **The machine:** the two microcontroller boards, the power and the grounds, the audio chain, the clock display, the dial needle and its sensors, the front-panel controls and lamps, and every cable.
- **The tube radio inside it**, a LLOYDS TM-838N, stage by stage. Its parts are given as fitted, with the original factory and service-data values beside them for reference.
- **The FM antenna** as built and installed, **the amplifier and its speakers**, and **the physical build**: the cabinet, the front panel, the tuning and needle mechanics, and the Faraday cages.
- **Datasheet limits** for the parts used here (Appendix A (p. 130)). They describe the catalogue part, not a measurement of the one fitted.

What it does not cover

- **The firmware.** What the two boards run, what you see on the radio, and how to update them are in the companion book, the **Ambersong Firmware Gospel**. This book states the wire and the part; the Gospel states what the code does with them.
- **The amplifier's own circuit.** Chapter 12 says what the amplifier is and how it connects, but not what is on its boards.
- **Build details:** wire colours (except the amplifier supply's four wires, chapter 5 (p. 27)), which side of a connector is pin 1, crimp and connector housings, the length of each cable (the only length facts: every cable inside is under 14 in, and the DAC-to-amplifier cable is under 12 in), cable routing, the screws of the printed plates, and the order of the contacts inside a connector.

Appendix C (p. 139) lists every open point and every gap.

How to use it

Read chapter 2 (p. 13) before you open the cabinet. Mains is present inside even with the front switch off, and the tube radio's metal chassis is wired to mains neutral.

Then find your job in the table below; it names the chapter to start at.

If you want to...	Start at
know what the machine is and how the parts fit together	chapter 1 (p. 10)
open the cabinet safely	chapters 2 and 3
find what a GPIO pin does	chapter 4 (p. 19)
trace a power or ground problem	chapter 5 (p. 27)

If you want to...	Start at
trace a sound problem	chapters 6 and 12
fix the clock display	chapter 7 (p. 40)
fix the dial needle or tuning sensors	chapter 8 (p. 45)
fix a front-panel control or lamp	chapter 9 (p. 50)
repair or remake a cable	chapter 10 (p. 56)
service the tube radio	chapter 13 (p. 93)

Most chapters end with a *When something is wrong* table. It holds only what this book knows: the wiring, and what the author has stated about the fitted parts.

Much of this book describes **one** machine: this cabinet, this radio, this amplifier and these boards. If you build your own, adapt it.

The drawings. The author drew the machine in KiCad to show a person what was built, not to make a circuit board. Some drawings place a part where it reads best rather than where it sits. You do not need them to use this book. If you do read them, this book is right where the two differ, and Appendix B (p. 136) lists the places where a drawing can mislead.

The firmware's own text. A few comments and names in the firmware describe wiring that is not what was built. Where one disagrees with this book, the hardware stated here is right and the firmware text is stale. Each known case is noted where its part is described (chapters 4, 6 and 9).

Notation

A few conventions run through the whole book. Learn them once here.

The two boards. The book calls them by what they do:

In this book	The board	Also written
main board	the ESP32-S3 development board: clock display, needle, tuning sensors, panel lamps	S3
audio board	the ESP32 development board: sound, Bluetooth, front controls, real-time clock	A32

Signal names are written in `code font`, exactly as on the drawings and the cable sheet: `GPIO5`, `A_I022`, `S3_to_A32`. On the main board a signal is usually named after its pin (`GPIO38`); on the audio board it carries an `A_` prefix (`A_I019` is the audio board's `GPIO19`). In the text, an audio-board pin is written with its board named, as in "the audio board's `GPIO25`".

The four grounds. The machine has four ground names, kept different on purpose. Three of them, Earth, the DC ground and the amplifier supply return, are all joined. The fourth, the radio chassis, is joined to none of them: it sits on mains neutral. Chapter 5, section 5.3 (p. 28), explains each one and how they are joined.

Name in this book	Written as
Earth	<code>Earth</code>
DC ground	<code>GND</code>
Amplifier supply return	<code>20V_GND</code>

Name in this book	Written as
Radio chassis	CHASSIS

Rails with two names. Some conductors carry two names on the drawings. The two names are the same wire. The last row is the one rail that is easy to confuse with them: it has one name and is a separate wire.

Name	Same as	What it is
5VDC_S	5VDC	the 5 V supply's output; 5VDC_S is its name before the bus
PSU_GND	GND	the 5 V supply's ground; PSU_GND is its name before the bus
A_VDC	A_3V3	the 3.3 V bus, fed from the audio board's 3.3 V output
S3_3V3	—	the main board's own 3.3 V output, a separate rail

CABLE S/FTP on a connector pin means the cable's shield and foil end there. Unless a cable's page says otherwise, they are joined to GND at the board end and left unconnected at the far end (chapter 10 (p. 56)).

Pin numbers on cables are the cable sheet's, counted along the drawn connector. They are not the numbers moulded on a real connector.

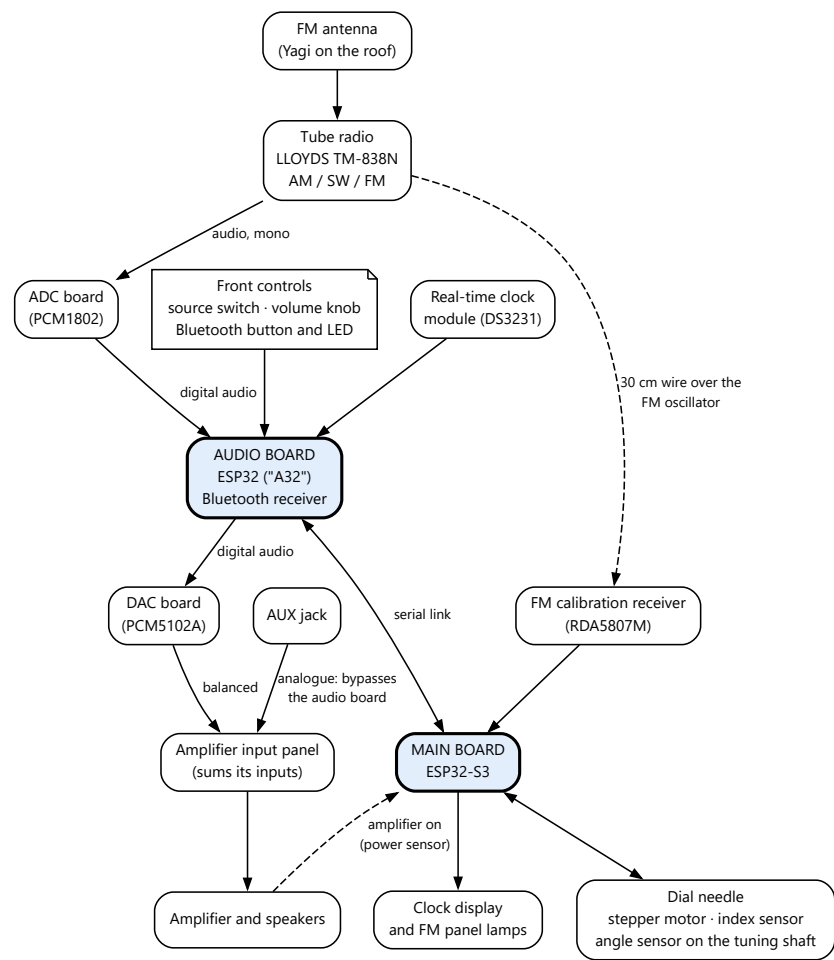
Parts are named by what they are. Where you need a reference to find a part on a drawing, it follows in brackets: "the safety capacitor between the antenna shield and the chassis (C51)". The tube radio's references clash with the main drawing's, so chapter 13 (p. 93) always gives a name with them.

Values are as fitted. Capacitors drawn as polarised are fitted the right way round as drawn; the others are film or ceramic.

1. What Ambersong is

Ambersong is a vintage vacuum-tube radio, a LLOYDS TM-838N AM, shortwave (SW) and FM set, joined to modern audio and control electronics. The tube radio still makes the sound when you listen to the radio. Two small ESP32 boards beside its chassis turn that sound into digital audio, run a clock, and move the dial needle with a motor. They also add Bluetooth and an auxiliary input. This chapter shows how the pieces fit together before the rest of the book takes them one at a time.

1.1 The machine at a glance



The whole machine: where the sound goes, and which board runs what.

Everything sits in one wooden cabinet: the tube radio's original chassis, the two boards, the converters, the amplifier with its speakers, and the power supplies. Chapter 3 describes the cabinet.

The table names each part, says what it does, and gives the chapters that cover it.

Part	What it does	Chapter
Tube radio (LLOYDS TM-838N)	Receives AM, SW and FM and makes the radio's audio. Its output transformer, loaded by a 7.5 Ω resistor, feeds the ADC board (below).	13

Part	What it does	Chapter
Main board (ESP32-S3)	Drives the clock display, the dial-needle motor and its index sensor, reads the angle sensor on the tuning shaft and the FM calibration receiver, dims the FM panel lamps, and senses when the amplifier is on.	4, 7, 8, 9
Audio board (ESP32, "A32")	Takes the radio's digital audio from the ADC, receives Bluetooth, and sends the chosen source to the DAC. Reads the source switch, the volume knob, the Bluetooth button, and the real-time clock; lights the Bluetooth LED.	4, 6, 9
ADC board (analogue-to-digital converter, PCM1802)	Turns the tube radio's audio into digital audio for the audio board.	6
DAC board (digital-to-analogue converter, PCM5102A)	Turns the audio board's digital audio back into an analogue signal for the amplifier.	6
Amplifier and speakers	A powered speaker pair's amplifier, with its two speakers. Its input panel adds up everything plugged into it.	12
Clock display	Four seven-segment digits and a colon, built for this front panel.	7
Dial needle	The radio's original brass needle, moved by a stepper motor, not by the dial cord. A magnet on its mount passes a Hall-effect index sensor, the needle's position reference.	8
Angle sensor (AS5600)	Reads the angle of the tuning capacitor's shaft through a magnet on the shaft.	8
FM calibration receiver (RDA5807M)	A small FM receiver chip, the main board's instrument for calibrating the dial. Its antenna is a 30 cm wire inside the radio's cage, bent over the tube radio's FM oscillator.	8
Real-time clock module (DS3231)	Keeps the time for the audio board.	4
FM antenna	A three-element Yagi on the roof, built for this machine, on the radio's antenna input.	11

The two boards talk to each other over one **serial link**: two signal wires and a ground, in a shielded cable (chapter 10 (p. 56)).

1.2 The sound path

The front **source switch** picks one of three sources: RADIO, BT (Bluetooth) or AUX. Each one reaches the speakers by its own path.

Source	Path to the speakers
RADIO	tube radio → its output transformer, loaded by a 7.5 Ω 5 W resistor → shielded cable → ADC → audio board → DAC → amplifier input panel → amplifier → speakers
BT	a paired Bluetooth device → audio board → DAC → amplifier input panel → amplifier → speakers
AUX	AUX jack → amplifier input panel → amplifier → speakers

AUX is analogue and bypasses the audio board entirely. It never goes through the DAC. On AUX the audio board stays silent, and nothing it does can change AUX loudness or mute it.

The radio feed is mono. One side of the radio's output reaches the ADC; the ADC's left and right inputs are fed from the same point (chapter 6 (p. 33)).

The amplifier's input panel sums its inputs; it does not choose between them. The DAC reaches it through its balanced input. The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn. If two of them carry sound at once, you hear both.

Three volume controls are in the chain. Chapter 6, section 6.8 (p. 38), sets out all three.

- The **tube radio's own**, kept at about one eighth of its travel.
- The front **volume knob**. The audio board reads it, so it acts on RADIO and BT only.
- The **amplifier's own**, reached only from the back.

1.3 The control side

The **main board** runs everything you see move or light: the clock digits and colon, the dial needle, and the four FM panel lamps. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor; nothing mechanical joins it to the needle. The angle sensor reads the capacitor's shaft, and the main board moves the needle to match. The FM calibration receiver is the main board's instrument for calibrating the dial. How the firmware uses these sensors is in the Firmware Gospel.

The **audio board** reads the front controls (source switch, volume knob, Bluetooth button), lights the Bluetooth LED, and keeps the real-time clock.

A small **amplifier power sensor**, an optocoupler lit by the amplifier's supply, tells the main board when the front power switch has turned the amplifier on (chapter 9 (p. 50)).

1.4 The power at a glance

One mains inlet feeds the whole machine. A **5 V supply** runs whenever the cord is plugged in. It powers both boards and everything digital. The 3.3 V parts on the audio side run from the audio board's own 3.3 V output. The **front power switch** turns on only the amplifier's ± 20 V supply and the tube radio. So the clock keeps running with the switch off, and **mains is still present inside the cabinet**.

Chapter 2 lists the dangers. Chapter 5 follows the power from the inlet to every board and explains the four grounds.

1.5 The front panel at a glance

From the front you reach the tuning knob, the volume knob, the source switch, the power switch and the Bluetooth LED, which is also the Bluetooth button. You also see the clock display and the four FM panel lamps. On the FM panel the needle shows the rough FM position against a blackened brass plate. What the needle shows on AM and SW is not recorded. Chapter 3, section 3.2 (p. 16), lists each one; where each sits on the panel is not recorded.

2. Safety before you open it

Ambersong runs on mains. It holds a tube radio built in the old "AC/DC" way: no mains transformer, and its metal chassis wired straight to the mains. This short chapter lists every point where mains or high voltage is present, and when. Chapter 5 explains the same wiring in detail.

Danger — unplug the cord. The front power switch does **not** remove mains from the cabinet. The only way to remove mains is to pull the cord out of the mains inlet. Do it before you open the back panel or the front panel.

2.1 Where the mains is, and when

The table lists each point that carries mains or high voltage, and what must be on for it to be live.

Point	Live when	Notes
Mains inlet (IEC C14) and its wiring	the cord is plugged in	Live, neutral and earth.
5 V power supply , mains side	the cord is plugged in	It is wired ahead of the front power switch, so it never switches off.
Front power switch , input side	the cord is plugged in	The switch breaks the live wire only.
Amplifier ± 20 V supply , mains side	the front switch is on	Its neutral wire stays connected with the switch off.
Mains line filter , in its own cage	the front switch is on	Its neutral wire stays connected with the switch off.
Tube radio : its mains terminals, its own power switch	the front switch is on	
Tube radio : its 0.75 A fuse, its heater string, and everything after its own power switch	the front switch and the radio's own power switch are on	The seven tube heaters are wired in series from the fused live to the chassis, which is on neutral.
Tube radio's high-voltage supply (B+)	the radio is on	Made from the mains by a half-wave rectifier inside the radio (chapter 13 (p. 93)).
Tube radio's chassis (CHASSIS)	always wired to mains neutral	See section 2.3 (p. 14).

B+ is the radio's high-voltage supply for its tubes. The radio makes it by rectifying the mains directly. The chain is a 28 Ω surge resistor and a silicon rectifier, then three 47 μF 315 V capacitors with two dropping resistors between them. No bleeder resistor is drawn across these capacitors. The numbers in the names of its points (B130, B115_A) are the original service data's nominal figures, not measured voltages.

2.2 The front power switch does not make it safe

The front power switch is a single-pole toggle. It breaks **only the live wire**, and only to two things: the amplifier's ± 20 V supply and the tube radio (through the line filter). Three things follow:

- The **5 V power supply** stays on. Both microcontroller boards, the clock and everything digital keep running.

- The **neutral wire** is never switched. It stays connected to the amplifier supply, the line filter and the tube radio's chassis.
- **Only unplugging the cord removes mains** from the cabinet.

2.3 The radio chassis is on mains neutral

The tube radio has no isolation transformer. Its circuit common, **CHASSIS** — the metal chassis and every part connected to it — is wired to mains neutral through the line filter. It sits near 0 V only while the wall socket's live and neutral are the right way round, and **nothing in the machine checks that**. If they are reversed, the chassis is on the live wire, and it stays there with the front switch off, because the neutral wire is never switched.

CHASSIS is never joined to Earth, to the Faraday cages, to the DC ground, or to the amplifier supply return. The machine uses four ground names. Earth, the DC ground and the amplifier supply return are all joined; the radio chassis is joined to none of them:

Name	What it is
Earth (Earth)	the mains protective earth; the Faraday cages and the antenna connector's shell are on it
DC ground (GND)	the return of both boards and everything on 5 V and 3.3 V; reaches the amplifier's ground and so on Earth at almost 0 Ω . By which path it reaches the amplifier's ground is not recorded
Amplifier supply return (20V_GND)	the 0 V output of the amplifier's ± 20 V supply; its two middle black wires go to the amplifier's ground, which sits on the earth star point of the amplifier's back plate, so it too is on Earth
Radio chassis (CHASSIS)	the tube radio's circuit common, on mains neutral

Chapter 5, section 5.3 (p. 28), says what each one is joined to.

The DC ground is on Earth, at almost 0 Ω , through the amplifier's ground (a washer on its Bass pot, chapter 5 (p. 27), section 5.3 (p. 28)). By which path the DC ground reaches the amplifier's ground is not recorded. Anything you connect to the DC ground is connected to Earth.

2.4 Safety-rated parts

A few capacitors in the tube radio, on the mains side or on the antenna input, are safety-rated types. The table lists them, with the safety class of each.

Part	Where	Fitted
The antenna coupling capacitor (C3)	between the antenna's centre conductor and the radio's FM input	100 pF, class X1Y1
The antenna shield capacitor (C51)	between the antenna coax shield, which is on Earth, and the chassis: the shield's only path to the chassis	0.001 μ F, class X1Y2
The mains bypass capacitor (C44)	between the fused live and the chassis	0.01 μ F, class X1Y2
Three of the four heater-string bypass capacitors (C47, C49, C52)	from points on the heater string to the chassis	0.0022 μ F each, class X1Y2. The fourth (C48) is the original 0.002 μ F ceramic, not a safety type.

Fuses. No fuse is recorded on the machine's mains side; the tube radio has its own 0.75 A fuse. That fuse sits in the radio's switched live, between its own power switch and the rest of the set. Its type and voltage rating are not recorded.

2.5 The low-voltage side

The rest of the machine runs at low voltage. This is what each part of it is referenced to.

- The **amplifier supply** gives +20 V and −20 V (± 19.33 V measured with no load) around its own return, `20V_GND`. It is isolated from the mains and held off the amplifier's back plate by plastic screws and standoffs.
- The **5 V** and **3.3 V** buses, and both boards, are on the DC ground, which is on Earth (section 2.3 (p. 14)).

Caution — two power sources on one board. Each board has a USB-C port on the back panel, carried straight through to the board, power wire included. The boards are also powered from the 5 V bus on their 5V pin whenever the cord is plugged in. Espressif's guide for the ESP32-S3-DevKitC-1, the board the main board is compatible with, calls the USB port, the 5V pin and the 3V3 pin "three mutually exclusive ways to provide power to the board". Espressif's guide for its original ESP32 DevKitC V4 board says the same, and adds that power must come from one and only one of them, "otherwise the board and/or the power supply source can be damaged". Whether the fitted boards match that V4 board is not established. These are the guides' rules; the author gives no rule of his own. Plugging a USB cable into a back-panel port while the cord is in gives the board two of these inputs at once. Chapter 4, section 4.6 (p. 24), describes the ports.

3. The physical build

This chapter says where things are and how they are held, so that you can open the cabinet, find a part, and repair or rebuild it. It describes this cabinet; yours will differ. Screw details for the printed plates, cable routing and the length of each cable are not covered.

Danger — unplug the cord first. The front power switch does not remove mains from the cabinet. Read chapter 2 (p. 13) before you open it.

3.1 The cabinet and how to open it

A **wooden cabinet** holds everything. The tube radio's original case has been stripped; its original chassis sits in the cabinet. The table says how each main piece is held, so you know what to undo.

What	How it is held
Back panel	An inverted-T panel of ½ in plywood behind the radio, held by ten countersunk 3 mm screws.
Front panel	Held from the inside by five 3 mm screws.
Tube radio	Two #8 ½ in wood screws through its tabs, one on each side.
Everything else	Almost every board and mechanical part is on a small perfboard or module in its own printed PLA case, or on a printed PLA carrying plate, screwed to the cabinet with #8 screws.

Every cable inside is under 14 in long, long enough to service the machine with the panels open and nothing unplugged.

The order in which the back panel and the front panel come off is not recorded.

Beyond this, the book does not describe where each module sits inside the cabinet.

3.2 The front panel

The table lists the controls and displays on the front panel, with the chapter that covers each one. Where each one sits on the panel is not recorded.

Control or display	What it is	Chapter
Tuning knob	A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor. Clockwise raises the frequency.	8
Volume knob	The audio board's volume control, read digitally. It is not the amplifier's volume control, which is reached only from the back.	9
Source switch	A three-position switch (double-pole, on-off-on): RADIO and BT at the two ends, AUX in the centre.	9
Power switch	The front power switch. It turns on the amplifier and the tube radio, but does not remove mains from the cabinet.	2, 5
Bluetooth LED	A single part that is both the Bluetooth LED and the Bluetooth button.	9

Control or display	What it is	Chapter
Clock display	Four seven-segment digits and a colon, made for this panel.	7
FM panel lamps	Four LEDs, dimmed together by the main board.	9
FM panel	The needle shows the rough FM position against a blackened brass plate. What it shows on AM and SW is not recorded.	8

The source switch and the power switch are fixed in the wooden front panel itself, each by its own nut and then plenty of hot glue.

The tube radio's own power and tone switch (S2) and its own volume control (R1) are **not** on the front panel. Where on the radio they sit, and which position S2 is left in, are not recorded. The radio's volume control sits at about one eighth of its travel (chapter 13 (p. 93), section 13.11 (p. 128)).

The AUX jack is not in this list either. Where it sits is not recorded (chapter 6 (p. 33), section 6.7 (p. 38)).

3.3 Tuning, needle and index

The printed front knob, on a printed shaft extension, turns the radio's original tuning knob, which drives the dial cord and the tuning capacitor. A magnetic angle sensor (AS5600) reads the capacitor's shaft. The needle sits on a printed mount on a small stepper motor, which alone moves it. A magnet on that mount passes a Hall-effect switch, the needle's only position reference. Chapter 8, section 8.1 (p. 45), describes the mechanics and how each part is mounted.

3.4 Cooling

Two fans move air (chapter 5 (p. 27), section 5.6 (p. 31)):

- The **box fan** sits at the top of the back panel and blows air **out** of the cabinet. Fresh air comes in through two small louvred printed vents at the other two ends of the back panel's inverted T.
- The **cage fan** is screwed to the tube radio's Faraday cage and blows air **into** it.

3.5 The Faraday cages

Three earthed cages shield the tube radio, the angle sensor and the mains line filter. The cages over the tube radio and the angle sensor are printed PLA boxes lined with aluminium tape. The tape is covered with Kapton tape. The cage over the tube radio stands well clear of it. The line filter's cage is layers of tape over the circuit. Chapter 5, section 5.4 (p. 30), describes each cage and how it is earthed.

3.6 Connectors and labels

This section lists the main connectors, and how the cables are marked.

- **Mains inlet:** a standard IEC C14 inlet, the one that came on the amplifier's aluminium back plate. That plate is at the back of the cabinet. The amplifier's own volume control is reached only from the back. Where the amplifier's input panel sits is not recorded.

- **Connectors that fit only one place.** Some connector types are used once only, so that a cable cannot be plugged where it would break something. The mains feed to the front power switch is one of them.
- **Labels.** Each cable is marked at both ends, with an evident name or with its name on the cable sheet. This book uses the cable sheet's names (chapter 10 (p. 56)).
- **Power sockets.** The 5 V and 3.3 V buses are rows of small 2-pin JST XH sockets (chapter 5 (p. 27), section 5.5 (p. 30)).
- **Back-panel USB-C ports.** One for each microcontroller board (chapter 4 (p. 19), section 4.6 (p. 24)).
- **Antenna connector:** a BNC (J10) on the back panel, where the FM antenna's coax arrives (chapter 11 (p. 87)).

3.7 Printed parts

Every printed part of this build is listed in Appendix D (p. 143). All are PLA except the membrane of the passive radiator, which is TPU.

4. The two microcontroller boards

Two ESP32 development boards run the machine. The **main board**, an ESP32-S3, runs everything you see move or light: the clock display, the dial needle and its sensors, the FM calibration receiver and the panel lamps. The **audio board**, an ESP32 called "A32", handles the sound, Bluetooth, the front controls and the real-time clock. This chapter gives each board's identity, every pin it uses, the small parts around those pins, the back-panel USB-C ports and the real-time clock module. What the firmware does with each pin is in the Firmware Gospel.

4.1 The boards

Both are development boards: a module on a small board with pin headers and USB. The table gives what each one is, so you can identify it.

	Main board	Audio board
Also written	S3	A32
Board marking	ESP32-S3 N16R8 YD-ESP32-23 2022-v1.3	FCC IDL 2BB77-ESP32-32X (exactly as marked)
Module	marked ESP32-S3-N16R8, FCC ID 2AD6X-ESP32S3N16R8	meant to carry an ESP32-WROOM-32E; whether it matches that module is not established
Chip	ESP32-S3 (QFN56), revision v0.2	ESP32-D0WD-V3, revision v3.1; dual core, 240 MHz
Radio	Wi-Fi, Bluetooth LE	Wi-Fi, Bluetooth
Flash	16 MB, quad; flash voltage 3.3 V, set by eFuse (a one-time setting burned into the chip)	4 MB; flash voltage 3.3 V, set by a strapping pin (a pin the chip reads at power-up)
PSRAM (extra memory)	8 MB, inside the chip	—
Crystal	40 MHz	40 MHz
USB	the chip's own USB port (USB-Serial/JTAG); USB ID 303A:1001	a Silicon Labs CP210x USB-to-serial bridge; USB ID 10C4:EA60
On-board LED	an RGB LED on GPIO48	—

The main board is compatible with Espressif's ESP32-S3-DevKitC-1 in firmware, and pin-compatible with it for every pin this machine uses. The pins it does not use have not been checked. Neither module's maker has been identified (Appendix C (p. 139)).

4.2 Main board pin map

	row 1	row 2	
3.3 V out: main board 3.3 V	3V3	GND	ground
second 3.3 V pin	3V3	43	not wired
not wired	RST	44	not wired
old limit input: pull-up and cap only	4	1	amplifier-on sensor
needle index sensor	5	2	digit 2 select
segment driver input 1	6	42	digit 1 select
segment driver input 2	7	41	digit 4 select
segment driver input 3	15	40	digit 3 select
segment driver input 4	16	39	angle sensor clock (SCL)
segment driver input 5	17	38	angle sensor data (SDA)
segment driver input 7	18	37	not wired (memory)
segment driver input 6	8	36	not wired (memory)
not wired	3	35	not wired (memory)
not wired	46	0	not wired
stepper driver IN1	9	45	not wired
stepper driver IN2	10	48	calibration receiver clock + RGB LED
stepper driver IN3	11	47	calibration receiver data
stepper driver IN4	12	21	FM panel lamps (PWM)
serial link, to audio board	13	20	USB D+ (back-panel port)
serial link, from audio board	14	19	USB D- (back-panel port)
5 V in, from the 5 V bus	5V	GND	ground
ground	GND	GND	ground

 power and ground	 clock display	 needle and tuning	 link to the audio board
 lamps and amplifier sensor	 USB	 not wired	

Numbers are GPIO numbers. Row order as on Espressif's ESP32-S3-DevKitC-1; positions of unused pins not checked on this board.

The main board's two pin rows, with the job of each pin.

The table lists every main-board pin this machine uses. *Signal* is the name on the drawings. *Cable* names the cable's connectors as on the cable sheet (chapter 10 (p.56)); an arrow joins a cable's two ends.

Pin	Signal	What it does	Connects to	Cable
GPIO1	GPIO1	Amplifier-on input	The output transistor (collector) of the optocoupler in the amplifier power sensor. 10 kΩ pull-up to the main board's 3.3 V. Low when the amplifier is on.	S3_PWR_DET → POWER_DET
GPIO2	GPIO2	Digit 2 select	Input 2 of the digit driver, then the P-channel MOSFET that switches digit 2	S3_DIGIT → DIGIT_IN
GPIO4	GPIO4	Nothing (old limit-switch input)	10 kΩ pull-up and 0.1 μF to ground. Wired to the main board's S3_LIMITS socket, but its contact is not carried in the cable.	—
GPIO5	GPIO5	Needle index	The output of the Hall-effect index switch (A3144). 10 kΩ pull-up to the main board's 3.3 V and 0.1 μF to ground at the pin.	S3_LIMITS → LIMIT
GPIO6	GPIO6	Segment drive 1	Segment driver input 1	S3_DISPLAY → DISPLAY_IN
GPIO7	GPIO7	Segment drive 2	Segment driver input 2	same

Pin	Signal	What it does	Connects to	Cable
GPIO15	<code>GPIO15</code>	Segment drive 3	Segment driver input 3	same
GPIO16	<code>GPIO16</code>	Segment drive 4	Segment driver input 4	same
GPIO17	<code>GPIO17</code>	Segment drive 5	Segment driver input 5	same
GPIO8	<code>GPIO8</code>	Segment drive 6	Segment driver input 6	same
GPIO18	<code>GPIO18</code>	Segment drive 7	Segment driver input 7	same
GPIO9	<code>GPIO9</code>	Stepper coil drive	Stepper driver input IN1	<code>S3_STEPPER → STEPPER_IN</code>
GPIO10	<code>GPIO10</code>	Stepper coil drive	Stepper driver input IN2	same
GPIO11	<code>GPIO11</code>	Stepper coil drive	Stepper driver input IN3	same
GPIO12	<code>GPIO12</code>	Stepper coil drive	Stepper driver input IN4	same
GPIO13	<code>S3_to_A32</code>	Serial link, transmit	The audio board's GPIO26	<code>S3_UART → A32_UART</code>
GPIO14	<code>A32_to_S3</code>	Serial link, receive	The audio board's GPIO27	same
GPIO19	—	USB D−	The board's own USB port, which reaches a back-panel USB-C port (section 4.6 (p. 24))	—
GPIO20	—	USB D+	The same	—
GPIO21	<code>GPIO21</code>	FM panel lamps, dimming	Through 100 Ω to the gate of the lamps' MOSFET (chapter 9 (p. 50))	<code>S3_FM_LED → FM_LED_IN</code>
GPIO38	<code>GPIO38</code>	Angle sensor data (SDA)	10 kΩ pull-up to the main board's 3.3 V at the pin, then 220 Ω in series, then the AS5600's SDA	<code>S3_TUNER</code>
GPIO39	<code>GPIO39</code>	Angle sensor clock (SCL)	10 kΩ pull-up to the main board's 3.3 V at the pin, then 220 Ω in series, then the AS5600's SCL	<code>S3_TUNER</code>
GPIO40	<code>GPIO40</code>	Digit 3 select	Input 3 of the digit driver	<code>S3_DIGIT → DIGIT_IN</code>
GPIO41	<code>GPIO41</code>	Digit 4 select	Input 4 of the digit driver	same
GPIO42	<code>GPIO42</code>	Digit 1 select	Input 1 of the digit driver	same
GPIO47	<code>GPIO47</code>	Calibration receiver data (SDIO)	The RDA5807M's SDIO. No pull-up resistors.	<code>S3_RDA → RDA_I2C</code>
GPIO48	<code>GPIO48</code>	Calibration receiver clock (SCLK)	The RDA5807M's SCLK. No pull-up resistors. Also drives the board's own RGB LED.	<code>S3_RDA → RDA_I2C</code>
3V3	<code>S3_3V3</code>	The main board's 3.3 V output	The pull-ups above; the angle sensor and the FM calibration receiver, through their cables. Not joined to the 3.3 V bus.	<code>S3_TUNER, S3_3V3 → RDA_3V3</code>
3V3 (the second one)	—	Nothing	Nothing is wired to it. Both 3V3 pins are the same on-board 3.3 V rail.	—
5V	<code>5VDC</code>	Power in	The 5 V bus	5 V bus → <code>S3_5VIN</code>

Pin	Signal	What it does	Connects to	Cable
GND	GND	Ground	The DC ground	with the 5 V cable

Not used: GPIO0, GPIO3, GPIO35, GPIO36, GPIO37, GPIO43, GPIO44, GPIO45, GPIO46, and the reset pin (RST). GPIO35 to GPIO37 are taken by the 8 MB PSRAM and cannot be used, according to Espressif's datasheet.

Digit numbering. Digit 1 is the rightmost digit, the units of minutes; digits 2, 3 and 4 run to its left. The digit pins are therefore not in number order: GPIO42, GPIO2, GPIO40 and GPIO41 select digits 1 to 4. Which segment each segment drive lights is in chapter 7 (p. 40).

The amplifier-on input. The optocoupler's output transistor pulls GPIO1 to ground while the amplifier's supply lights its LED; the pull-up holds GPIO1 high otherwise. Chapter 9 describes the sensor.

Pins with a second job on the chip. Some used pins have other duties on the ESP32-S3; Appendix A (p. 130) lists the chip's limits.

Pins	Other duty on the chip	Here
GPIO19, GPIO20	USB D– and D+ of the chip's own USB port	used for USB only
GPIO39 to GPIO42	the chip's JTAG debug interface	used for the angle sensor clock and three digit selects
GPIO0, GPIO3, GPIO45, GPIO46	strapping pins, read at power-up	not used

4.3 Audio board pin map

The table lists every audio-board pin this machine uses, with the same columns as section 4.2 (p. 20). The two converters are the ADC and the DAC (chapter 6 (p. 33)).

Pin	Signal	What it does	Connects to	Cable
GPIO0	A_I00A, A_I00D	Master clock (SCK) for both converters	Two 33 Ω resistors at the pin, one per converter: A_I00A to the ADC's SCK, A_I00D to the DAC's SCK	A32_I2S (chapter 10 (p. 56), section 10.15 (p. 69))
GPIO4	A_I04	Audio data to the DAC (DIN)	The DAC's DIN, with a 100 k Ω pull-down to ground	A32_I2S
GPIO17	A_I017	Word clock (LRCK)	The LRCK of both the ADC and the DAC	A32_I2S
GPIO18	A_I018	Bit clock (BCK)	The BCK of both the ADC and the DAC	A32_I2S
GPIO19	A_I019	Audio data from the ADC (DOUT)	The ADC's DOUT	A32_I2S
GPIO21	A_I021	Real-time clock, clock line (SCL)	The clock module's SCL	A32_RTC → RTC_I2C
GPIO22	A_I022	Real-time clock, data line (SDA)	The clock module's SDA	A32_RTC → RTC_I2C

Pin	Signal	What it does	Connects to	Cable
GPIO25	A_I025	Bluetooth button	One side of the button; the other side is on ground. Active low: pressing it pulls the pin to ground. No pull-up resistor is drawn; the firmware uses the chip's internal one.	A32_BT_LED → BT_LED
GPIO26	S3_to_A32	Serial link, receive	The main board's GPIO13	A32_UART → S3_UART
GPIO27	A32_to_S3	Serial link, transmit	The main board's GPIO14	same
GPIO33	A_I033	Bluetooth LED	The LED's cathode; its anode is on 3.3 V (A_3V3). Active low: the pin lights the LED by pulling low. No series resistor is drawn.	A32_BT_LED → BT_LED
GPIO35	A_I035	Volume knob	The wiper of the front volume knob, a potentiometer with its ends on 3.3 V (A_VDC) and ground	A32_VOL → VOLUME
GPIO36	A_I036	Source switch	Through 1 kΩ to the source switch's resistor ladder; 0.1 μF to ground at the pin. Reads 3.3 V on RADIO, 0.81 V on BT, 0 V on AUX.	A32_SWITCH → LISTENER
3V3	A_3V3	The audio board's 3.3 V output	Feeds the 3.3 V bus (A_VDC)	A32_3V3
5V	5VDC	Power in	The 5 V bus	5 V bus → A32_5VIN
GND	GND	Ground	The DC ground	with the power cables

Not used: GPIO1, GPIO2, GPIO3, GPIO5, GPIO6 to GPIO11, GPIO12, GPIO13, GPIO14, GPIO15, GPIO16, GPIO23, GPIO32, GPIO34, GPIO39, and the enable pin (EN). GPIO6 to GPIO11 belong to the module's flash memory. GPIO34 and GPIO39 are not joined to GPIO36 or to each other on this board. An older comment in the firmware's pin file says these three pins were measured continuous; that comment is stale.

One clock, two names. A_I00A and A_I00B are both the audio board's GPIO0. Each converter gets its own 33 Ω series resistor, fitted at the audio board's pin; the two names only make each converter's wiring read clearly. Chapter 6 explains the audio bus.

GPIO36 is the selector signal only. On AUX the audio itself never reaches the audio board. The source switch, its ladder and the firmware's thresholds for the three levels are in chapter 9 (p. 50), section 9.3 (p. 52).

GPIO34, GPIO35, GPIO36 and GPIO39 are input-only on this chip. The volume knob (GPIO35) and the source switch (GPIO36) use two of them (Appendix A (p. 130)).

4.4 Main board support parts

These are the small parts fitted around the main board's pins, and why each is there.

Pin	Parts	Why
3V3	0.1 μF and 10 μF to ground	decoupling of the main board's 3.3 V
5V	100 μF and 0.1 μF to ground	decoupling of the 5 V input
GPIO1	10 kΩ pull-up to the main board's 3.3 V	the optocoupler only pulls the line low

Pin	Parts	Why
GPIO4	10 k Ω pull-up to the main board's 3.3 V, 0.1 μ F to ground	left from the old limit switch; the pin goes nowhere else
GPIO5	10 k Ω pull-up to the main board's 3.3 V, 0.1 μ F to ground	the Hall switch's output only pulls low; the pull-up sets its high level at 3.3 V
GPIO38, GPIO39	10 k Ω pull-up to the main board's 3.3 V at the pin, then 220 Ω in series toward the sensor	the angle sensor's two-wire bus
GPIO47, GPIO48	none	the calibration receiver's bus has no external pull-ups

Wattage is not recorded for any of these pull-up or series resistors.

The shields of the `S3_LIMITS` and `S3_TUNER` cables are on `GND` at the main board. The GPIO4 and GPIO5 capacitors return to the `S3_LIMITS` cable's ground and shield contact.

Firmware text that disagrees. Two items in the firmware's pin file describe this board differently. A comment speaks of "4.7k pull-ups to S3_3V3"; the angle sensor's pull-ups are 10 k Ω . A define, `S3_LIMIT_RIGHT 4`, names GPIO4, but no firmware code uses it, and GPIO4 goes nowhere beyond its socket. In both cases the hardware stated here is right.

4.5 Audio board support parts

These are the small parts fitted around the audio board's pins, and why each is there.

Pin	Parts	Why
3V3	0.1 μ F, 10 μ F and 10 μ F to ground	decoupling of the 3.3 V that feeds the 3.3 V bus
5V	0.1 μ F and 100 μ F to ground	decoupling of the 5 V input
GPIO0	two 33 Ω in series, at the pin, one per converter	one clock line for each converter
GPIO4	100 k Ω to ground	pull-down on the DAC's data input
GPIO36	1 k Ω in series from the ladder; 0.1 μ F to ground on the pin side	filters the source switch's level
GPIO36 ladder	22 k Ω from the ladder node to ground; 68 k Ω in the BT leg	sets the three levels (chapter 9 (p. 50))

The 22 k Ω and 68 k Ω ladder resistors are **on the audio board**, not at the switch. How they and the other parts in this table are mounted at the audio board is not recorded. Where the 100 k Ω pull-down sits along the DAC data line is not recorded.

4.6 The back-panel USB-C ports

The back panel has two USB-C sockets, one for each board. Each passes all four USB wires straight through to its board's USB-C connector: the 5 V power wire (VBUS), D $-$, D $+$ and ground. Each socket's other three ground pins are joined together. Nothing else on the sockets is recorded, including the USB-C configuration pins (CC).

Back-panel port	Goes to
main (P2)	the main board's own USB port: the chip's USB-Serial/JTAG, USB ID <code>303A:1001</code>
audio (P1)	the audio board's USB-C: the CP210x USB-to-serial bridge, USB ID <code>10C4:EA60</code>

How the firmware uses the ports is in the Firmware Gospel.

Caution — two power sources. VBUS reaches each board from the back panel while the 5 V bus also feeds the board's 5V pin. Espressif's guides treat these as mutually exclusive ways to power the board (chapter 2 (p. 13), section 2.5 (p. 15)).

4.7 The real-time clock module

The real-time clock is a **DS3231 module of the common ZS-042 style**, read by the audio board. The same module carries an **AT24C32 EEPROM** at I2C address 0x57, and a **CR2032** backup cell. Most modules of this kind carry such a cell, but not all, so a replacement module may not. Whether the module charges its cell is not recorded.

The table shows how the module is wired.

Module connection	Goes to	Cable
VCC	the 3.3 V bus (A_VDC)	A32_VDC pin 1 → RTC_3V3 pin 1
GND	DC ground	A32_VDC pin 2 → RTC_3V3 pin 2
SDA	audio board GPIO22 (A_I022)	A32_RTC → RTC_I2C
SCL	audio board GPIO21 (A_I021)	A32_RTC → RTC_I2C

The pin numbers are the cable sheet's. The cable sheet uses the name `A32_VDC` for a second jack with the opposite pin order (chapter 5 (p. 27), section 5.5 (p. 30)).

The module's I2C socket (`RTC_I2C`) is a 4-pin JST with only two pins connected, SDA and SCL. Which physical positions are the two unused ones is not recorded. The signal cable's shield ends on ground at the audio board only.

The DS3231's datasheet limits are not included in this book.

4.8 When something is wrong

Find what you see in the left column; the right column says what to check.

You see	Check
The main board's RGB LED is lit brightly	Nothing: it is normal. GPIO48 drives both that LED and the calibration receiver's clock.
The needle cannot find its index	The index switch on GPIO5: its cable (<code>S3_LIMITS</code> → <code>LIMIT</code>), its 5 V supply, and the 10 kΩ pull-up. GPIO4 on the same socket goes nowhere and is not the fault.
The angle sensor does not answer	The <code>S3_TUNER</code> cable, the main board's 3.3 V on it, and the two 220 Ω series resistors and 10 kΩ pull-ups at GPIO38 and GPIO39.

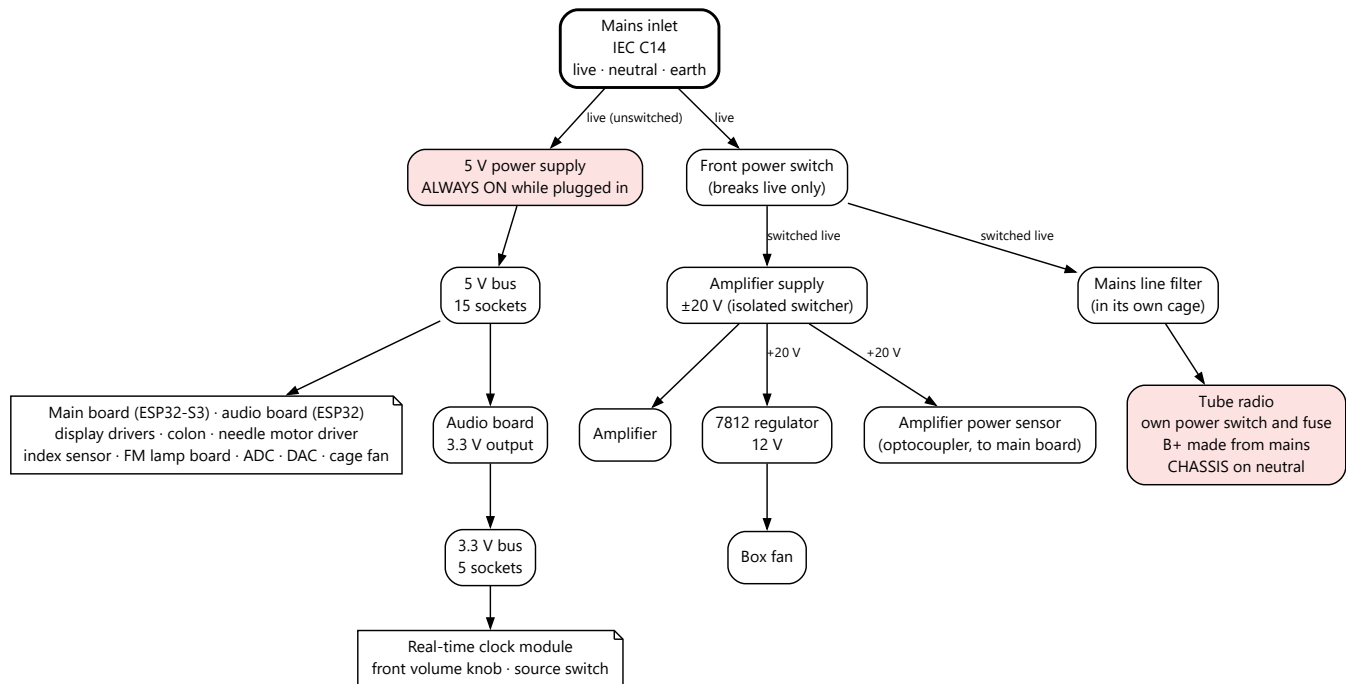
You see	Check
The calibration receiver does not answer	The <code>S3_RDA</code> signal cable and the <code>S3_3V3</code> power cable to the receiver. Its bus has no pull-up resistors; that is as built.
The two boards do not talk	The <code>S3_UART</code> cable: main board GPIO13 to audio board GPIO26, audio board GPIO27 to main board GPIO14, and its ground.
The clock forgets the time when the cord is pulled	The module's CR2032 backup cell.
A board is dead while the other runs	Its own 5 V cable from the bus (<code>S3_5VIN</code> or <code>A32_5VIN</code>), and its polarity at both ends.

5. Power, earth and grounds

This chapter follows the power from the wall socket to every board. It also explains the four different "grounds" in the machine. One of them, the tube radio's chassis, sits on mains neutral, so read section 5.3 (p. 28) before you touch anything.

Danger — mains inside the box. The front power switch does **not** remove mains from the cabinet. The 5 V power supply is always live while the cord is plugged in. The tube radio's metal chassis sits on mains **neutral**, and it is never joined to Earth. **Unplug the cord before you open the cabinet.** Chapter 2 lists every live point.

5.1 The power at a glance



The power tree: from the mains inlet to every board.

Mains enters through one socket: a standard **IEC C14 inlet**, the same kind as a computer's. It is the inlet that came on the amplifier's aluminium back plate, at the back of the cabinet. From there the power splits in two:

- **Before the front power switch:** the **5 V power supply**. It runs whenever the cord is plugged in. It feeds the 5 V bus, which powers both microcontroller boards and everything digital.
- **After the front power switch:** the **amplifier's ±20 V power supply**, and the **tube radio** (through the mains line filter). Switching the front switch off silences the radio and the amplifier, but not the clock or the computers.

The table shows where each part takes its power from, and when it is on.

What	Where its power comes from	On when
5 V power supply	mains live, before the front switch	the cord is plugged in
Both microcontroller boards, display, needle motor, sensors, DAC, ADC, cage fan	the 5 V bus	the cord is plugged in
3.3 V parts (real-time clock module, front volume knob, source switch)	the audio board's own 3.3 V output	the cord is plugged in
Amplifier's ± 20 V supply	mains live, after the front switch	the front switch is on
Amplifier, box fan, amplifier power sensor	the ± 20 V supply	the front switch is on
Tube radio	mains live after the front switch, through the line filter	the front switch is on and the radio's own power switch is on

5.2 The mains side

The table follows the mains wiring point by point, from the inlet to the tube radio.

Point	What it is
Mains inlet	IEC C14, three pins: live, neutral, earth.
Front power switch	A single-pole on/off toggle on the front panel. It breaks only the live wire , and only to the amplifier supply and the tube radio. The mains to it uses a connector type found nowhere else in the cabinet, so no other cable can be plugged into it by mistake.
Live, always on	Inlet live → 5 V power supply, and → the input side of the front switch.
Live, switched	Front switch output → amplifier supply, and → line filter → tube radio's live terminal.
Neutral	Inlet neutral → 5 V power supply, amplifier supply, and line filter → tube radio's neutral terminal.
Mains line filter	An external EMI filter between the switched mains and the tube radio. It took the place of two of the radio's original coils, which were removed by accident. It sits under its own small Faraday cage (section 5.4 (p. 30)).
Fuses	No fuse is recorded on the machine's mains side; the tube radio has its own 0.75 A fuse (chapter 2 (p. 13), section 2.4 (p. 14)).

The ratings of the front power switch and of the line filter are not recorded. Neither is the 5 V power supply's make, model or current rating.

The tube radio has no isolation transformer. Its circuit common — the metal chassis and every part connected to it — is wired straight to mains neutral. It sits near 0 V only while the wall socket's live and neutral are the right way round, and **nothing in the machine checks that**. The radio also makes its high-voltage supply (B+) by rectifying the mains directly (chapter 13 (p. 93)), and that supply is present whenever the radio is on.

5.3 The four grounds

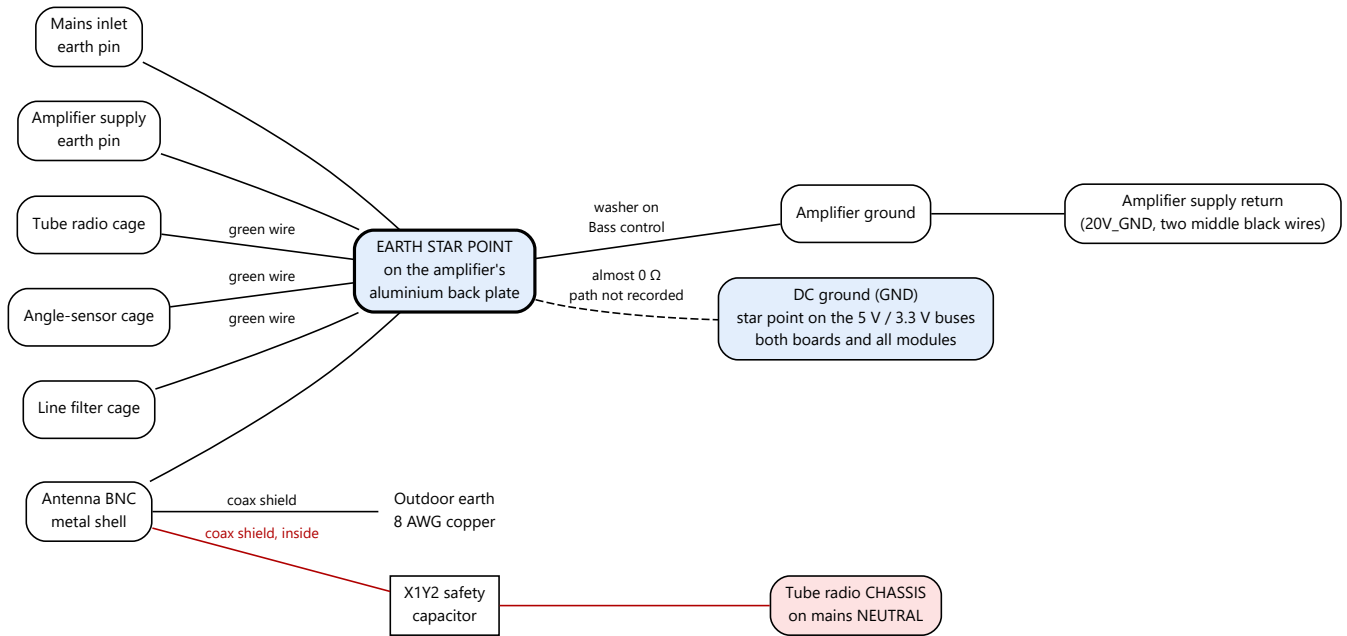
The machine has four ground names, kept different on purpose. They are not four separate grounds. Three of them, Earth, the DC ground and the amplifier supply return, are all joined. The fourth, the radio chassis, is joined to none of them. The table says what each one is joined to.

Name in this book	Also written	What it is	Joined to
Earth	Earth	The mains protective earth, from the inlet's earth pin.	The amplifier supply's earth pin, all three Faraday cages, the antenna connector's metal shell, and the amplifier's aluminium back plate, whose earth star point carries the amplifier's ground (below). The DC ground and the amplifier supply return are on it too.
DC ground	GND	The common return of everything on the 5 V and 3.3 V side: both microcontroller boards, the display, the motor, the sensors, the ADC and DAC.	All its wires meet at one star point: the ground pins of the 5 V and 3.3 V buses, and from there reach the amplifier's ground (below), so it is on Earth at almost 0 Ω . By which path it reaches the amplifier's ground is not recorded.
Amplifier supply return	20V_GND	The middle (0 V) output of the amplifier's ± 20 V supply.	The amplifier's ground, through the supply's two middle black wires (section 5.6 (p. 31)), and so Earth. Also the amplifier power sensor's LED cathode, the box-fan regulator's return and the box fan's –.
Radio chassis	CHASSIS	The tube radio's circuit common. On mains neutral.	Mains neutral only. It is not joined to Earth, to the Faraday cages, to the DC ground, or to the amplifier supply return. The antenna's shield reaches it only through one safety-rated capacitor (section 5.4 (p. 30)).

How the three are joined. Earth, GND and 20V_GND are all joined. The amplifier's ground sits on the earth star point of its aluminium back plate, through a washer on the Bass pot (the amplifier's bass control). This joint is part of the amplifier's original design; it is not on any drawing. The amplifier supply's ground wires, the two middle black ones (20V_GND on the drawings), go to that amplifier ground. The DC side's GND, both boards and the rest, reaches that same amplifier ground and so Earth, at almost 0 Ω . By which path the GND buses reach the amplifier's ground is not recorded. The amplifier supply itself is insulated from the plate by plastic screws and standoffs, and the amplifier's volume and treble controls do not touch the plate. CHASSIS is joined to none of them.

The amplifier power sensor. The sensor is an optocoupler. Its LED is lit from the amplifier supply (the 20V_GND side) and its transistor is read by the main board (the GND side). Inside the sensor no wire joins the two sides; the "amplifier is on" signal crosses by light. The two returns are still joined outside it, as described above. The sensor itself is in chapter 9 (p. 50).

5.4 Earth, cages and the antenna



Where Earth goes, and the one path from the antenna to the radio's chassis.

The three **Faraday cages** shield the noisiest or most sensitive parts:

Cage	What it covers	How it is built
Tube radio cage	the whole tube radio, well clear of it	A PLA box lined with aluminium tape, covered with Kapton tape. A cage fan on the 5 V bus blows air into it.
Sensor cage	the magnetic angle sensor on the tuning shaft (AS5600)	Built the same way. The sensor is mounted on the cage's base, clear of the aluminium and copper tape.
Line filter cage	the mains line filter	Kapton tape over the circuit, then aluminium tape, then Kapton again.

Each cage is earthed the same way: a stranded **green** wire, held against the aluminium by copper tape with conductive adhesive.

The antenna. The FM antenna's coax arrives at the antenna connector, a BNC on the back panel. The connector's shell, and so the coax shield, is on **Earth**. Outside, an 8 AWG stranded copper run goes from the antenna's grounding block to a lug on the meter base (chapter 11 (p. 87)). Inside, the shield reaches the radio's **chassis** only through one capacitor: a safety-rated X1Y2 type inside the tube radio.

5.5 The 5 V and 3.3 V buses

All the low-voltage power is shared out from two **star-point buses**: rows of small 2-pin JST XH sockets, held in their own small PLA cases. Devices plug their power cables into these sockets. Each socket carries power on one pin and DC ground on the other. Every ground pin is the same wire: this is the "single star point" of the DC ground.

Bus	Positions	Fed from	What runs on it
5 V bus	15 sockets	the 5 V power supply	main board (ESP32-S3), audio board (ESP32), clock display colon, display driver board, needle motor driver, needle index sensor, FM panel-lamp board, ADC board, DAC board, cage fan
3.3 V bus	5 sockets, plus the two jacks named for the audio board (A32_VDC, A32_3V3) on the same wire; where those two sit is not recorded	the audio board's own 3.3 V output	the same 3.3 V wire also feeds the real-time clock module, the front volume knob and the source switch; which of them plug into bus sockets is not recorded

The 3.3 V bus is fed from the audio board's 3.3 V pin, which is decoupled there by one 0.1 μ F and two 10 μ F capacitors. The main board's own 3.3 V output does not go to the bus. It feeds only the two parts on its own cables: the FM calibration receiver (RDA5807M) and the angle sensor (AS5600).

Two jacks named A32_VDC. The cable sheet gives the name A32_VDC to two different jacks, and draws them with opposite pin orders:

Jack named A32_VDC	Pin 1	Pin 2	Its cable goes to
the real-time clock's supply	A_VDC (3.3 V)	GND	the clock module's RTC_3V3
the audio board's 3.3 V cable	GND	A_3V3 (3.3 V)	the audio board's A32_3V3

The pin numbers are the cable sheet's and settle nothing physical. Which 3.3 V jacks are bus positions and which are on the audio board is not recorded.

Caution — the two A32_VDC jacks. As drawn, the two jacks named A32_VDC carry 3.3 V and ground on opposite pins. The name alone does not tell which of the two jacks a cable belongs to.

Caution — the colon's power cable. The 5 V cable to the clock display's colon has its polarity reversed at the COLON_IN side, compared with the other 5 V cables.

A JST XH socket is rated 3 A, for wire from 30 to 22 AWG.

5.6 The amplifier supply and the box fan

The amplifier supply is the one that came with the amplifier: an isolated mains switching supply, measured at ± 19.33 V with no load. It reaches the amplifier on four wires:

Wire colour	Carries
red	+20 V
black (the two middle wires)	0 V, the amplifier supply return
yellow	−20 V

These colours are the author's reading, not verified inside the supply. The two middle black wires go to the amplifier's ground (section 5.3 (p. 28)). The same supply also feeds the box fan and the amplifier power sensor, from its +20 V output.

The box fan sits at the top of the back panel and blows air **out** of the cabinet. Fresh air comes in through two small louvred PLA vents at the other two ends of the back panel's inverted T. The fan runs on 12 V, made from the +20 V output by an LM7812 regulator. It therefore turns only while the front switch is on. The regulator's parts:

Part	Value
Regulator	LM7812, input from +20 V, output 12 V
Input capacitors	0.1 μ F and 100 μ F, to the supply return
Output capacitors	0.1 μ F and 100 μ F, to the supply return
Protection diode	one diode across the 12 V output; its type and which way round it is fitted are not recorded

The fan harness has two 2-pin plugs: **FAN_IN** brings +20 V and its return, and **FAN_OUT** takes 12 V and its return to the fan. Which board they sit on is not recorded.

The cage fan is a small 5 V fan screwed to the tube radio's Faraday cage, blowing air into it. It plugs into the 5 V bus, so it runs whenever the cord is plugged in.

The size and make of the two fans are not recorded.

5.7 When something is wrong

Find what you see in the left column; the right column says what to check.

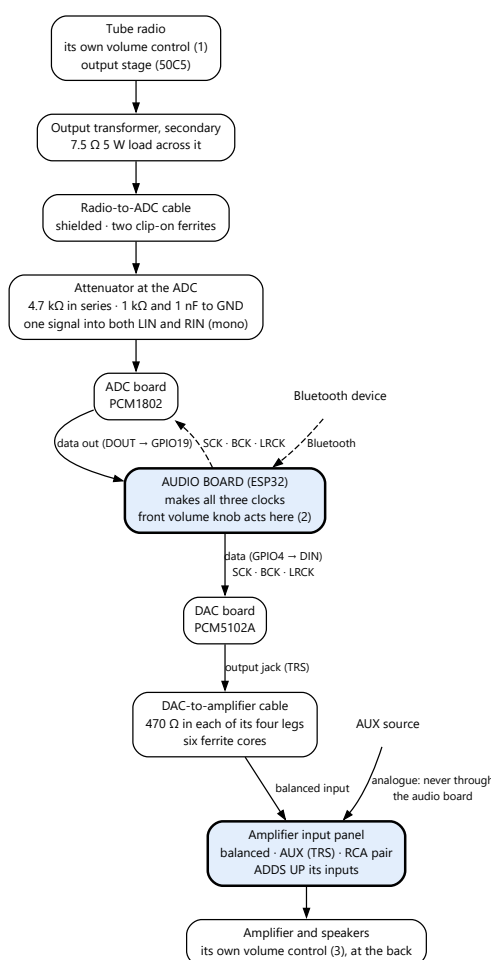
You see	Check
Nothing at all works: no clock, no sound	The cord and the wall socket. The 5 V supply has no switch; if it is dead, everything digital is dead.
Clock and needle work, but no sound from radio or amplifier	The front power switch. Then the amplifier supply: +20 V on the red wire, -20 V on the yellow.
Radio silent but Bluetooth plays	The tube radio's own power switch, and its fuse (chapter 13 (p. 93)).
One board or module dead, the others fine	Its own power cable to the bus, and the polarity at both ends.
Box fan stopped while the radio plays	The FAN_IN and FAN_OUT plugs, then 12 V at the regulator's output.

6. The audio chain

This chapter follows the sound from each source to the amplifier. The tube radio's audio is turned into digital audio by an ADC (analogue-to-digital converter) and passed through the audio board. A DAC (digital-to-analogue converter) then turns it back into an analogue signal. Bluetooth joins at the audio board. AUX, the analogue input you plug a cable into, goes straight to the amplifier. When the sound is missing, distorted or noisy, this chapter tells you which piece to look at.

What the audio board's firmware does with the sound (sample rate, volume, muting, source changes) is in the Firmware Gospel, chapter 10.

6.1 The path at a glance



The audio chain: from each source to the speakers, with the three volume controls numbered.

The machine has three sources, and each takes its own road:

Source	What carries it	Converted by
RADIO	the tube radio's output transformer → the radio-to-ADC cable → the audio board	the ADC, then the DAC
BT	Bluetooth → the audio board	the DAC

Source	What carries it	Converted by
AUX	an analogue run to the amplifier's input panel	nothing: it never reaches the audio board

Every source ends at the **amplifier's input panel**, which adds up whatever reaches it (section 6.7 (p. 38)). The amplifier and its speakers are in chapter 12 (p. 90).

6.2 The digital audio bus (I2S)

I2S is a simple way to send digital audio between chips over a few wires: a **bit clock** (BCK) that ticks once per data bit, a **word clock** (LRCK) that marks left and right samples, one **data** line per direction, and a **master clock** (SCK) that the converters run their insides from.

The audio board makes all three clocks. The ADC runs in slave mode: it takes its clocks from outside. The DAC is given its master clock. So neither converter makes its own timing. Both converters share the same bit clock and word clock, so they always run at the same rate.

The table gives each clock and data line, from the audio board's pin to each converter's pin:

Signal	Audio board pin	ADC (PCM1802) pin	DAC (PCM5102A) pin	Notes
Master clock	GPIO0	SCKI, as A_I00A	SCK, as A_I00D	One pin, two branches, each through its own 33 Ω at the audio board's pin
Bit clock	GPIO18 (A_I018)	BCK	BCK	shared
Word clock	GPIO17 (A_I017)	LRCK	LRCK	shared
Data from the ADC	GPIO19 (A_I019)	DOUT	—	
Data to the DAC	GPIO4 (A_I04)	—	DIN	100 k Ω pull-down to ground

A_I00A and A_I00D are the same GPIO0. They carry two names only so that each converter's wiring reads clearly (chapter 4 (p. 19), section 4.3 (p. 22)). GPIO0 is also one of the chip's strapping pins, read at power-up (Appendix A (p. 130)).

All of these signals travel in one shielded harness from the audio board. The harness splits into an ADC branch and a DAC branch. The shield is on ground at the audio board end only. Chapter 10, section 10.15 (p. 69), gives it wire by wire.

6.3 The ADC (PCM1802)

The **ADC** (analogue-to-digital converter) is a PCM1802 on a small module board. It turns the tube radio's audio into digital samples for the audio board.

Power. The module's 5V pin is on the 5 V bus, through its own cable (chapter 10 (p. 56), section 10.25 (p. 78)), with 100 μ F and 0.1 μ F to ground. The module has its own 3.3 V pin, named P_3V3 on the drawing. Only one thing is wired to that pin: the FMT0 setting pin (below).

Caution — two unconnected nodes share the name P_3V3 . The module's own 3.3 V pin is named P_3V3 , and only FMT0 is wired to it. The drawing also uses the same name, P_3V3 , for a separate label that carries two other setting pins (PDWN and FSYNC, below); that label appears on no other drawing and is not joined to the module's pin. The name alone does not tell you which node you are looking at.

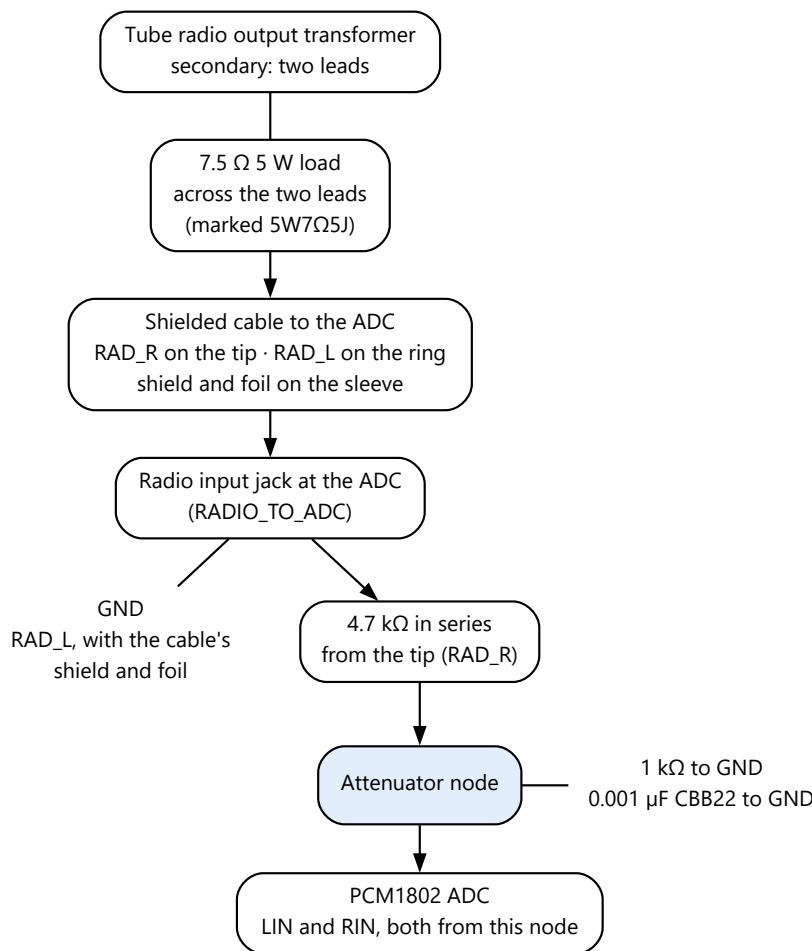
Setting pins. The PCM1802 is configured by tying pins high or low. The table gives each pin, what it is tied to, and what that means by the datasheet (Appendix A (p. 130)).

Pin	Tied to	Meaning
MODE0, MODE1	ground	Slave mode: the clocks come from the audio board
FMT1	ground	with FMT0 high: I2S, 24-bit
FMT0	the module's own 3.3 V pin (P_3V3), and nothing else	high (see FMT1)
OSR	ground	oversampling ×64
BYPAS	ground	high-pass filter on : DC is cut from the input
PDWN (power-down, active low) and FSYNC (frame sync)	joined together, as drawn, on the separate P_3V3 label. That label goes nowhere else, and it is not joined to the module's own 3.3 V pin.	see below

By the chip's datasheet, PDWN, BYPAS, OSR, FMT0, FMT1, MODE0 and MODE1 have internal pull-downs (about 50 kΩ). In slave mode the PCM1802 sends data only while FSYNC is high, and it runs only while PDWN is high. These are facts about the chip. On the fitted module the converter runs, so PDWN is not held low. How the module itself ties PDWN and FSYNC is not recorded.

Data and clocks reach the module on a 6-pin header: data out, bit clock, word clock and master clock; two of its pins are unused. Its pin order is in chapter 10 (p. 56), section 10.15 (p. 69).

6.4 The tube radio into the ADC



The radio's output, from the transformer to the ADC's inputs.

The load. The tube radio's output transformer has a two-lead secondary. In the original set it fed the radio's speakers; here it feeds the ADC. A **7.5 Ω 5 W resistor** sits across the two leads as the transformer's load (it is marked 5W7Ω5J). The transformer's primary belongs to the radio's output stage, a 50C5 tube (chapter 13 (p. 93)).

The cable. The two secondary leads, `RAD_L` and `RAD_R`, run in a shielded cable to the **radio input jack at the ADC** (J51, `RADIO_TO_ADC`), on a three-pole (TRS: tip, ring, sleeve) plug:

Plug contact	Carries
tip	<code>RAD_R</code> , one transformer lead
ring	<code>RAD_L</code> , the other transformer lead
sleeve	the cable's shield and foil

Tip and ring are interchangeable: which transformer lead goes to which does not matter. Two clip-on ferrite cores sit on this cable, one near the transformer and one near the ADC. How the wires end at the transformer is not recorded. Chapter 10, section 10.22 (p. 76), has the cable and the cores.

The jack's output side. Toward the ADC, the jack's sleeve and ring are on ground and its tip goes into a **4.7 kΩ** resistor. So `RAD_L` goes to ground, with the cable's shield and foil, and `RAD_R` goes through the 4.7 kΩ.

The attenuator. The 4.7 k Ω leads to the attenuator node. From that node a **1 k Ω** resistor and a **0.001 μ F CBB22 capacitor** (C59) go to ground. The node feeds **both** the ADC's left input (LIN) and its right input (RIN).

The radio feed is mono. One signal reaches both ADC inputs.

6.5 The DAC (PCM5102A)

The **DAC** (digital-to-analogue converter) is a PCM5102A on a small module board. It turns the audio board's digital audio, from the radio or from Bluetooth, back into an analogue signal.

Power. The module's VIN is on the 5 V bus. Its 5 V lead joins the audio harness's DAC branch (chapter 10 (p. 56), section 10.15 (p. 69)). At VIN there are three capacitors to ground, all fitted: **100 μ F**, **10 μ F ceramic** and **0.1 μ F**.

Data input. The DAC's DIN is the audio board's GPIO4, with a **100 k Ω pull-down** to ground. Where that resistor sits along the line is not recorded (chapter 4 (p. 19), section 4.5 (p. 24)).

Setting pins. Like the ADC, the DAC is configured by tying pins high or low:

Pin	Tied to	Meaning
FLT	ground	normal-latency filter
DEMP	ground	de-emphasis off
FMT	ground	I2S format
XSMT (soft mute)	pulled up to 3.3 V on the module itself	un-muted. Nothing outside the module reaches it.

The module's own 3.3 V pin (A3V3) is not connected to anything. The audio board's GPIO16 is unused: no wire runs from it to the DAC.

The firmware's pin file still defines `A32_PCM5102_XSMT` as 16, and the audio board's code sets GPIO16 as an output and drives it high. Since GPIO16 is not wired and XSMT is pulled up on the module, that code drives nothing. The hardware stated here is right; the firmware text is stale.

The DAC mutes itself on silence. By its datasheet, after 1024 word-clock periods of zero data on both channels (21 ms at 48 kHz) the PCM5102A puts its output in full analogue mute (Appendix A (p. 130)).

Output. The module has a three-pole (TRS: tip, ring, sleeve) output jack:

Jack contact	Carries
tip	left output (OUTL)
ring	right output (OUTR)
sleeve	the DAC's analogue ground (AGND)

6.6 From the DAC to the amplifier

A plug in the DAC's output jack feeds a short cable to the **balanced input** on the amplifier's input panel. A balanced input takes each channel on two wires, a plus and a minus, plus a ground. The cable puts a **470 Ω resistor in each of its four signal legs**. The two on the plus legs are at the DAC end. Each amplifier input is fed like this:

Amplifier input	Comes from
AMP_L+	the plug's tip (left output), through 470 Ω
AMP_R+	the plug's ring (right output), through 470 Ω
AMP_G	the plug's sleeve (DAC analogue ground), directly
AMP_L-	the plug's sleeve, through its own 470 Ω
AMP_R-	the plug's sleeve, through its own 470 Ω

So each minus leg is the DAC's ground seen through 470 Ω . The cable is under 12 inches long and carries **six ferrite cores**, three at each end. Each end has one HDMI-type sleeve, one small toroid and one large toroid, and the cable passes once through each. Chapter 10, section 10.21 (p. 75), gives the cable and the cores; chapter 12 (p. 90) gives what was measured at the amplifier's input.

What was observed. A frying noise was heard at the amplifier. It went away when this cable was unplugged at the amplifier, and almost went away when it was unplugged at the DAC.

Separately, once the clip-on ferrites were on the radio-to-ADC cable, no pops were audible. That was before the 470 Ω resistors on the plus legs and this cable's HDMI-type ferrite sleeves were added.

Whether the frying noise is gone now is not recorded, and neither is the reason for adding the 470 Ω resistors and the ferrite cores to this cable.

6.7 The AUX path and the amplifier's input panel

AUX is analogue and bypasses the audio board entirely. It never goes through the ADC or the DAC. The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn. Where the AUX jack sits is not recorded. On AUX the audio board sends silence to the DAC. Nothing the audio board does can change the AUX loudness or mute it.

The front source switch only tells the audio board which source you chose (chapter 9 (p. 50), section 9.3 (p. 52)). It does not switch any audio.

The input panel adds up its inputs; it does not choose between them. It has three inputs:

Input	What reaches it	With nothing plugged in
Balanced (L+, L-, ground, R-, R+)	the DAC, by the cable of section 6.6 (p. 37): RADIO and BT	not grounded
AUX (TRS jack)	see below	grounded
RCA pair	nothing recorded	grounded

Because the run from the AUX jack is not drawn, this book does not say which of the panel's inputs it lands on. Nothing is recorded on the RCA pair.

If two inputs carry sound at the same time, you hear both. On RADIO or BT, anything playing into AUX is still heard.

6.8 The three volume controls

Three volume controls sit in the chain, one at each stage. Their numbers match the figure in section 6.1 (p. 33):

No.	Control	Where	Acts on
1	The tube radio's own volume control	on the tube radio	RADIO only, before the ADC. In normal use it sits at about one eighth of its travel; above that, the sound is too loud further down the chain.
2	The front volume knob	front panel; read by the audio board (chapter 9 (p. 50), section 9.4 (p. 53))	RADIO and BT, inside the audio board. Not AUX.
3	The amplifier's own volume control	on the amplifier's board, reached only from the back	everything. It does not reach silence at its minimum.

The front knob is not the amplifier's volume control: it is read digitally by the audio board and does nothing to the analogue audio. The amplifier's own control is a B10k dual-gang potentiometer (chapter 12 (p. 90)).

6.9 When something is wrong

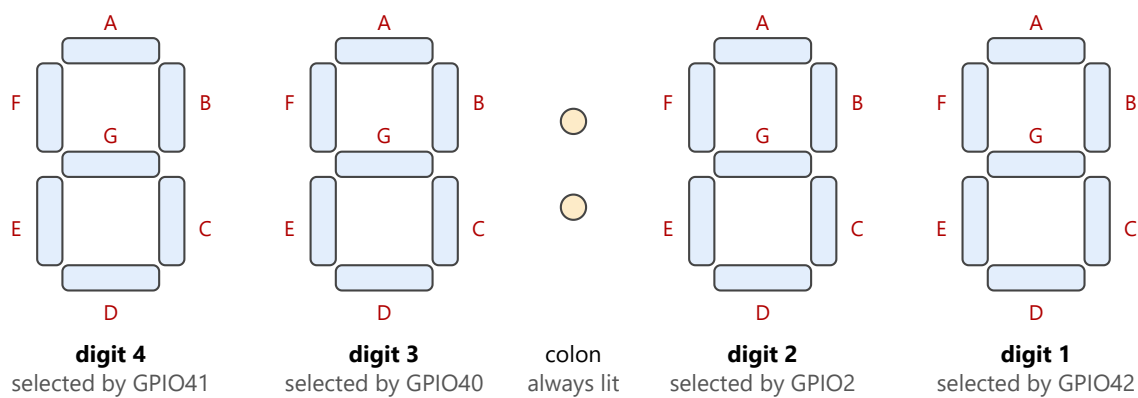
You see	Check
No radio sound, but Bluetooth plays	The radio side only. The tube radio itself: its power switch and its own volume control (chapter 13 (p. 93)). The radio-to-ADC cable and its plug in the radio input jack. The ADC branch of the audio harness and the ADC's 5 V cable (chapter 10 (p. 56), sections 10.15 (p. 69) and 10.25). By the chip's datasheet, PDWN and FSYNC must both be high for the ADC to send data (section 6.3 (p. 34)).
No sound from RADIO or BT, but AUX plays	The DAC side. The DAC branch of the audio harness and its 5 V lead (chapter 10 (p. 56), section 10.15 (p. 69)), the DAC-to-amplifier cable, and the front volume knob.
No sound from any source	The amplifier: the front power switch and the amplifier supply (chapter 5 (p. 27)), and the amplifier's own volume control.
Two sources heard at once	Normal: the input panel adds its inputs. Unplug whatever is playing into AUX.
Radio sound too loud further down the chain	The tube radio's own volume control: keep it near one eighth of its travel.
A frying noise at the amplifier	The DAC-to-amplifier cable: its four 470 Ω resistors and its ferrites. Unplugging it at the amplifier tells you whether it is the path.
The amplifier is never fully silent	Normal: its volume control does not reach silence at its minimum.

7. The clock display

The clock display shows the time on the front panel: four seven-segment digits and a colon, made for this front panel from single LEDs. The display is called the Leditron. The main board drives it through a small driver board. This chapter gives the digit numbering, which LED is which segment, how each digit and segment is switched, and the colon's own simple circuit.

What the main board shows on the display, and how fast it switches the digits, is in the Firmware Gospel, chapter 4 (p.19).

7.1 The display at a glance



Seen from the front. Each segment is one LED; red letters name the segments.

The clock display from the front: digit numbers, the pin that selects each digit, and the segment letters.

Digit numbering. Digit 1 is the **rightmost** digit, the units of minutes. Digits 2, 3 and 4 run to its left. The main board selects them with GPIO42, GPIO2, GPIO40 and GPIO41, in that order. So the select pins are not in number order. This order is exact: it is the order the firmware drives, and the order the drawings show (section 7.4 (p.42)).

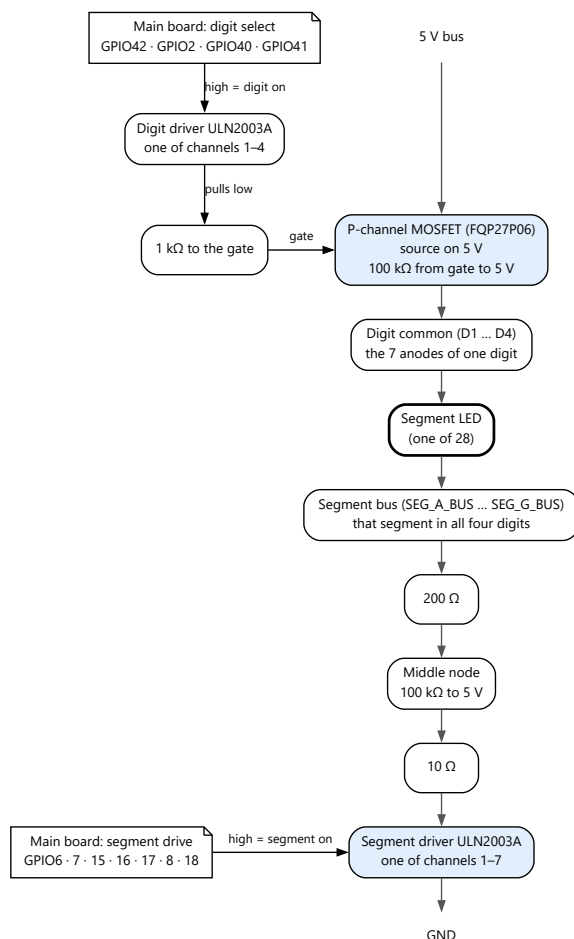
What it is made of. Each segment is one LED: 28 segment LEDs in all, seven per digit. In each digit the seven LEDs share one **common anode** (their plus side) and each has its own cathode. The colon is one more LED with its own supply (section 7.6 (p.43)).

The display is built from four parts:

Part	What it does
Four digits	Seven segment LEDs each, on an 8-pin header: pins 1 to 7 are the cathodes of segments A to G, pin 8 is the common anode.
Display bus board	It takes the seven segment lines and the four digit commons from the driver board, and shares them out on four 8-pin digit outputs. It also carries the colon's supply. It sits in its own printed case.
Display driver board	A perfboard that switches the digits and the segments: one ULN2003A for the four digit selects, four P-channel MOSFETs, and one ULN2003A for the seven segments.
Colon	One LED with a 33 Ω resistor and a variable resistor, fed straight from the 5 V bus.

A **ULN2003A** is a chip of seven switches to ground (Darlington transistor pairs). When an input is driven high, its output pulls down toward ground. It accepts the main board's 3.3 V logic directly.

7.2 How it is driven



One digit channel and one segment channel. A segment lights when both its digit and its segment are switched on.

The four digits share the **seven segment lines**: segment A of every digit is on one wire, segment B on another, and so on. Each digit has its **own common**, switched to 5 V by its own MOSFET. A segment LED lights only when its digit's common is on 5 V **and** its segment line is pulled low.

So, to show four different numbers, the main board lights one digit at a time and changes the segment lines between digits, fast enough that all four look lit. In short:

To light	The main board drives high
a digit	that digit's select pin (section 7.4 (p. 42))
a segment of that digit	that segment's drive pin (section 7.5 (p. 43))

7.3 The segment map

This table gives, for each segment: the main board pin that drives it, its pin on every digit's 8-pin header, and which LED is that segment in each digit.

Segment	Drive pin	Header pin on each digit	LED in digit 1	LED in digit 2	LED in digit 3	LED in digit 4
A	GPIO6	1	D1	D24	D17	D10
B	GPIO7	2	D3	D27	D20	D13
C	GPIO15	3	D5	D28	D21	D14
D	GPIO16	4	D6	D26	D19	D12
E	GPIO17	5	D4	D23	D16	D9
F	GPIO8	6	D2	D22	D15	D8
G	GPIO18	7	D7	D25	D18	D11
common anode	—	8	all seven	all seven	all seven	all seven

The LED references (D1 to D28) are only there to find a LED on a drawing. Do not confuse them with the four digit commons, whose signal names are **D1** to **D4** (section 7.4 (p. 42)).

Each digit's header plugs into its own 8-pin socket, pin for pin: pin 1 of all four sockets is the segment A line, pin 2 the segment B line, and so on to pin 7; pin 8 of each socket is that digit's own common. The display bus board's four 8-pin digit outputs carry the same signals on the same pin numbers. Whether those outputs are the sockets the digits plug into is not recorded (chapter 10 (p. 56), section 10.34 (p. 84)).

7.4 The digit drivers

Each digit's common anode is switched to 5 V by a **P-channel MOSFET** (FQP27P06). One ULN2003A, the digit driver, sits between the main board's select pins and the four MOSFET gates. Each digit has its own channel. The table is the chain as drawn:

Digit	Select pin	Digit driver input	Drives	Digit common
1 (rightmost)	GPIO42	1	the first MOSFET	D1
2	GPIO2	2	the second MOSFET	D2
3	GPIO40	3	the third MOSFET	D3
4 (leftmost)	GPIO41	4	the fourth MOSFET	D4

One channel, step by step:

1. The select pin drives the digit driver's input.
2. The driver's output (**PFET1** to **PFET4**) goes through a **1 kΩ** resistor to the MOSFET's gate.
3. A **100 kΩ** resistor pulls the gate up to 5 V, the MOSFET's source. With the select pin low, this holds the MOSFET off.
4. With the select pin high, the driver pulls the gate low, the MOSFET turns on, and its drain puts 5 V on the digit's common.

Per digit, a **0.1 μF** and a **10 μF** capacitor go from the 5 V at the MOSFET's source to ground.

The digit driver's ground pin is on the DC ground. Its COM pin (the common of its built-in protection diodes) is not connected, and its channels 5 to 7 are unused.

The digit order itself is certain (section 7.1 (p. 40)). The drawing does not settle where along the physical chain, from select pin to digit, the wiring sets that order, and it is not recorded. The author believes it is at the digit driver board.

7.5 The segment drivers

The second ULN2003A (the segment driver) pulls each segment line low. Each segment has its own channel:

Segment	Drive pin	Segment driver input	Segment line
A	GPIO6	1	SEG_A_BUS
B	GPIO7	2	SEG_B_BUS
C	GPIO15	3	SEG_C_BUS
D	GPIO16	4	SEG_D_BUS
E	GPIO17	5	SEG_E_BUS
F	GPIO8	6	SEG_F_BUS
G	GPIO18	7	SEG_G_BUS

Each channel runs: driver output → **10 Ω** → a middle node → **200 Ω** → the segment line. The middle node has its own **100 kΩ** to 5 V, which holds the line high (segment off) while the driver is off. The seven channels are fully separate: seven lines, one 100 kΩ each.

The segment driver's ground pin is on the DC ground. Its COM pin is not connected.

7.6 The colon

The colon is not switched by either board. It is lit whenever the 5 V bus is on.

Its circuit, in order: **5 V** → **33 Ω** → **the colon LED** → **a variable resistor** → **ground**. The variable resistor sits at the colon module. It is glued at **1.5 kΩ**.

The colon has its own 2-pin power cable from the 5 V bus, COLON_IN. It is the one 5 V cable whose polarity is reversed at the COLON_IN side (chapter 5 (p. 27), section 5.5 (p. 30)).

7.7 Boards and cables

The display hangs on seven cables. Chapter 10 draws each one wire by wire.

Cable	From → to	Carries	Cable page
S3_DISPLAY → DISPLAY_IN	main board → driver board	the seven segment drive pins	section 10.5 (p. 61)
S3_DIGIT → DIGIT_IN	main board → driver board	the four digit select pins	section 10.8 (p. 63)
5 V bus → DRIVERS_5VIN	5 V bus → driver board	the driver board's 5 V and ground	section 10.26 (p. 78)

Cable	From → to	Carries	Cable page
DIGIT_OUT → DRIVER_IN	driver board → display bus board	the four digit commons D1 to D4; one shielded cable over two plugs	section 10.27 (p. 79)
DISPLAY_OUT → LEDITRON_IN	driver board → display bus board	the seven segment lines	section 10.28 (p. 80)
5 V bus → COLON_IN	5 V bus → display bus board	the colon's 5 V and ground, reversed at COLON_IN	section 10.23 (p. 76)
colon output	display bus board → colon	the two colon LED leads	section 10.34 (p. 84)

7.8 When something is wrong

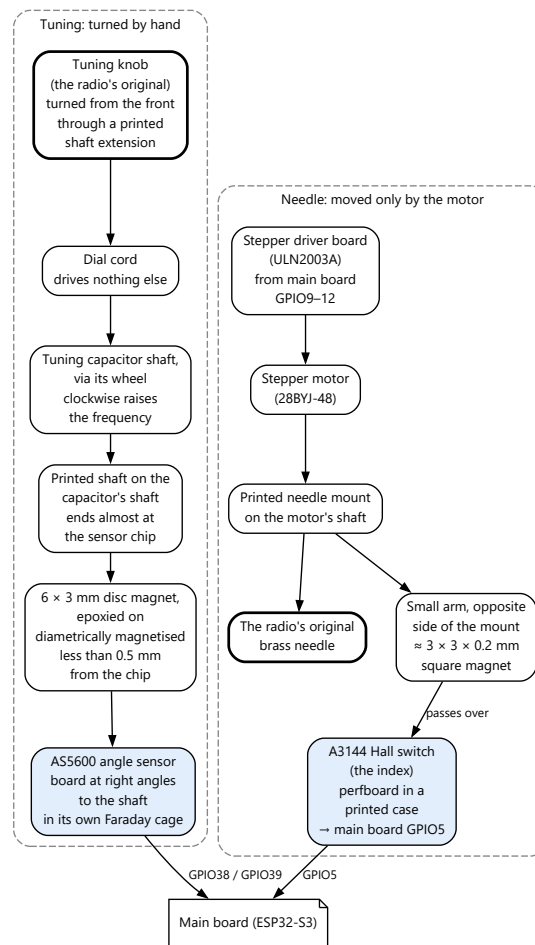
You see	Check
One segment dark in all four digits	That segment's channel: its drive pin, its segment driver channel, the 10 Ω and 200 Ω , its 100 k Ω , and its wire in the segment-line cable (DISPLAY_OUT → LEDITRON_IN).
One segment dark in one digit only	That segment's LED (section 7.3 (p. 42)), and its pin on that digit's 8-pin header.
One whole digit dark	That digit's select chain: its select pin, its digit driver channel, the 1 k Ω , the MOSFET, its wire in the digit-rail cable, and pin 8 of the digit's header.
All digits dark, colon lit	The driver board's 5 V cable (DRIVERS_5VIN), then the main board itself (chapter 4 (p. 19)). The digits also go dark while the main board takes a firmware update (Firmware Gospel).
All digits and the colon dark	The 5 V bus and the 5 V supply (chapter 5 (p. 27)).
Colon dark, digits fine	The colon's cable, and its polarity at COLON_IN; the 33 Ω , the variable resistor and the colon LED.
Digits in the wrong places after a rebuild	The digit order is GPIO42, GPIO2, GPIO40, GPIO41 for digits 1 to 4, rightmost first. Check each select pin's chain against section 7.4 (p. 42).

8. Dial, needle and tuning sensors

The tuning knob and the dial needle are not joined. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob, which drives the dial cord and the tuning capacitor as it always did. A magnetic angle sensor reads that capacitor's shaft, and a small stepper motor moves the needle to match. A magnetic index switch gives the needle its one fixed reference point. A small FM receiver chip serves as the instrument that calibrates the dial. This chapter describes that hardware: the parts, how they are mounted, and how they are wired.

What the main board does with them (homing, following the knob, the calibration) is in the Firmware Gospel: chapter 5 (p. 27) for the needle, chapter 6 (p. 33) for tuning and calibration.

8.1 The mechanics



The two halves of the dial: the tuning path, turned by hand, and the needle path, moved only by the motor.

Tuning. A printed front knob, on a printed shaft extension, turns the tube radio's original tuning knob. The original knob drives the dial cord, and the dial cord turns the tuning capacitor's shaft through its wheel. The cord drives nothing else. **Turning the knob clockwise raises the frequency.** The printed shaft extension is for tuning only.

The angle sensor's magnet. Another printed shaft, the angle sensor's own, extends the tuning capacitor's shaft almost to the angle sensor's chip. A disc magnet is epoxied to its end. The sensor's board stands at right angles to the shaft, with the shaft's axis through the centre of the chip (section 8.4 (p. 47)).

The needle. The tube radio's original brass needle sits on a printed mount on the stepper motor's shaft. The motor alone moves it; it has never been on the dial cord.

The index. A small arm on the opposite side of the same mount carries a tiny magnet. As the needle moves, the magnet passes over the index Hall switch (section 8.3 (p. 46)).

The angle sensor sits under its own earthed Faraday cage (chapter 5 (p. 27), section 5.4 (p. 30)).

8.2 The stepper motor and its driver

The needle motor is a **28BYJ-48** stepper motor. It is driven by a ULN2003A on a driver board; the motor's own lead plugs into the 5-pin socket that comes on the common ULN2003A boards. A ULN2003A is a chip of seven switches to ground (chapter 7 (p. 40), section 7.1 (p. 40)). The four coil lines run like this:

Main board pin	Driver input	Driver output	Motor socket pin	Coil, as named on the drawing
GPIO9	IN1	OUT1	1	coil 4
GPIO10	IN2	OUT2	2	coil 3
GPIO11	IN3	OUT3	3	coil 2
GPIO12	IN4	OUT4	4	coil 1
—	COM	—	5	motor common, on 5 V

Power. The driver board takes 5 V and ground from the 5 V bus on its own 2-pin cable (`STEPPER_5VIN`). The ULN2003A's COM pin, the common of its built-in protection diodes, and the motor's common (socket pin 5) are both on 5 V. A **0.1 μ F** and a **47 μ F** capacitor go from that 5 V to ground. The ULN2003A's ground pin is on the DC ground. Its channels 5 to 7 are unused.

The step sequence, speeds and microstepping are the firmware's (Firmware Gospel, chapter 5).

8.3 The index sensor (A3144) and its magnet

The **index** is the needle's only position reference. It is an **A3144 Hall-effect switch**: a sensor whose output switches on when a magnet comes close enough.

What	Detail
Where	On a small perfboard in its own printed case, under the path of the needle mount's small arm.
Magnet	A square neodymium magnet, about 3 × 3 mm and 0.2 mm thick, on the arm. Which pole faces the A3144 is not recorded.
Supply	5 V from the 5 V bus, with a 0.1 μF capacitor across the sensor's supply.
Output	Main board GPIO5 . A 10 kΩ pull-up to the main board's 3.3 V and a 0.1 μF capacitor to ground sit at the main board's pin.

How it switches. The A3144's output is open-collector: it can only pull the line low. By its datasheet, it pulls low when the magnetic field rises above its operate point, and lets go when the field falls below its release point. The pull-up sets the high level at 3.3 V, the main board's own level, even though the sensor runs on 5 V.

Pins (datasheet, viewed from the branded side): 1 supply, 2 ground, 3 output. On the sensor's cable (`LIMIT`): wire 1 is 5 V to the supply pin, wire 2 is ground, wire 3 is the output to GPIO5.

Replacing it. The A3144 is discontinued; its maker names the **A1104** as the replacement (Appendix A (p. 130)).

The main board's GPIO4, an old limit-switch input, is on the same main board socket but goes nowhere (chapter 4 (p. 19), section 4.2 (p. 20)).

8.4 The angle sensor (AS5600)

The **AS5600** is a magnetic angle sensor: a chip that reads the direction of a magnet spinning above it. It reads the tuning capacitor's shaft, through the magnet on the printed shaft that extends that shaft (section 8.1 (p. 45)).

What	Detail
Magnet	A 6 × 3 mm disc neodymium magnet, epoxied to the end of the printed shaft that extends the capacitor's shaft. It must be diametrically magnetised (its north and south poles on opposite sides of the disc, not on its faces). Less than 0.5 mm from the chip.
Mounting	Its board at right angles to the tuning capacitor's shaft, with the shaft's axis through the centre of the chip; the printed shaft reaches almost to the chip. Inside its own Faraday cage, on the cage's base, clear of the aluminium and copper tape.
Supply	The main board's own 3.3 V (<code>S3_3V3</code>) and ground. Not the 3.3 V bus. By the datasheet, the chip's VDD5V and VDD3V3 pins must be tied together for 3.3 V operation (Appendix A (p. 130)); whether the fitted module does so is not recorded.
Direction pin (DIR)	On ground. By the datasheet, the reading then increases clockwise.
Analogue output (OUT)	Not connected.
Bus	I2C, a two-wire bus: data (SDA) and clock (SCL).

Each bus line has a series resistor and a pull-up:

Sensor pin	Main board pin	Parts
SDA	GPIO38	220 Ω in series; 10 kΩ pull-up to the main board's 3.3 V at the main board's pin
SCL	GPIO39	the same: 220 Ω in series, 10 kΩ pull-up at the pin

The AS5600 answers at I2C address 0x36 (Appendix A (p. 130)). Its cable (`S3_TUNER`) carries ground with the shield, 3.3 V, SDA and SCL; the shield is on ground at the main board end. The connector at the sensor end is not recorded.

8.5 The FM calibration receiver (RDA5807M)

The **RDA5807M** is a single-chip FM receiver on a small module board. Here it plays no audio: it is the main board's instrument for calibrating the dial. Its antenna wire listens to the tube radio's own FM oscillator. That oscillator runs one IF (the radio's intermediate frequency, 10.6 MHz) below the station.

What	Detail
Bus	Data (SDIO) on main board GPIO47 , clock (SCLK) on GPIO48 . No pull-up resistors : that is as built. GPIO48 also drives the main board's own RGB LED (chapter 4 (p. 19), section 4.2 (p. 20)).
Address	The fitted chip answers at I2C address 0x11 . Its datasheet documents only 0x10.
Supply	The main board's own 3.3 V (<code>S3_3V3</code>) and ground. Across the supply, at the module: 100 μF electrolytic, 10 μF ceramic and 0.1 μF ceramic.
Unused pins	The clock input (RCLK), both audio outputs (LOUT, ROUT), and IO1 and IO2 are not connected.
Antenna	A 30 cm length of wire inside the tube radio's Faraday cage, bent in a U over the radio's FM oscillator section.
Antenna plug	The module reaches the wire through a 2 × 3 pin grid, so the module can be unplugged from its antenna there during a repair.

On the receiver's signal cable the shield ends at the **receiver** end, on ground, and is left open at the main board: the reverse of most cables (chapter 10 (p. 56), section 10.3 (p. 60)).

8.6 Cables

The dial hardware hangs on seven cables. Chapter 10 draws each one wire by wire.

Cable	Carries	Cable page
<code>S3_STEPPER</code> → <code>STEPPER_IN</code>	GPIO9 to GPIO12 to the driver's IN1 to IN4. The pin order reverses end to end; follow the signal names.	section 10.6 (p. 62)
5 V bus → <code>STEPPER_5VIN</code>	the motor driver's 5 V and ground	section 10.24 (p. 77)
motor lead → motor socket	the motor's four coils and its common	section 10.32 (p. 83)
<code>S3_LIMITS</code> + 5 V bus → <code>LIMIT</code>	the index sensor's output, ground and 5 V; one harness that joins a bus jack	section 10.9 (p. 64)
<code>S3_TUNER</code>	the angle sensor's ground, 3.3 V, SDA and SCL	section 10.10 (p. 65)
<code>S3_3V3</code> → <code>RDA_3V3</code>	the calibration receiver's 3.3 V and ground	section 10.2 (p. 59)
<code>S3_RDA</code> → <code>RDA_I2C</code>	the calibration receiver's SDIO and SCLK	section 10.3 (p. 60)

8.7 When something is wrong

You see	Check
The needle never moves	The motor driver board's 5 V cable, the stepper input cable, and the motor's lead in its socket.
The needle moves erratically or the wrong way	That each of GPIO9 to GPIO12 reaches IN1 to IN4 in order, by signal name, and that the motor lead is in its socket the right way.
The needle cannot find its index (the firmware reports NEEDLE FAULT; Firmware Gospel, chapter 5)	The index sensor: its 5 V from the bus jack on its harness, its cable to GPIO5, the 10 k Ω pull-up, and the magnet on the needle mount's arm.

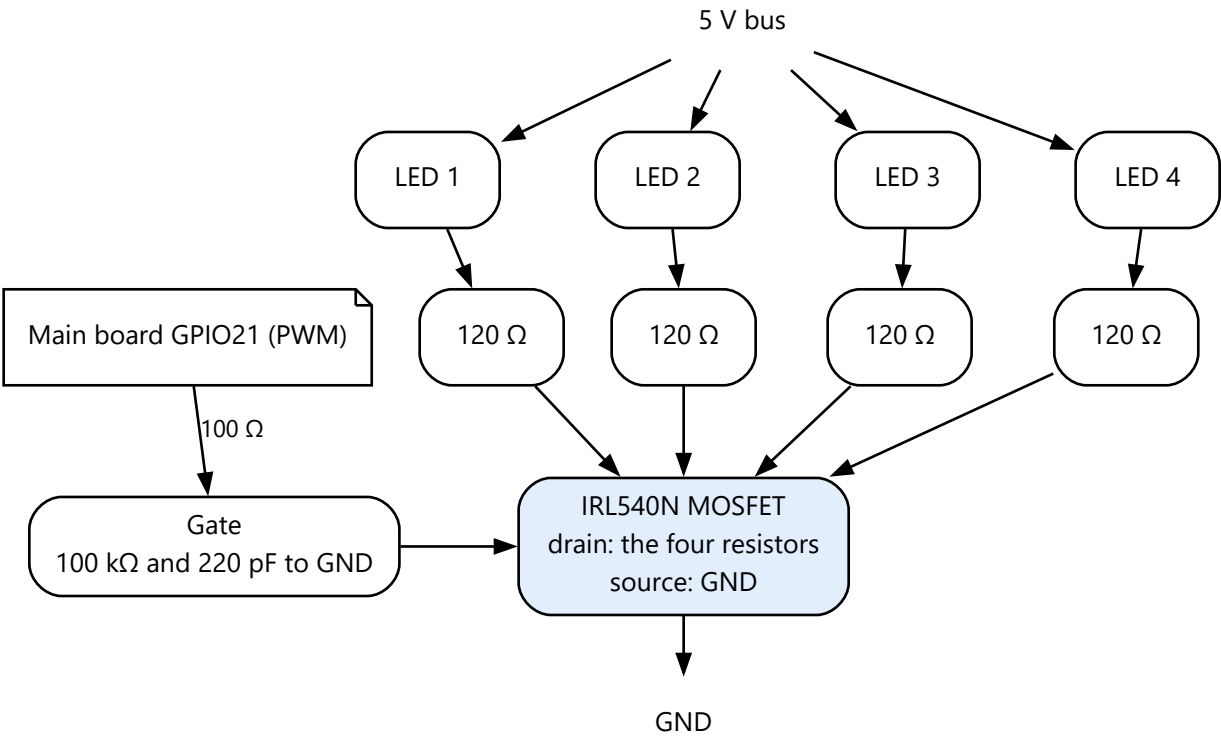
You see	Check
The needle does not follow the knob	The angle sensor: its cable, the main board's 3.3 V on it, the two 220 Ω resistors and the two 10 k Ω pull-ups. Then the magnet: diametrically magnetised, and within 0.5 mm of the chip.
The calibration receiver does not answer	Its two cables. It answers at 0x11, not the 0x10 of its datasheet. Its bus has no pull-ups; that is as built.
The receiver answers but finds no signal	Its antenna wire, the 2 $\times$ 3 pin grid it plugs through, and the wire's place over the radio's FM oscillator. The tube radio must be on.

9. Front-panel controls and lamps

The front panel carries the controls you touch and the lamps you see. The main board dims the four FM panel lamps and senses when the amplifier is switched on. The audio board reads the source switch, the volume knob and the Bluetooth button, and lights the Bluetooth LED. This chapter gives the circuit behind each one. Chapter 3, section 3.2 (p. 16), lists them; where each sits on the panel is not recorded. The tuning knob is in chapter 8 (p. 45).

What the firmware does with these controls (dimming, pairing, the volume law) is in the Firmware Gospel: chapter 4 for the lamps, chapter 10 for the audio board's controls. The source switch's thresholds are given here, in section 9.3 (p. 52).

9.1 The FM panel lamps



The FM panel lamps: four LEDs, each with its own resistor, switched together by one MOSFET.

The four FM panel lamps are LEDs, dimmed together by the main board. The main board dims them with **PWM** (pulse-width modulation): GPIO21 switches the lamps on and off very fast, and the share of "on" time sets the brightness. The lamp circuit is:

Part	Value	What it does
LED anodes	on 5 V	all four, from the 5 V bus
LED resistors	120 Ω , one per LED	each LED's cathode returns through its own resistor. The value was chosen for maximum brightness control.

Part	Value	What it does
MOSFET	IRL540N	its drain takes the four resistors, its source is on ground. A logic-level MOSFET: its gate threshold is 1 to 2 V (Appendix A (p. 130)).
Gate resistor	100 Ω	from GPIO21 to the gate
Gate pull-down	100 kΩ to ground	holds the lamps off while GPIO21 is not driven
Gate capacitor	220 pF to ground	across the pull-down

Wattage is not recorded for any resistor in this circuit.

The lamps light and dim with the PWM.

A comment in the firmware's pin file still speaks of "680 ohm on the 5 V rail". The lamp resistors are 120 Ω ; the comment is stale.

The panel-lamp board. The MOSFET and its parts sit on a small board of their own. It has three connections:

Connector	Carries	Cable page
FM_LED_IN, 3 pins	GPIO21 from the main board (shielded, shield on ground at the main board end), and 5 V and ground from a 5 V bus jack	section 10.11 (p. 66)
5 V out (5VOUT), 2 pins	as drawn, pin 1 on the board's 5 V and pin 2 on the four LED anodes; whether the two are one conductor or have a part between them is not recorded	section 10.33 (p. 83)
lamp returns, 4 pins	the four LED cathodes, each to its own 120 Ω on the board	section 10.33 (p. 83)

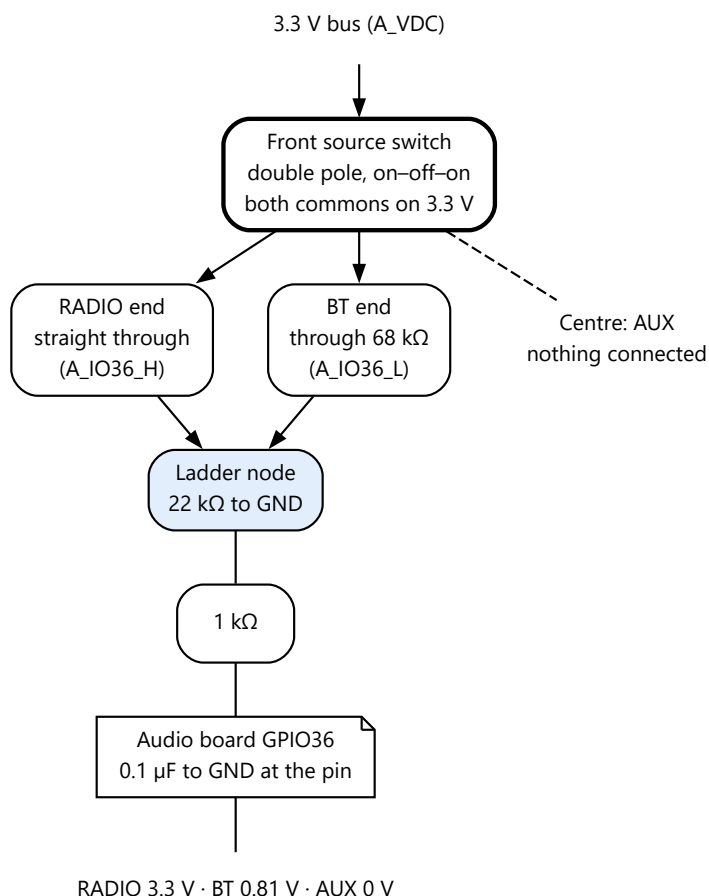
9.2 The Bluetooth button and LED

On the front panel, **one part is both the Bluetooth LED and the Bluetooth button**. Both sides are **active low**: pressing the button pulls its pin to ground, and the audio board lights the LED by pulling its pin low. Each side has its own audio board pin:

Side	Audio board pin	Wiring
Button	GPIO25 (A_I025)	between the pin and ground. No pull-up resistor is drawn; the firmware turns on the chip's internal one.
LED	GPIO33 (A_I033)	anode on the audio board's 3.3 V (A_3V3), cathode to the pin. No series resistor is drawn. Whether the LED part has a resistor of its own is not recorded.

One 4-wire cable, with no shield, carries 3.3 V, GPIO33, ground and GPIO25 to the front panel. Its pin order crosses from one end to the other (chapter 10 (p. 56), section 10.14 (p. 68)).

9.3 The source switch



The source switch's resistor ladder: one pin reads three voltages.

The front **source switch** chooses RADIO, BT or AUX. It is a double-pole, on-off-on switch: RADIO and BT at the two ends, AUX in the centre. **It switches no audio.** It only puts one of three voltages on the audio board's GPIO36, and the audio board picks its source from that. AUX audio never passes through the audio board at all (chapter 6 (p. 33), section 6.7 (p. 38)).

The ladder. A resistor ladder turns the switch position into one of three voltages on one pin:

- Both of the switch's commons are on the 3.3 V bus (A_VDC).
- At one end, one pole connects 3.3 V **straight** to the ladder node (the leg named A_IO36_H).
- At the other end, the other pole connects 3.3 V through **68 kΩ** to the ladder node (A_IO36_L).
- The two remaining end contacts are not connected.
- A **22 kΩ** resistor holds the ladder node to ground.
- The node reaches GPIO36 through **1 kΩ**, with **0.1 μF** to ground on the pin side.

Each position puts its own voltage on the pin:

Position	What reaches the node	Voltage on GPIO36
RADIO (one end)	3.3 V directly	3.3 V

Position	What reaches the node	Voltage on GPIO36
BT (other end)	3.3 V through 68 k Ω , against the 22 k Ω to ground	0.81 V
AUX (centre, off)	nothing; the 22 k Ω pulls the node down	0 V

The 22 k Ω and 68 k Ω ladder resistors are **on the audio board**, not at the switch, and the 0.1 μ F is at the audio board's pin.

The thresholds. The audio board's firmware reads GPIO36 in counts of its analogue input and sorts the reading like this. These numbers are one of the few facts this book takes from the firmware.

Reading on GPIO36	Source chosen	Measured in that position
below 424 counts	AUX	0 counts (0 V)
from 424 up to 2470 counts	BT	848 counts (0.81 V)
2471 counts and up	RADIO	4095 counts (3.3 V)

The wiring. Two cables meet on a small source-switch board, which is wired on to the switch. One comes from the audio board, and one from a 3.3 V jack. Which of the 3.3 V jacks are on the audio board and which are bus positions is not recorded.

Cable	Carries	Cable page
A32_SWITCH	the two legs, A_I036_L and A_I036_H, in a shielded cable (shield on ground at the audio board end only)	section 10.20 (p. 73)
A32_VDC_LISTENER	3.3 V (A_VDC) to the switch's commons	section 10.20 (p. 73)

No ground wire goes to the switch.

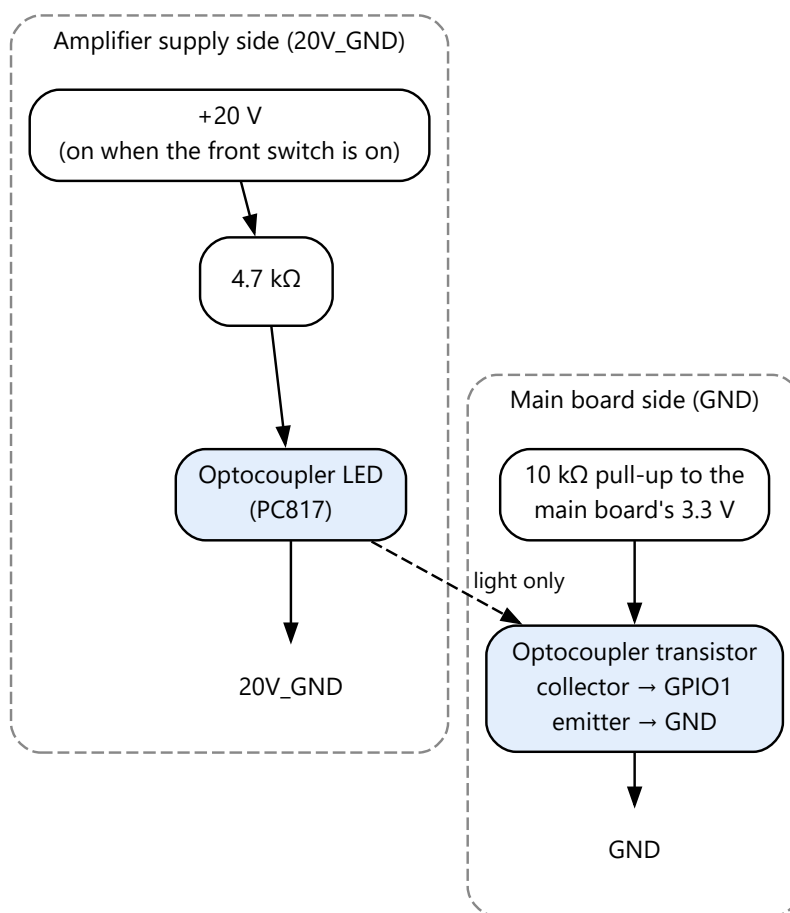
9.4 The front volume knob

The front **volume knob** turns a potentiometer read by the audio board. It is **not** the amplifier's volume control, which is reached only from the back (chapter 6 (p. 33), section 6.8 (p. 38)). The knob does nothing to the analogue audio: the audio board reads its position and sets the volume digitally, so it acts on RADIO and BT only, never on AUX. The potentiometer (RV4) is drawn as B10K. Its three pins go to:

Potentiometer pin	Goes to
one end	the 3.3 V bus (A_VDC)
wiper	the audio board's GPIO35 (A_I035), an input-only pin
other end	ground

Two cables meet in the potentiometer's 3-pin connector (VOLUME). One carries the wiper, shielded, with the shield on ground at the audio board end only. The other carries 3.3 V and ground (chapter 10 (p. 56), section 10.13 (p. 67)). The knob itself is a printed part (Appendix D (p. 143)).

9.5 The amplifier power sensor



The amplifier power sensor: the signal crosses by light; no wire inside the sensor joins its two sides.

The main board needs to know when the front power switch has turned the amplifier on. A small board tells it, through an **optocoupler** (PC817): an LED and a light-sensitive transistor in one package, with no electrical path between them. Its two sides are wired like this:

Side	Wiring
LED	+20 V from the amplifier supply → 4.7 kΩ → the LED's anode; the LED's cathode → the amplifier supply return (20V_GND)
Transistor	collector → main board GPIO1 , with a 10 kΩ pull-up to the main board's 3.3 V; emitter → DC ground (GND)

How it reads. The +20 V exists only while the front power switch is on (chapter 5 (p. 27)). Then the LED lights, the transistor conducts, and GPIO1 reads **low**. With the amplifier off, the pull-up holds GPIO1 **high**. Inside the sensor no wire joins its two returns, 20V_GND and GND; outside it they are joined, as Earth, GND and 20V_GND all are (chapter 5 (p. 27), section 5.3 (p. 28)).

Pins (datasheet): 1 anode, 2 cathode, 3 emitter, 4 collector.

Its cable to the main board carries GPIO1 and ground, shielded, with the shield on ground at the main board end (chapter 10 (p. 56), section 10.7 (p. 62)). What the firmware does when the amplifier goes on or off is in the Firmware Gospel.

9.6 When something is wrong

You see	Check
The FM panel lamps never light	The panel-lamp board's cable: 5 V and ground from its bus jack, and GPIO21. Then the MOSFET, and the 5 V lead to the LEDs.
One FM panel lamp dark	That LED, its 120 Ω , and its return lead to the board.
The Bluetooth button does nothing	Its wires to GPIO25 and to ground in the Bluetooth cable.
The Bluetooth LED never lights	3.3 V (A_3V3) and GPIO33 in the Bluetooth cable, and the LED's polarity.
The radio picks the wrong source	The voltage on GPIO36 in each position: 3.3 V, 0.81 V, 0 V. Then 3.3 V on A32_VDC_LISTENER, the two legs in A32_SWITCH, and the 68 k Ω and 22 k Ω on the audio board.
The volume knob does nothing	The wiper cable to GPIO35, and 3.3 V and ground on the knob's other cable. On AUX this is normal: the knob never acts on AUX.
The main board does not see the amplifier come on	GPIO1 should read low with the amplifier on and 3.3 V with it off. Check +20 V at the sensor board, the 4.7 k Ω , and the sensor's cable.

10. Cables and harnesses

Every board, sensor and control in the machine is joined to the others by a cable that plugs in at both ends. This chapter draws each of those cables, wire by wire, so that you can test one, trace a fault to it, or make a replacement. It is the only chapter that draws the cables themselves, wire by wire; other chapters draw circuits.

10.1 How to read the cable pages

Most of this chapter follows the **cable sheet**: the drawing that defines the important and less evident cables of the machine. Sections 10.2 to 10.36 each describe one cable, or a few wires that belong together. Every page uses the same headings in the same order, so you always know where to look. A heading is left out when there is nothing under it: a cable that uses all its pins has no *Not used*, for example.

Heading	What it tells you
Carries	What the cable is for: the signals or the power on it, and as much of the circuit at each end as you need to understand them.
Shape	How the cable is built: how many wires, whether it is shielded, the connector at each end with its name on the cable sheet, and whether the pin numbers cross from one end to the other.
<i>(drawing)</i>	The cable, drawn by signal name with the cable sheet's pin numbers. How to read it is at the end of this section.
Wire by wire	A table with one row per wire: the pin at one end, the signal, what it is, and the pin at the other end. A dash (—) means that the wire has no pin at that end. Where two cables meet in one connector, each cable has its own table.
Shield	Where the shield and foil end, or "none". A missing ground wire is noted here too, with where the board gets its ground instead.
Not used	Pins that carry no wire.
<i>(notes and cautions)</i>	Anything else you must know before you work on the cable: a warning box, parts built into the cable, ferrite cores.
If it fails	What you see when a wire in this cable breaks. Section 10.37 turns this round: it goes from what you see to the cable to check.

The cables in sections 10.29 (p. 80) to 10.36 are not on the cable sheet. Their pages are short, and they draw only what is recorded.

Pin numbers are the cable sheet's. The cable sheet numbers each connector's pins from the top of the connector's symbol, as that symbol is defined in the drawing's parts library: pin 1, pin 2, and so on. That number is **not** a number moulded on the connector. The order of the contacts inside a real connector is not recorded, so a pin number says nothing about where a wire sits in the real connector: **each wire is identified by its signal name, not by its position.** Where this chapter says that pins cross or run in opposite order, it speaks of that numbering only, not of the real connector.

Signals are named as on the cable sheet. The names you will meet:

Name	What it is
<code>GPIO_n</code>	In a signal name, pin <i>n</i> of the main board (ESP32-S3, "S3"). In the text, "the audio board's GPIO _n " is pin <i>n</i> of the audio board, whose signal names are <code>A_IOn</code> .

Name	What it is
A_IOn	Pin n of the audio board (ESP32, "A32")
5VDC	The 5 V bus (chapter 5 (p. 27))
S3_3V3	The main board's own 3.3 V output
A_VDC , A_3V3	The audio side's 3.3 V: the audio board's 3.3 V output, shared out by the 3.3 V bus. The two names are one rail.
GND	DC ground
CABLE S/FTP	The cable's shield and foil (below)

The shield rule. Where it is possible and useful, the cables are twisted pairs, shielded, foiled, or both. A shielded cable here is S/FTP: a braided shield (S) and a foil (F) around twisted pairs (TP). `CABLE S/FTP` on a pin means that the cable's shield and foil end on that pin, joined to ground on that board, and are **left unconnected at the cable's other end**, unless the page says otherwise. Where a pin reads `GND + CABLE S/FTP`, both a ground wire and the shield end there.

Length and labels. Every cable inside the cabinet is under 14 inches: long enough to service the box with its panels open and nothing unplugged. Each cable is marked at both ends, with a name that makes its job evident or with its name on the cable sheet.

What is not recorded. For the cables in this chapter, these are not recorded: wire colours (only the amplifier supply's four are recorded, as the author's reading, section 10.30 (p. 81)), which side is pin 1, the connector housings, each cable's exact length and route, and the contact order inside a connector.

Connectors. The 5 V bus jacks are JST XH; their rating is in chapter 5 (p. 27), section 5.5 (p. 30). Some connector types are used for one cable only, so that no cable can be plugged where it would break something. The mains feed to the front power switch is one of them.

In the drawings, each connector box gives its name, its pins and, at the bottom, what it belongs to. The middle box is the cable. Each of its lines follows one wire, left to right: the connector, pin and signal at one end, written `connector:pin:signal`; the wire's number and its signal; then the connector, pin and signal at the other end, written the same way. It repeats what the end boxes show, so that you can follow one wire along one line. `+ S` means the cable has a shield. A pin marked `(not used)` carries no wire. A wire that ends in the cable box with no connector after it goes to an end that is not recorded.

All the cables

This table lists every page in the chapter. In the **Wires** column, "+ shield" means that the cable is shielded. A second "+" adds the wires of another cable or connector described on the same page.

Section	Cable	From → to	Wires	Carries
10.2	FM receiver power	main board → FM calibration receiver	2	3.3 V and ground
10.3	FM receiver control	main board → FM calibration receiver	2 + shield	control bus (I2C)
10.4	Main board 5 V	5 V bus → main board	2	5 V and ground

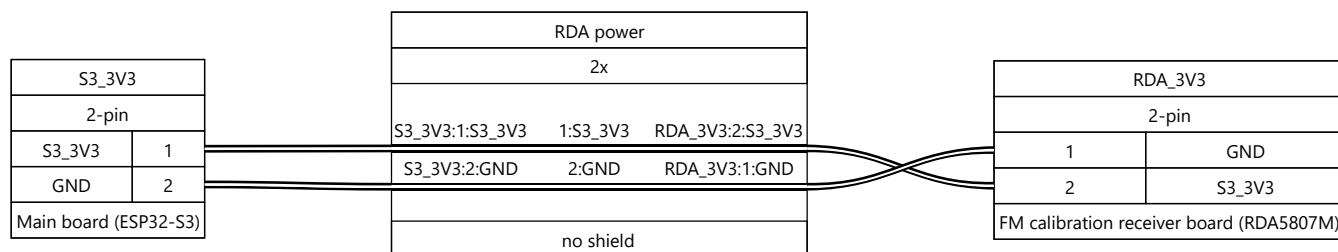
Section	Cable	From → to	Wires	Carries
10.5	Segment drive	main board → display driver board	7	which segments light
10.6	Needle motor control	main board → needle motor driver	4	the four coil signals
10.7	Amplifier power sensor	main board → power sensor board	2 + shield	"amplifier is on"
10.8	Digit select	main board → display driver board	4	which digit lights
10.9	Needle index sensor	main board and 5 V bus → index sensor	2 + shield, + 1	sensor output, ground, 5 V
10.10	Tuning angle sensor	main board → angle sensor	4 + shield	3.3 V, ground, control bus (I2C)
10.11	Panel-lamp control	main board and 5 V bus → panel-lamp board	1 + shield, + 2	lamp brightness, the board's 5 V
10.12	Board link	main board ↔ audio board	3 + shield	serial link both ways, ground
10.13	Front volume knob	audio board and 3.3 V → volume knob	1 + shield, + 2	knob position, 3.3 V, ground
10.14	Bluetooth button and LED	audio board → front panel	4	button and LED
10.15	Audio harness	audio board → ADC and DAC; 5 V bus → DAC	6 + shield, + 2	digital audio, the DAC's 5 V
10.16	Clock module power	3.3 V jack → real-time clock module	2	3.3 V and ground
10.17	Clock module control	audio board → real-time clock module	2 + shield	control bus (I2C)
10.18	Audio board 5 V	5 V bus → audio board	2	5 V and ground
10.19	Audio board 3.3 V	audio board → 3.3 V jack	2	the audio board's 3.3 V output
10.20	Source switch	audio board and 3.3 V → source-switch board	2 + shield, + 1	the RADIO / BT / AUX position
10.21	DAC to amplifier	DAC board → amplifier input	3 at the plug, 5 at the amplifier	analogue audio, balanced
10.22	Radio to ADC	tube radio's output transformer → ADC	2 + shield	the radio's audio, mono
10.23	Colon 5 V	5 V bus → display bus board, for the colon	2	5 V and ground, reversed at COLON_IN
10.24	Motor driver 5 V	5 V bus → needle motor driver	2	5 V and ground
10.25	ADC 5 V	5 V bus → ADC board	2	5 V and ground
10.26	Driver board 5 V	5 V bus → display driver board	2	5 V and ground
10.27	Digit rails	display driver board → display bus board	4 + shield	each digit's switched 5 V

Section	Cable	From → to	Wires	Carries
10.28	Segment buses	display driver board → display bus board	7	the seven segment lines
10.29	Box-fan harness	amplifier supply → fan regulator → box fan	2 + 2	+20 V in, 12 V out
10.30	Amplifier supply wires	amplifier supply → amplifier	4	±20 V and ground
10.31	USB	back panel → each board's USB port	4 each	USB, straight through
10.32	Needle motor lead	needle motor driver → needle motor	5	four coils and their common
10.33	Panel-lamp leads	panel-lamp board → the four FM lamps	1 + 4	5 V out, four returns
10.34	Inside the clock display	display bus board → colon and digits	2; 8 per digit	the colon LED, the digits
10.35	FM receiver antenna and the antenna coax	—	—	the calibration receiver's antenna wire; the FM antenna's coax (neither drawn as a cable)
10.36	Mains and earth wiring	—	—	see chapter 5 (p. 27)

10.2 FM receiver power — main board to FM calibration receiver

Carries: the main board's own 3.3 V (`S3_3V3`) and ground to the FM calibration receiver, an RDA5807M FM receiver chip on its own small module.

Shape: one 2-wire cable, with a 2-pin connector at each end: `S3_3V3` at the main board, `RDA_3V3` at the receiver board. The pins cross in the cable sheet's numbering: pin 1 at one end is pin 2 at the other.



The FM receiver power cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Receiver end, pin
1	<code>S3_3V3</code>	The main board's 3.3 V output	2
2	<code>GND</code>	Ground	1

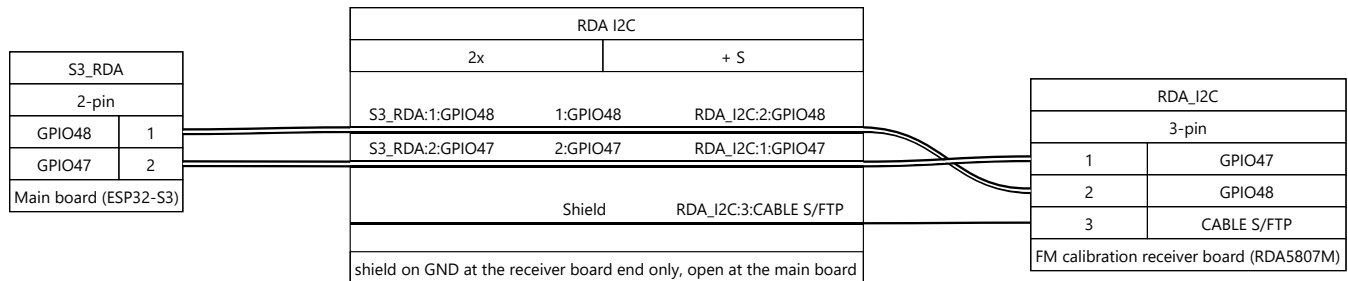
Shield: none.

If it fails: the receiver loses its power and stops answering the main board.

10.3 FM receiver control — main board to FM calibration receiver

Carries: the two-wire control bus (I2C, a clock line and a data line) between the main board and the FM calibration receiver: `GPI047` is the data line, `GPI048` the clock line. Neither line has an external pull-up resistor.

Shape: one shielded cable with two wires. A 2-pin connector `S3_RDA` at the main board, a 3-pin connector `RDA_I2C` at the receiver board; the third pin takes the shield.



The FM receiver control cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Receiver end, pin
1	<code>GPI048</code>	Clock line (the receiver's SCLK)	2
2	<code>GPI047</code>	Data line (the receiver's SDIO)	1
—	<code>CABLE S/FTP</code>	Shield and foil	3

Shield: at the receiver end, unlike most cables here. The shield and foil are joined to ground at the **receiver board end**, on pin 3. They are left unconnected at the main board end, whose connector has only two pins. There is no ground wire: the receiver's ground comes on its power cable (section 10.2 (p. 59)).

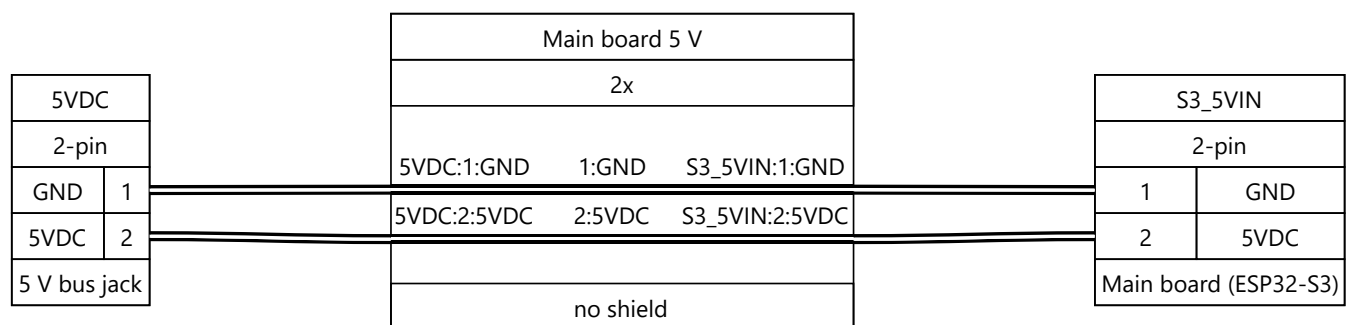
`GPI048` also drives the main board's own RGB LED, so that LED is lit while the machine runs.

If it fails: the main board can no longer talk to the receiver.

10.4 Main board 5 V — 5 V bus to main board

Carries: 5 V and ground from the 5 V bus to the main board's 5 V pin.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to `S3_5VIN` on the main board. Which bus jack it uses is not recorded.



The main board's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	Main board end, pin
1	GND	1
2	5VDC	2

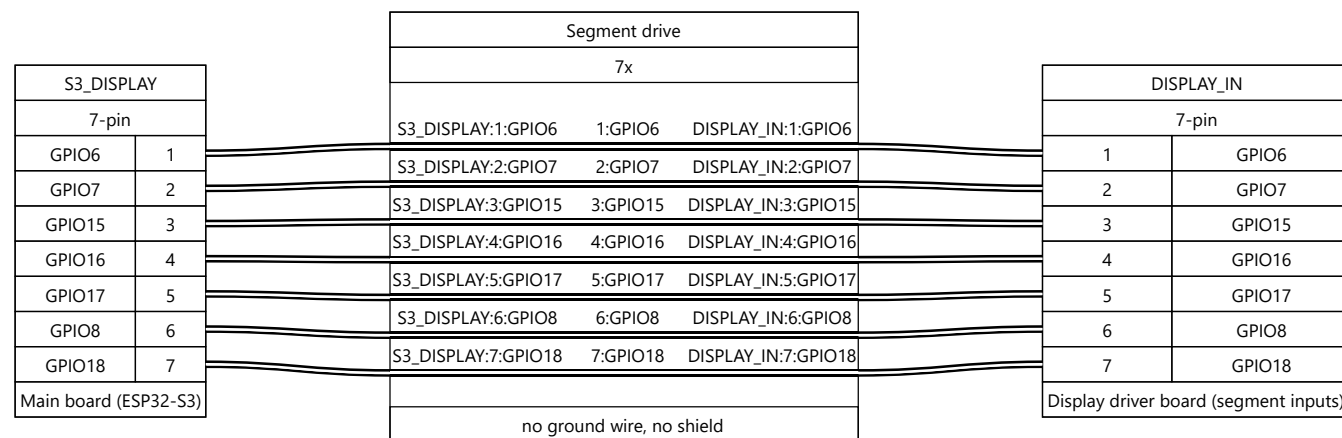
Shield: none.

If it fails: the main board goes dead: no clock display, no needle, no panel lamps. The audio board has its own 5 V cable (section 10.18 (p. 72)).

10.5 Segment drive — main board to display driver board

Carries: the seven signals that choose which segments of the clock display light. Each one drives one input of the segment driver on the display driver board, a ULN2003A (a chip of seven transistor switches that sink current to ground). Each segment line is shared by all four digits.

Shape: one 7-wire cable, pin for pin: S3_DISPLAY at the main board, DISPLAY_IN at the driver board.



The segment drive cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Driver board end, pin
1	GPIO6	Segment A (driver input 1)	1
2	GPIO7	Segment B (driver input 2)	2
3	GPIO15	Segment C (driver input 3)	3
4	GPIO16	Segment D (driver input 4)	4
5	GPIO17	Segment E (driver input 5)	5
6	GPIO8	Segment F (driver input 6)	6
7	GPIO18	Segment G (driver input 7)	7

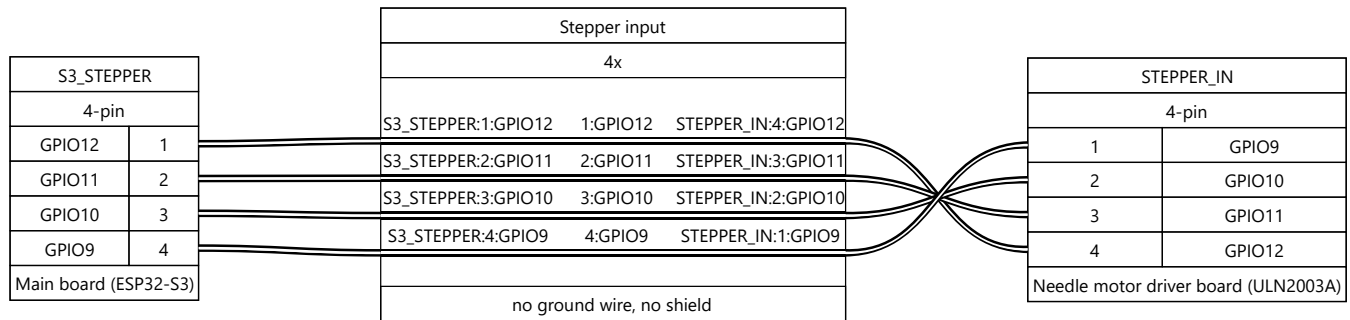
Shield: none. There is no ground wire either: the driver board takes its ground from its own power cable (section 10.26 (p. 78)).

If it fails: a broken wire leaves one segment dark on all four digits.

10.6 Needle motor control — main board to needle motor driver

Carries: the four signals that step the needle motor. They enter the needle motor driver board, a ULN2003A board, on its inputs IN1 to IN4, which drive the motor's four coils (section 10.32 (p. 83)).

Shape: one 4-wire cable, `S3_STEPPER` at the main board, `STEPPER_IN` at the driver board. The pin order runs opposite at the two ends.



The needle motor control cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Driver board end, pin
1	GPIO12	Driver input IN4	4
2	GPIO11	Driver input IN3	3
3	GPIO10	Driver input IN2	2
4	GPIO9	Driver input IN1	1

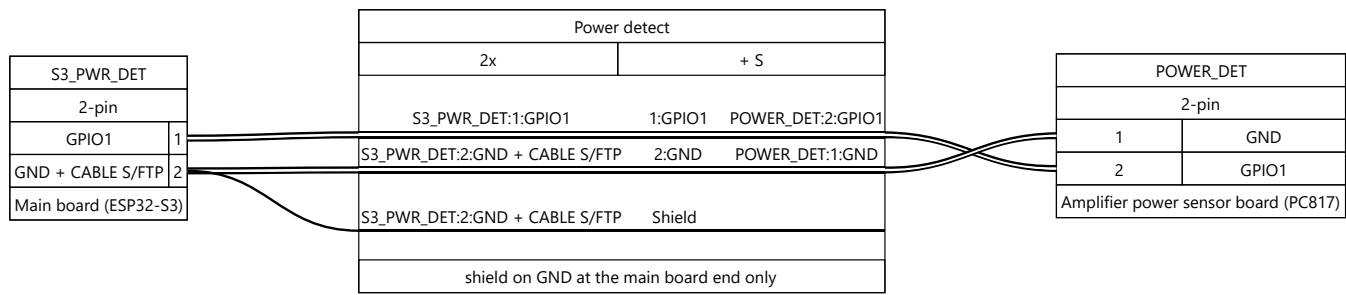
Shield: none. There is no ground wire either: the driver board takes its ground from its own power cable (section 10.24 (p. 77)).

If it fails: one coil is no longer driven, so the needle cannot step properly.

10.7 Amplifier power sensor — main board to power sensor board

Carries: the "amplifier is on" signal. The sensor board holds an optocoupler (PC817): an LED and a light-sensitive transistor in one package, so that a signal crosses by light with no wire between the two sides. The LED is lit from the amplifier's +20 V supply. The transistor's collector is on `GPIO1`, which has a 10 kΩ pull-up to `S3_3V3` on the main board; its emitter is on ground. While the amplifier supply is on, the transistor conducts and pulls `GPIO1` low. Chapter 9 has the sensor.

Shape: one shielded 2-wire cable, `S3_PWR_DET` at the main board, `POWER_DET` at the sensor board.



The amplifier power sensor cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Sensor end, pin
1	GPIO1	The optocoupler's collector	2
2	GND + CABLE S/FTP	Ground to the optocoupler's emitter; the shield and foil end here too	1

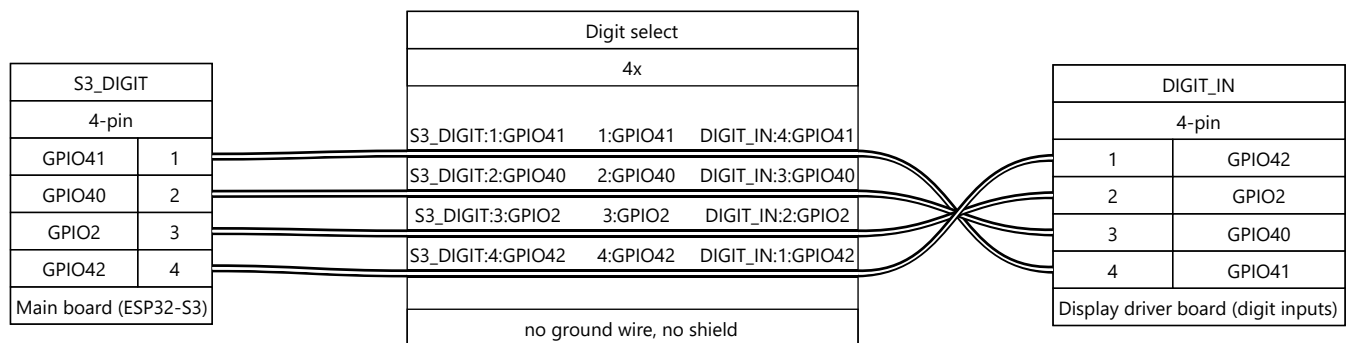
Shield: joined to ground at the main board end only. The ground wire itself runs end to end.

If it fails: with either wire broken, nothing can pull GPIO1 low, so the main board reads "amplifier off" all the time.

10.8 Digit select — main board to display driver board

Carries: the four signals that choose which digit of the clock display lights. Digit 1 is the rightmost digit (units of minutes); digits 2, 3 and 4 run leftwards from it. On the driver board each signal goes through a ULN2003A channel and a 1 kΩ resistor to the gate of a P-channel MOSFET (FQP27P06), which switches 5 V onto that digit (section 10.27 (p. 79)). Chapter 7 has the circuit.

Shape: one 4-wire cable, S3_DIGIT at the main board, DIGIT_IN at the driver board. The pin order runs opposite at the two ends.



The digit select cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Driver board end, pin
1	GPIO41	Selects digit 4 (leftmost)	4

Main board end, pin	Signal	What it is	Driver board end, pin
2	GPI040	Selects digit 3	3
3	GPI02	Selects digit 2	2
4	GPI042	Selects digit 1 (rightmost)	1

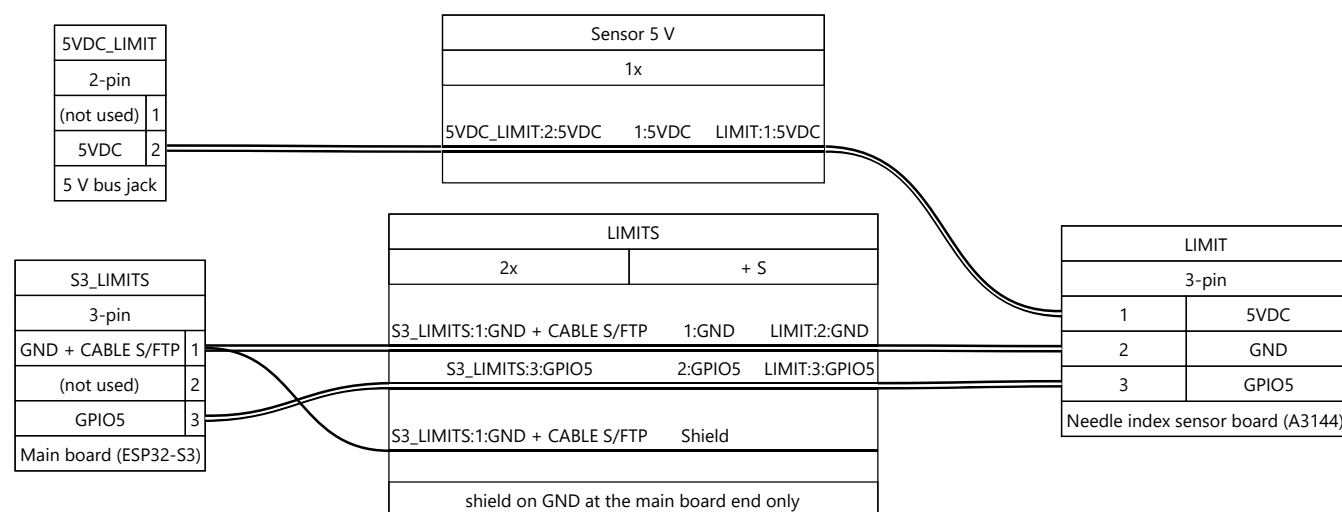
Shield: none is recorded, and no ground wire: the cable has four wires only.

If it fails: a broken wire leaves one digit dark.

10.9 Needle index sensor — main board and 5 V bus to the index sensor

Carries: the index sensor's output, its ground and its 5 V. The sensor is an A3144 Hall switch (it reacts to a magnet) on a small perfboard in its own PLA case. A small magnet on the needle mount passes over it, and its output switches low when the magnet's field is strong enough. That gives the needle a fixed reference point. The output is open-collector: on the main board, GPI05 has a 10 kΩ pull-up to S3_3V3 and 0.1 μF to ground.

Shape: a Y harness. A shielded 2-wire cable from S3_LIMITS at the main board, and a single 5 V wire from a bus jack 5VDC_LIMIT, meet in the sensor's 3-pin connector LIMIT. Which bus jack is not recorded.



The needle index sensor harness, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Sensor end, pin
1	GND + CABLE S/FTP	Ground for the sensor; the shield and foil end here too	2
2	(not used)	—	—
3	GPI05	The sensor's output	3

5 V bus jack, pin	Signal	Sensor end, pin
1	(not used)	—
2	5VDC	1

Shield: joined to ground at the main board end only.

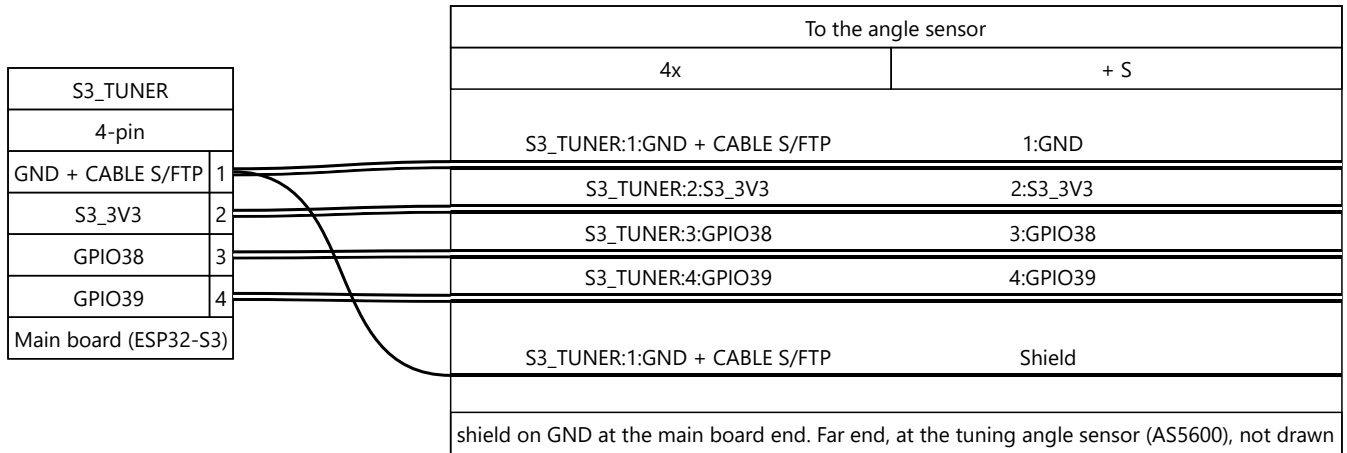
Not used: pin 2 at the main board end, and the bus jack's ground pin. The sensor's ground comes from the main board, its 5 V from the bus. At the main board, `GPIO4` is still wired to pin 2, with its own 10 k Ω pull-up and 0.1 μ F, but the cable carries nothing on that pin.

If it fails: if any of the three wires breaks, `GPIO5` stays high and the main board never sees the index magnet, so the needle cannot find its reference point.

10.10 Tuning angle sensor — main board to the angle sensor

Carries: power and the control bus (I2C) for the tuning angle sensor, an AS5600 magnetic angle sensor. It sits inside its own Faraday cage, its board perpendicular to the tuning capacitor's shaft, with the shaft's axis through the centre of the chip. It reads the angle of a magnet glued to the end of a PLA extension of that shaft. At the main board, each bus line has a 10 k Ω pull-up to `S3_3V3` at the board's pin, then a 220 Ω resistor in series before the connector.

Shape: one shielded 4-wire cable from `S3_TUNER` at the main board. **Only the main board end is drawn:** the connector at the sensor, and its pin order, are not recorded. The table gives the sensor pin each wire reaches.



The tuning angle sensor cable. Only the main board end is drawn.

Wire by wire.

Main board end, pin	Signal	What it is	Sensor pin it reaches
1	GND + CABLE S/FTP	Ground; the shield and foil end here too	GND
2	S3_3V3	The main board's 3.3 V output	VCC
3	GPIO38	Data line (SDA)	SDA
4	GPIO39	Clock line (SCL)	SCL

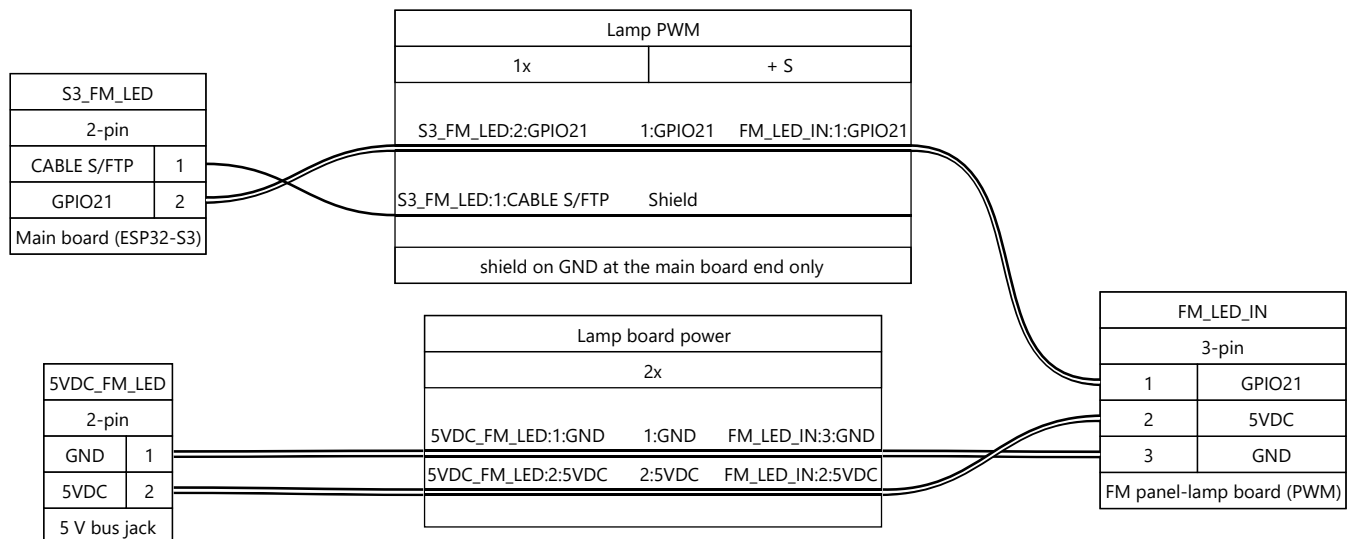
Shield: joined to ground at the main board end.

If it fails: the main board can no longer read where the tube radio is tuned.

10.11 Panel-lamp control — main board and 5 V bus to the panel-lamp board

Carries: the brightness signal for the four FM panel lamps, and the lamp board's power. `GPIO21` reaches the gate of a logic-level MOSFET (IRL540N) through $100\ \Omega$; the gate also has $100\ \text{k}\Omega$ and $220\ \text{pF}$ to ground. The MOSFET sinks the four lamps' current. The main board dims the lamps by PWM (pulse-width modulation: switching on and off fast, with a varying on-time). Chapter 9 has the circuit.

Shape: two cables into one 3-pin connector, `FM_LED_IN`, on the panel-lamp board: a shielded 1-wire cable from `S3_FM_LED` on the main board, and a 2-wire power cable from a 5 V bus jack `5VDC_FM_LED`.



The panel-lamp control cables, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Lamp board end, pin
1	CABLE S/FTP	Shield and foil	—
2	GPIO21	The lamp brightness signal (PWM)	1

5 V bus jack, pin	Signal	Lamp board end, pin
1	GND	3
2	5VDC	2

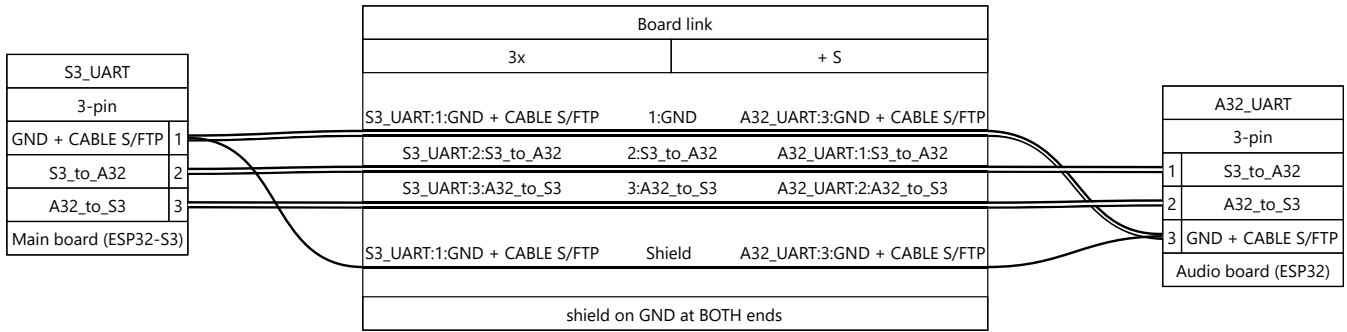
Shield: joined to ground at the main board end, left unconnected at the lamp board. There is no ground wire from the main board: the lamp board takes its ground from the bus jack.

If it fails: if the `GPIO21` wire breaks, the gate's $100\ \text{k}\Omega$ holds the MOSFET off and the lamps stay dark. A broken power wire also leaves them dark.

10.12 Board link — main board to audio board

Carries: the serial link (UART) between the two boards, one wire each way, and ground. `S3_to_A32` runs from the main board's `GPIO13` to the audio board's `GPIO26`. `A32_to_S3` runs from the audio board's `GPIO27` to the main board's `GPIO14`.

Shape: one shielded 3-wire cable, S3_UART at the main board, A32_UART at the audio board.



The board link cable, by signal name, as on the cable sheet.

Wire by wire.

Main board end, pin	Signal	What it is	Audio board end, pin
1	GND + CABLE S/FTP	Ground; the shield and foil end here too	3
2	S3_to_A32	Main board GPIO13 → audio board GPIO26	1
3	A32_to_S3	Audio board GPIO27 → main board GPIO14	2

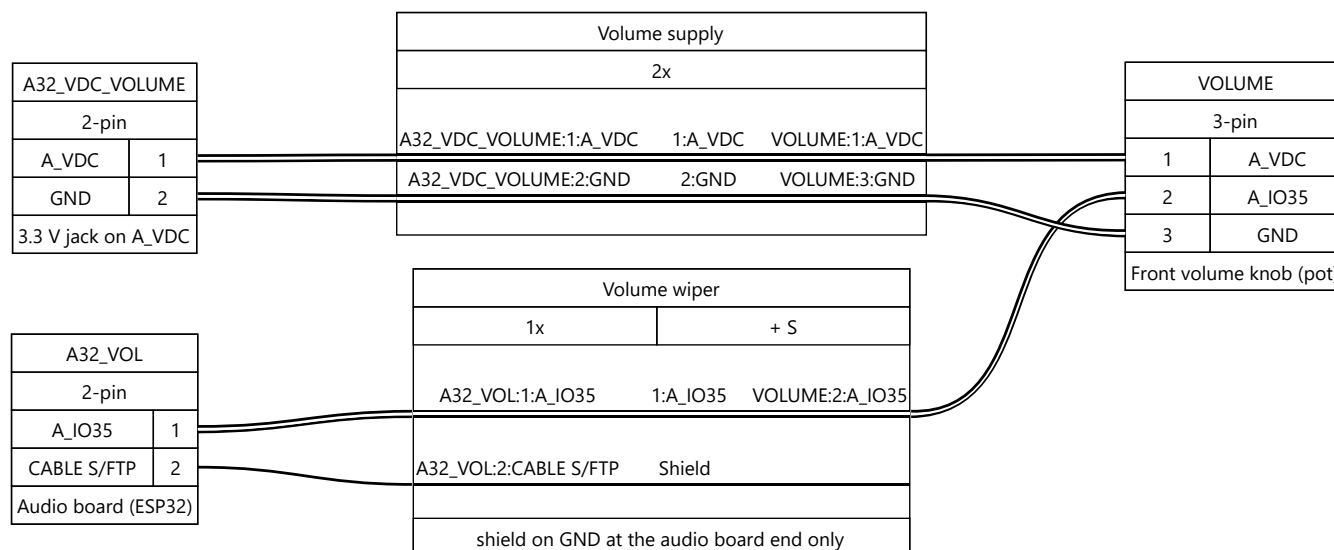
Shield: joined at both ends. Unlike the other shielded cables, this one has its shield and foil joined to ground at **both** boards.

If it fails: the two boards can no longer talk to each other.

10.13 Front volume knob — audio board and 3.3 V to the volume knob

Carries: the front volume knob's position to the audio board. The knob turns a potentiometer (RV4, drawn as B10K) whose two ends sit on 3.3 V and ground; its wiper goes to the audio board's GPIO35. This is the audio board's own volume, set digitally. The amplifier's own volume control is a different pot, on the amplifier's board, reached only from the back.

Shape: two cables into one 3-pin connector, VOLUME, at the knob. A shielded 1-wire cable from A32_VOL on the audio board carries the wiper. A 2-wire cable from A32_VDC_VOLUME brings 3.3 V and ground. A32_VDC_VOLUME is a jack on the audio side's 3.3 V (A_VDC); which jack is not recorded.



The front volume knob cables, by signal name, as on the cable sheet.

Wire by wire.

Audio board end, pin	Signal	What it is	Knob end, pin
1	A_IO35	The pot's wiper	2
2	CABLE S/FTP	Shield and foil	—

3.3 V jack A32_VDC_VOLUME, pin	Signal	What it is	Knob end, pin
1	A_VDC	3.3 V, to the pot's top end	1
2	GND	Ground, to the pot's bottom end	3

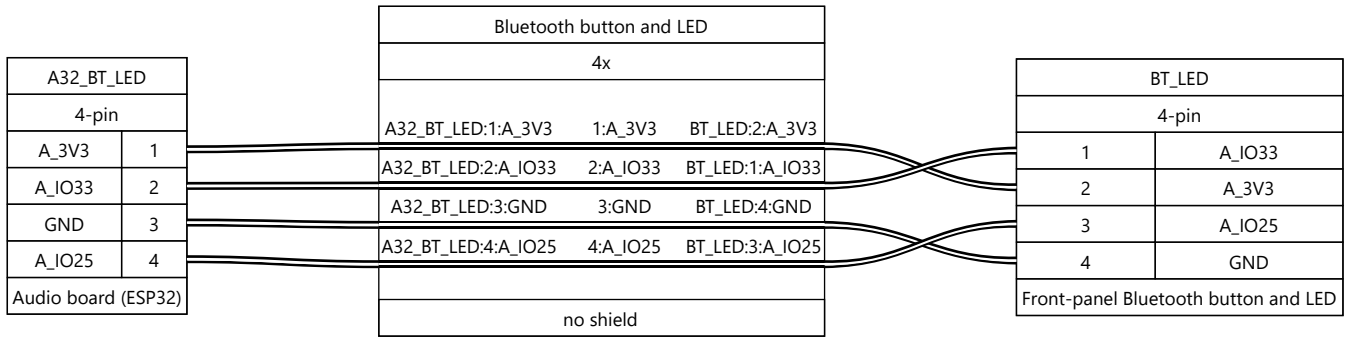
Shield: joined to ground at the audio board end, left unconnected at the knob.

If it fails: the knob no longer changes the volume.

10.14 Bluetooth button and LED — audio board to front panel

Carries: the front-panel Bluetooth LED and the Bluetooth pairing button. On the front panel the LED is also the button. The LED's anode is on 3.3 V and its cathode on `GPI033`: the audio board lights it by pulling `GPI033` low. No series resistor is drawn. The button connects `GPI025` to ground when pressed; `GPI025` uses the chip's internal pull-up, so pressed reads low.

Shape: one 4-wire cable, `A32_BT_LED` at the audio board, `BT_LED` at the front panel. The pins cross in pairs: 1 and 2 swap, 3 and 4 swap.



The Bluetooth button and LED cable, by signal name, as on the cable sheet.

Wire by wire.

Audio board end, pin	Signal	What it is	Front panel end, pin
1	A_3V3	3.3 V to the LED's anode	2
2	A_IO33	The LED's cathode; low lights it	1
3	GND	The button's ground side	4
4	A_IO25	The button; pressed pulls it low	3

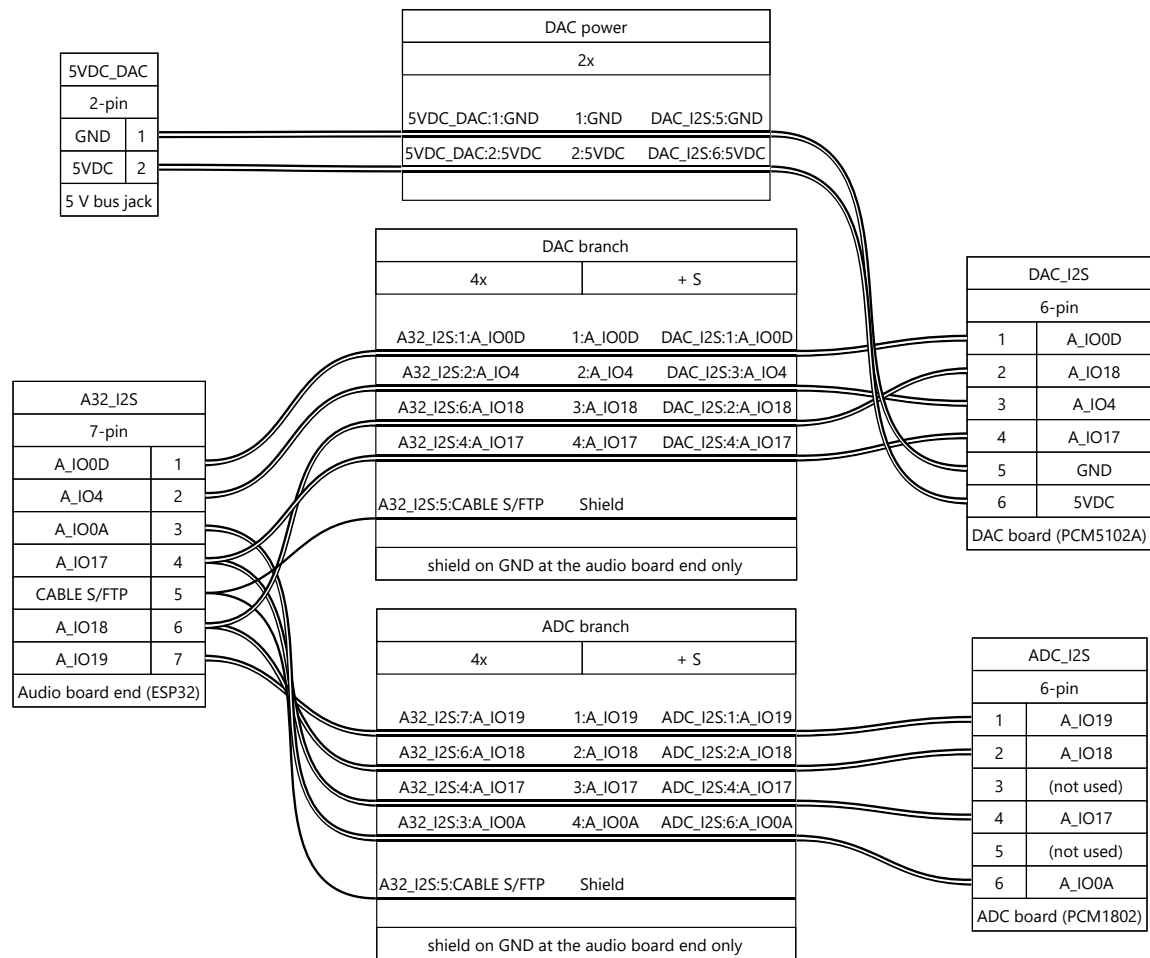
Shield: none.

If it fails: a broken pin 1 or 2 wire leaves the LED dark; a broken pin 3 or 4 wire makes the button do nothing.

10.15 The audio harness — audio board to ADC and DAC

Carries: the digital audio (I2S) between the audio board (ESP32, "A32") and its two converters: the ADC that digitises the tube radio, and the DAC that feeds the amplifier. The DAC's 5 V power rides in the same harness.

Shape: one shielded harness that starts at the audio board and splits in two: one branch to the ADC board, one to the DAC board. A short power lead from the 5 V bus joins the DAC branch.



The audio harness, by signal name, as on the cable sheet.

Wire by wire. Pin numbers are the cable sheet's.

Audio board end, pin	Signal	What it is	ADC end, pin	DAC end, pin
1	A_I00D	Master clock to the DAC: the audio board's GPIO0, through its own 33 Ω resistor at the board	—	1
2	A_IO4	Audio data to the DAC (GPIO4)	—	3
3	A_I00A	Master clock to the ADC: the same GPIO0, through its other 33 Ω resistor	6	—
4	A_IO17	Word clock, left/right (LRCK), shared by both converters (GPIO17)	4	4
5	CABLE S/FTP	The harness's shield and foil, on ground at the audio board	—	—
6	A_IO18	Bit clock (BCK), shared by both converters (GPIO18)	2	2
7	A_IO19	Audio data from the ADC (GPIO19)	1	—

5 V bus jack, pin	Signal	DAC end, pin
1	GND	5
2	5VDC	6

Shield: the shield and foil are joined to ground at the **audio board end only**, and left unconnected at both converter ends.

Not used: pins 3 and 5 at the ADC end carry nothing. The ADC takes its power from its own cable on the 5 V bus, not from this harness.

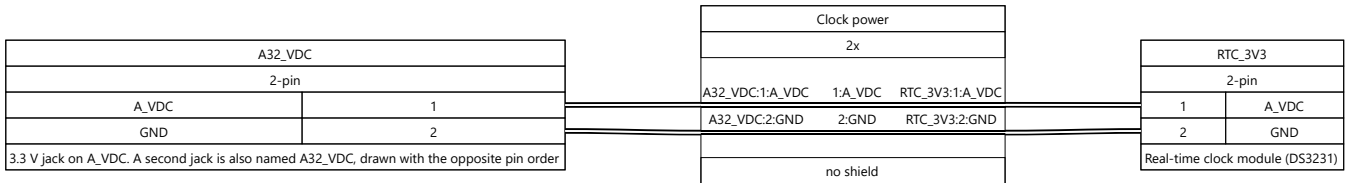
The two clock wires are the same signal. A_I00A and A_I00D both come from the audio board's GPIO0. They have two names only so that each converter's drawing reads clearly. Each has its own 33 Ω series resistor, at the audio board's pin.

If it fails: no radio sound points to the ADC branch (pins 3, 4, 6, 7); no radio and no Bluetooth sound, while AUX still plays, points to the DAC branch (pins 1, 2, 4, 6) or its power lead. AUX never passes through this harness. Bluetooth plays through the DAC but not the ADC, so it tells the two branches apart.

10.16 Clock module power — 3.3 V jack to the real-time clock module

Carries: 3.3 V and ground to the real-time clock module (a DS3231 module, chapter 4 (p. 19)).

Shape: one 2-wire cable, from a 3.3 V jack the cable sheet names A32_VDC , on the audio side's 3.3 V (A_VDC), to RTC_3V3 on the clock module. Which jack is not recorded. A second jack has the same name: the one that takes the audio board's 3.3 V cable (section 10.19 (p. 73)).



The clock module power cable, by signal name, as on the cable sheet.

Wire by wire.

3.3 V jack, pin	Signal	Clock module end, pin
1	A_VDC	1
2	GND	2

Shield: none.

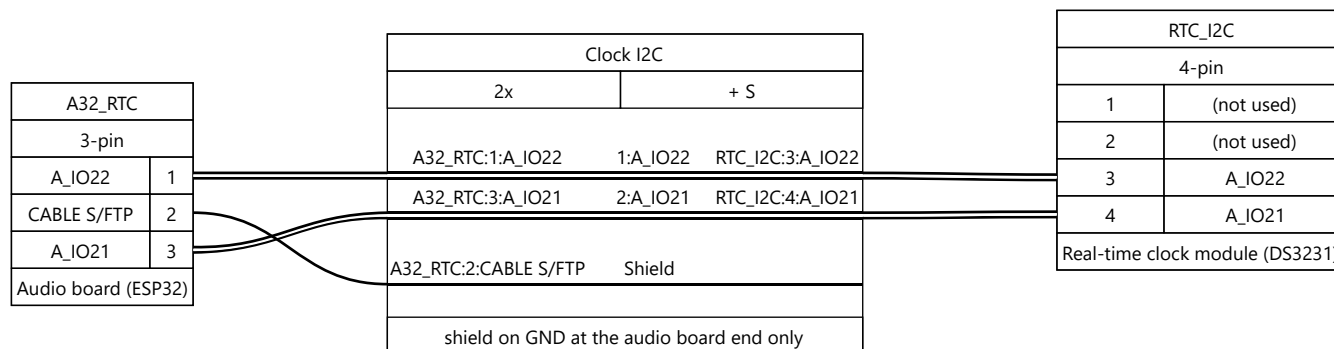
Caution — two jacks named A32_VDC , with opposite pin orders. As the cable sheet draws them, this jack has pin 1 on 3.3 V (A_VDC) and pin 2 on ground. The other A32_VDC jack, which takes the audio board's 3.3 V cable (section 10.19 (p. 73)), has pin 1 on ground and pin 2 on 3.3 V (A_3V3 , the same rail). The name alone does not tell you which jack you are holding, nor which way round its 3.3 V and ground are. These are the cable sheet's pin numbers; nothing more is recorded about the real connectors.

If it fails: the real-time clock module loses its 3.3 V supply.

10.17 Clock module control — audio board to the real-time clock module

Carries: the control bus (I2C) between the audio board and the real-time clock module: **A_IO22** is the data line (SDA), **A_IO21** the clock line (SCL).

Shape: one shielded 2-wire cable, **A32_RTC** at the audio board (3 pins) and **RTC_I2C** at the clock module (4 pins).



The clock module control cable, by signal name, as on the cable sheet.

Wire by wire.

Audio board end, pin	Signal	What it is	Clock module end, pin
1	A_IO22	Data line (SDA)	3
2	CABLE S/FTP	Shield and foil	—
3	A_IO21	Clock line (SCL)	4

Shield: joined to ground at the audio board end, left unconnected at the module.

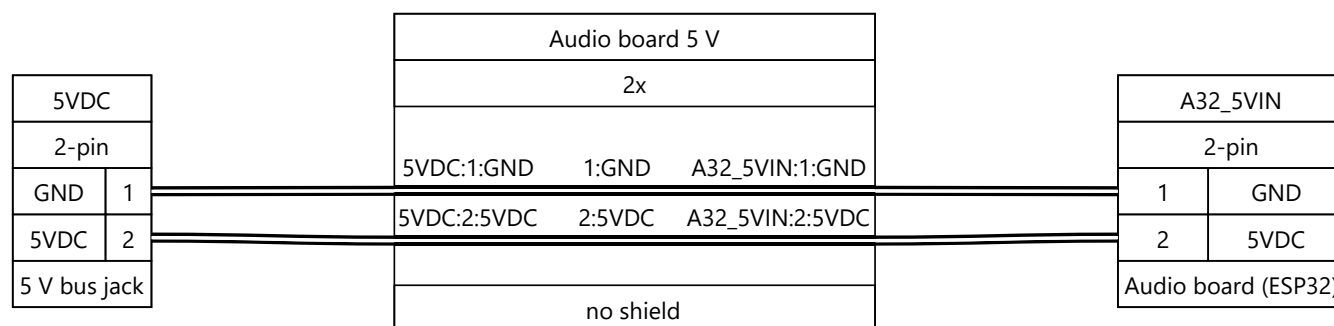
Not used: pins 1 and 2 at the clock module end. Only two of that connector's four positions carry wires. **Which two, on the real connector, is not recorded.**

If it fails: the audio board can no longer talk to the real-time clock module.

10.18 Audio board 5 V — 5 V bus to audio board

Carries: 5 V and ground from the 5 V bus to the audio board's 5 V pin.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to **A32_5VIN** on the audio board. Which bus jack is not recorded.



The audio board's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	Audio board end, pin
1	GND	1
2	5VDC	2

Shield: none.

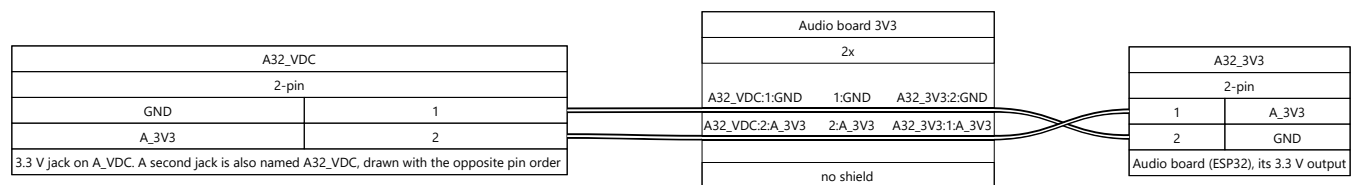
If it fails: the audio board goes dead, and with it the 3.3 V it makes (section 10.19 (p. 73)). There is no sound from any source but AUX.

10.19 Audio board 3.3 V — audio board to the 3.3 V supply

Carries: the audio board's own 3.3 V output (A_3V3) and ground out to the 3.3 V bus (chapter 5 (p. 27), section 5.5 (p. 30)). The bus's 3.3 V is one wire with the 3.3 V jacks that feed the real-time clock module, the front volume knob and the source switch. Which of those jacks are bus positions, and which sit on the audio board, is not recorded. The audio board decouples its 3.3 V pin with one 0.1 μ F and two 10 μ F capacitors.

Shape: one 2-wire cable, from A32_3V3 on the audio board to a jack the cable sheet names A32_VDC, on the 3.3 V supply. The pins cross in the cable sheet's numbering: pin 1 at one end is pin 2 at the other.

Two jacks are named A32_VDC, with opposite pin orders as drawn. This one has pin 1 on ground and pin 2 on 3.3 V. The other, which feeds the real-time clock module (section 10.16 (p. 71)), has pin 1 on 3.3 V and pin 2 on ground. Both are on the same 3.3 V rail and the same ground. The caution in section 10.16 (p. 71) applies to both.



The audio board's 3.3 V cable, by signal name, as on the cable sheet.

Wire by wire.

A32_VDC end, pin	Signal	Audio board end, pin
1	GND	2
2	A_3V3	1

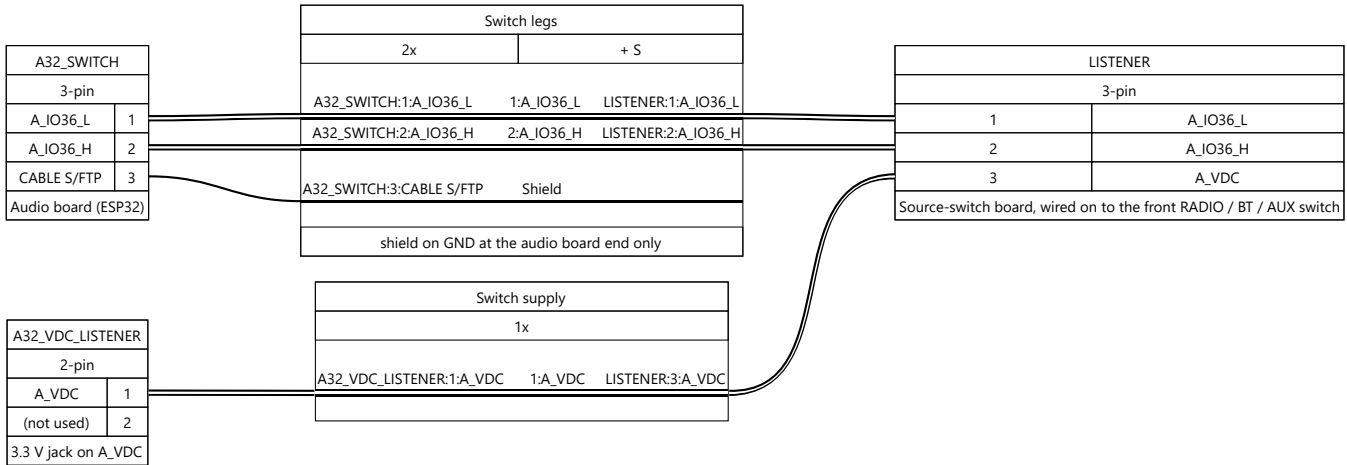
Shield: none.

If it fails: the 3.3 V bus loses its supply, and every part fed from a bus position loses its 3.3 V. Which of the clock module, the volume knob and the source switch that includes is not recorded.

10.20 Source switch — audio board and 3.3 V to the source-switch board

Carries: the position of the front RADIO / BT / AUX switch, as a voltage on the audio board's GPIO36. The switch puts 3.3 V on one of two legs: A_I036_H, the ladder node itself, on RADIO (3.3 V), or A_I036_L, which reaches the node through 68 k Ω , on BT (about 0.81 V). On AUX it feeds neither, and the node reads 0 V. The ladder's resistors are all on the audio board; chapter 9 (p. 50), section 9.3 (p. 52), has the circuit.

Shape: two cables into one 3-pin connector, `LISTENER`, on the source-switch board, a small board in its own printed case that the drawings call the IN/LISTENER/OUT board. A shielded 2-wire cable from `A32_SWITCH` on the audio board carries the two legs. A 1-wire cable from the 3.3 V jack `A32_VDC_LISTENER` brings 3.3 V; which jack is not recorded. From the source-switch board, wires go on to the front switch. That last leg is not drawn as a cable; it is given by signal at the end of *Wire by wire*.



The source switch cables, by signal name, as on the cable sheet.

Wire by wire.

Audio board end, pin	Signal	What it is	Switch board end, pin
1	A_IO36_L	The BT leg, through 68 kΩ	1
2	A_IO36_H	The RADIO leg, direct	2
3	CABLE S/FTP	Shield and foil	—

3.3 V jack A32_VDC_LISTENER, pin	Signal	What it is	Switch board end, pin
1	A_VDC	3.3 V for the switch	3
2	(not used)	—	—

The last leg, from the source-switch board to the front switch, by signal. The pins of the source-switch board's own connector on this leg are not recorded.

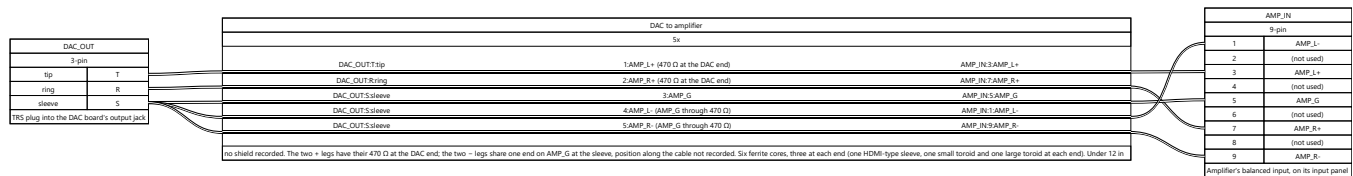
Signal	Front switch pin
A_VDC	2 and 5
A_IO36_H	1
A_IO36_L	6

- Shield:** joined to ground at the audio board end, left unconnected at the switch board.
- Not used:** the 3.3 V jack's ground pin (pin 2), so no ground goes to the switch; and the front switch's pins 3 and 4.
- If it fails:** with the `A_IO36_H` wire broken, RADIO reads as AUX. With the `A_IO36_L` wire broken, BT reads as AUX. With the 3.3 V wire broken, every position reads as AUX.

10.21 DAC to amplifier — DAC board to the amplifier's balanced input

Carries: the analogue audio from the DAC to the amplifier's balanced input (an input that takes each channel as a + and a – wire). The DAC gives left, right and ground; the cable feeds each channel to a + leg and ties each – leg to the DAC's ground through 470 Ω .

Shape: a 3-pole TRS plug, **DAC_OUT** (tip, ring, sleeve, like a headphone plug), goes into the DAC board's output jack. At the other end, a 9-pin connector **AMP_IN** plugs into the amplifier's input panel. The plug has three poles, and the amplifier end uses five pins. **The cable carries four 470 Ω resistors**, one in each signal leg. The two on the + legs sit at the DAC end. The two on the – legs share one end on **AMP_G**, at the plug's sleeve; where along the cable they sit is not recorded.



The DAC-to-amplifier cable, by signal name, as on the cable sheet.

Wire by wire.

Plug pole	DAC signal	Wire	In the cable	Amplifier end, pin
Tip	left out (OUTL)	AMP_L+	470 Ω in series, at the DAC end	3
Ring	right out (OUTR)	AMP_R+	470 Ω in series, at the DAC end	7
Sleeve	analogue ground (AGND)	AMP_G	direct	5
Sleeve	analogue ground	AMP_L-	through its own 470 Ω	1
Sleeve	analogue ground	AMP_R-	through its own 470 Ω	9

Shield: none is recorded.

Not used: pins 2, 4, 6 and 8 at the amplifier end.

Ferrite cores. Six cores sit on this cable, three at each end: at each end, one HDMI-type sleeve, one small toroid and one large toroid. The cable passes once through each; it is under 12 inches long, so it makes no turns. The table gives the size of each kind of core:

Core	Size
HDMI-type sleeve	17.4 mm outside, 9.7 mm inside, 28.5 mm long (the same type as the antenna's feed-point cores, chapter 11 (p. 87))
Small toroid	16 mm outside, 3 mm wall, 8 mm long
Large toroid	22 mm outside, 4 mm wall, 8 mm long

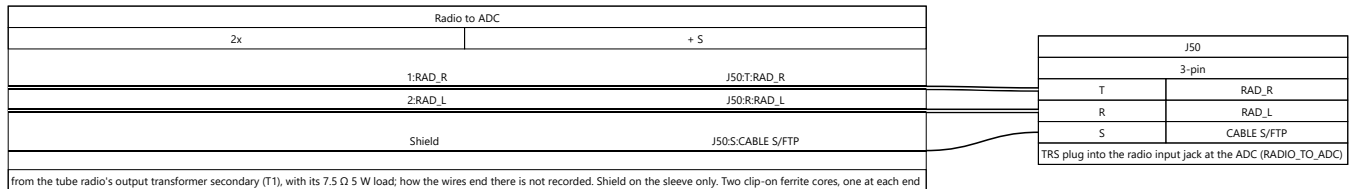
The toroids are salvaged and unidentified: no maker, part number, grade or magnetic data is recorded. Their material is not that of the radio cable's clip-on cores (section 10.22 (p. 76)). The four resistors and the six cores are part of this cable.

If it fails: radio and Bluetooth go silent at the amplifier while AUX still plays. AUX reaches the amplifier's input panel on its own path and never goes through this cable.

10.22 Radio to ADC — tube radio's output transformer to the ADC

Carries: the tube radio's audio, from the secondary of its output transformer (T1), to the ADC. A $7.5\ \Omega\ 5\ \text{W}$ resistor (marked 5W7Ω5J) is wired across that secondary as its load. The feed is mono.

Shape: one shielded 2-wire cable. At the ADC end, a 3-pole TRS plug, J50, goes into the radio input jack at the ADC (J51, RADIO_TO_ADC). At the radio end, the two wires come from the transformer secondary; **how they end there, soldered or plugged, is not recorded.**



The radio-to-ADC cable. Only the plug end is drawn.

Wire by wire.

Plug pole	Signal	What it is	Radio end
Tip	RAD_R	One side of the transformer secondary	T1 terminal SB
Ring	RAD_L	The other side	T1 terminal SA
Sleeve	CABLE S/FTP	Shield and foil	not connected

The jack at the ADC (J51). Cable side: shield and foil on the sleeve, RAD_L on the ring, RAD_R on the tip. Output side: sleeve and ring on ground, tip into the $4.7\ \text{k}\Omega$ of the input attenuator (chapter 6 (p. 33), section 6.4 (p. 36)).

Shield: on the plug's sleeve, which is ground at the ADC end. It is left unconnected at the radio end.

Tip and ring can be swapped. The feed is mono: at the ADC one side goes to ground, the other into the input attenuator.

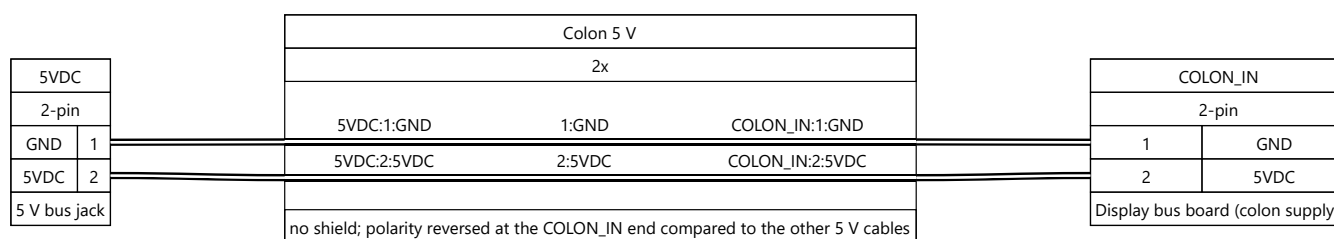
Ferrite cores. Two clip-on cores sit on this cable, one near the transformer end and one near the ADC end. The cable passes once through each. Measured: 10 mm outside, 3.5 mm inside, 3.5 mm walls, 20 mm long. One of these figures is rounded: the hole and the two walls make 10.5 mm, against 10 mm outside. No maker, part number or grade is recorded. Their material, as supplied: rectangle ratio 20, coercivity 16 A/m, remanence 200 mT, Curie temperature 130 °C, density $4.9\ \text{g}/\text{cm}^3$, "soft magnetic".

If it fails: the radio goes silent while Bluetooth still plays.

10.23 Colon 5 V — 5 V bus to the clock display's colon

Carries: 5 V and ground to the clock display's colon. At the colon, the 5 V goes through $33\ \Omega$, the colon LED and a variable resistor glued at $1.5\ \text{k}\Omega$ to ground (section 10.34 (p. 84)).

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to COLON_IN on the display bus board, which feeds the colon. Which bus jack is not recorded.



The colon's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	COLON_IN, pin
1	GND	1
2	5VDC	2

Shield: none.

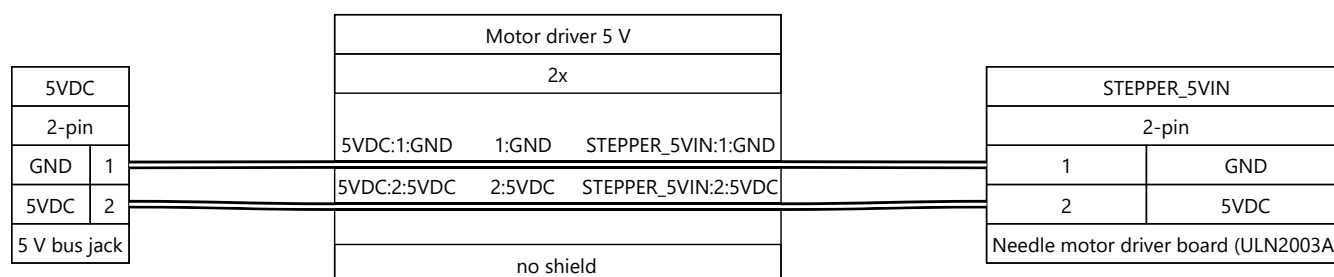
Caution — this cable's polarity is reversed at COLON_IN. The cable sheet marks this 5 V cable's polarity as reversed at the COLON_IN side, compared with the other 5 V cables. Where the reversal physically sits is not recorded.

If it fails: the colon stays dark while the digits light.

10.24 Motor driver 5 V — 5 V bus to the needle motor driver

Carries: 5 V and ground to the needle motor driver board. The 5 V feeds the ULN2003A's common pin and the motor's common wire, with 0.1 μ F and 47 μ F to ground.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to STEPPER_5VIN on the driver board. Which bus jack is not recorded.



The needle motor driver's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	Driver board end, pin
1	GND	1
2	5VDC	2

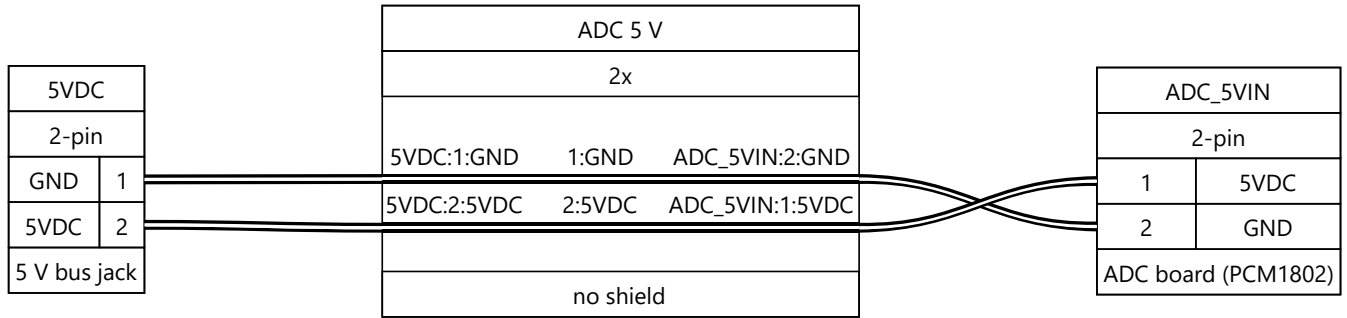
Shield: none.

If it fails: the needle motor has no power and the needle does not move.

10.25 ADC 5 V — 5 V bus to the ADC board

Carries: 5 V and ground to the ADC board (PCM1802). The ADC module's 5 V pin has 100 μ F and 0.1 μ F to ground.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to ADC_5VIN on the ADC board. Which bus jack is not recorded. The pin numbers differ at the two ends in the cable sheet's numbering: at the ADC end, pin 1 is 5 V and pin 2 is ground, reversed from the bus-jack end. This cable's pins do swap end to end, but the cable sheet carries no caution note for it. The Colon 5 V cable is the other way round: its pin numbers match end to end (section 10.23 (p. 76)), yet the author's note on the sheet calls its polarity reversed.



The ADC's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	ADC end, pin
1	GND	2
2	5VDC	1

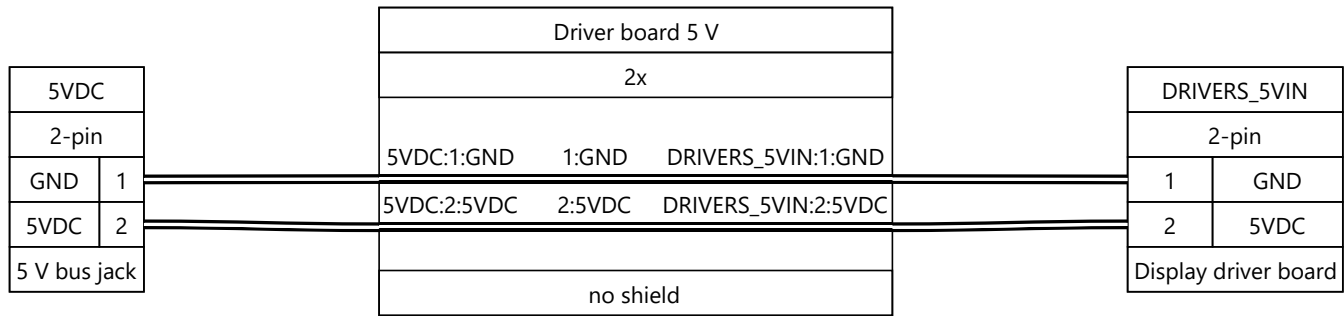
Shield: none.

If it fails: the radio goes silent while Bluetooth still plays.

10.26 Driver board 5 V — 5 V bus to the display driver board

Carries: 5 V and ground to the display driver board. The 5 V is what the four digit switches deliver to the digits (section 10.27 (p. 79)), and the ground is the return of both ULN2003A drivers.

Shape: one 2-wire cable, from a 2-pin jack on the 5 V bus to DRIVERS_5VIN on the driver board. Which bus jack is not recorded.



The display driver board's 5 V cable, by signal name, as on the cable sheet.

Wire by wire.

5 V bus jack, pin	Signal	Driver board end, pin
1	GND	1
2	5VDC	2

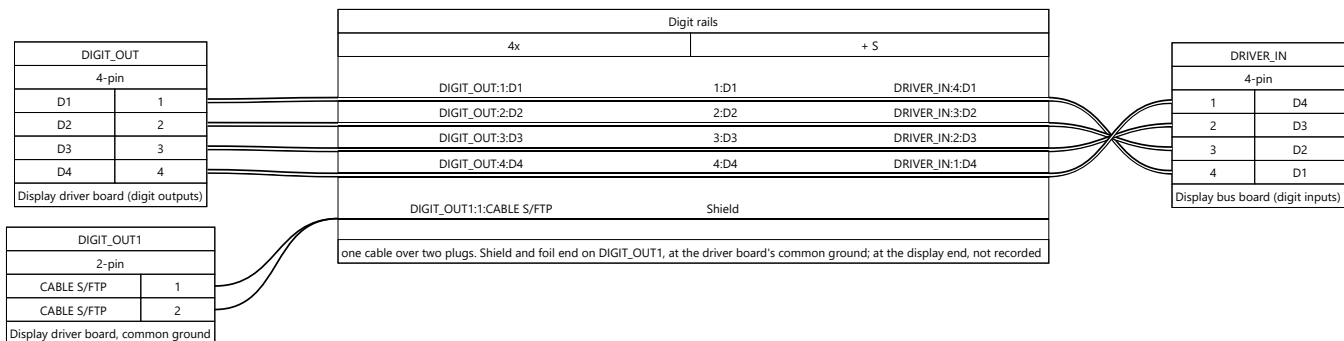
Shield: none.

If it fails: the whole clock display goes dark; the colon has its own cable (section 10.23 (p. 76)).

10.27 Digit rails — display driver board to the display bus board

Carries: each digit's switched 5 V. **D1** to **D4** are the drains of the four P-channel MOSFETs on the driver board; each feeds the common anode of one digit, digit 1 being the rightmost.

Shape: one shielded 4-wire cable that ends in **two** connectors at the driver board: **DIGIT_OUT** (4 pins) for the wires, and **DIGIT_OUT1** (2 pins) for the shield and the foil. At the display bus board it ends in **DRIVER_IN**. The pin order runs opposite at the two ends.



The digit rail cable, by signal name, as on the cable sheet.

Wire by wire.

Driver board end, pin	Signal	What it is	Display bus board end, pin
DIGIT_OUT 1	D1	5 V for digit 1 (rightmost)	4
DIGIT_OUT 2	D2	5 V for digit 2	3
DIGIT_OUT 3	D3	5 V for digit 3	2
DIGIT_OUT 4	D4	5 V for digit 4 (leftmost)	1

Driver board end, pin	Signal	What it is	Display bus board end, pin
DIGIT_OUT1 1 and 2	CABLE S/FTP	Shield and foil	—

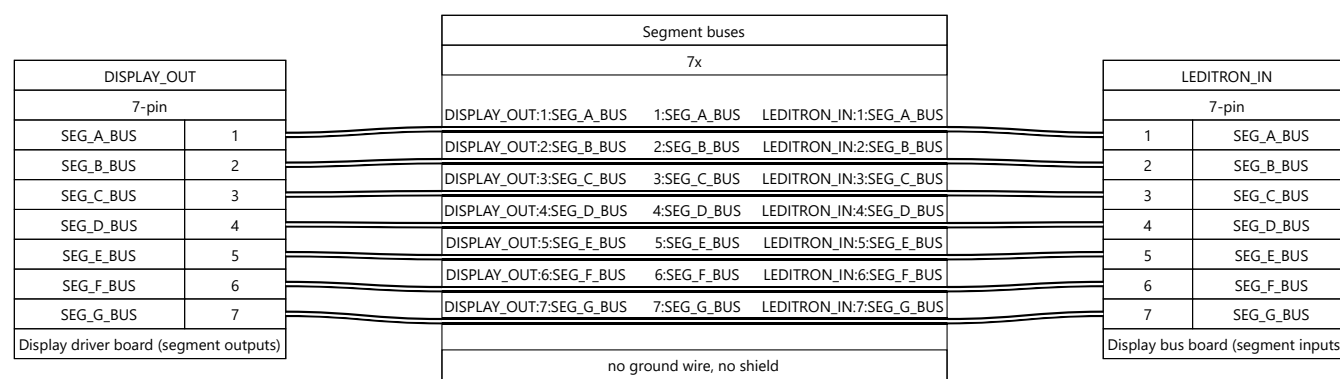
Shield: the shield and the foil end on the two pins of `DIGIT_OUT1`, at the driver board's common ground. What they do at the display bus board end is not recorded.

If it fails: a broken wire leaves one digit dark.

10.28 Segment buses — display driver board to the display bus board

Carries: the seven segment lines, A to G. Each is the cathode line of one segment on all four digits. The segment driver sinks it to ground through a 10 Ω and a 200 Ω resistor in series (chapter 7 (p. 40)).

Shape: one 7-wire cable, pin for pin: `DISPLAY_OUT` at the driver board, `LEDITRON_IN` at the display bus board.



The segment bus cable, by signal name, as on the cable sheet.

Wire by wire.

Driver board end, pin	Signal	Display bus board end, pin
1	SEG_A_BUS	1
2	SEG_B_BUS	2
3	SEG_C_BUS	3
4	SEG_D_BUS	4
5	SEG_E_BUS	5
6	SEG_F_BUS	6
7	SEG_G_BUS	7

Shield: none. There is no ground wire.

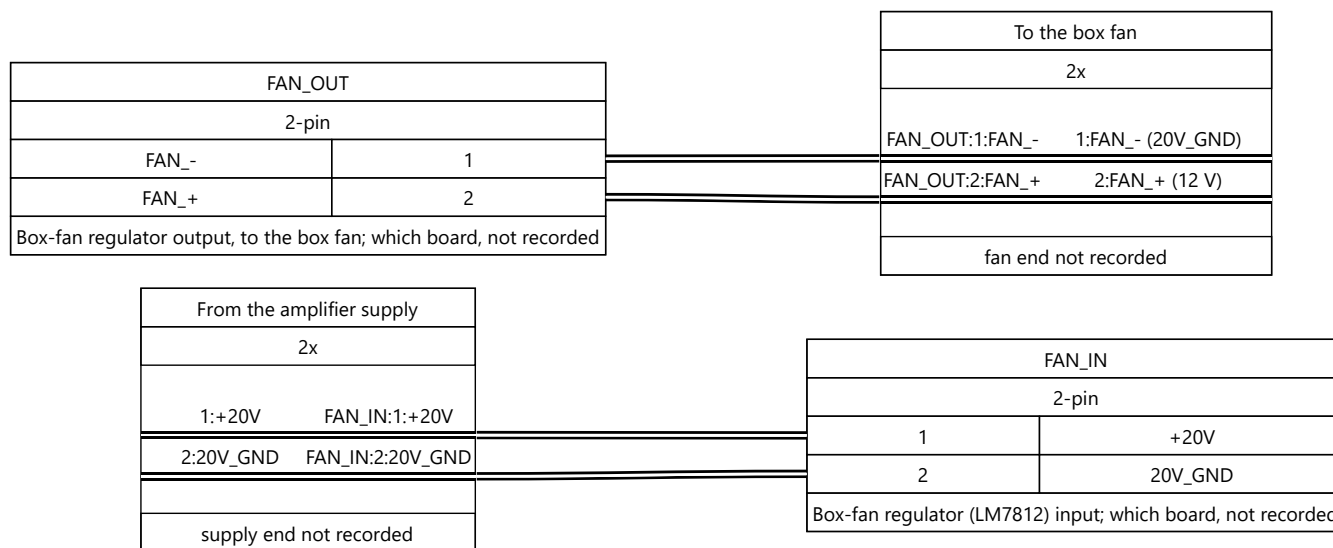
If it fails: a broken wire leaves one segment dark on all four digits, like a broken segment drive wire (section 10.5 (p. 61)).

10.29 Box-fan harness — amplifier supply to the box fan

This page and the ones after it describe wiring that is not on the cable sheet (section 10.1 (p. 56)).

Carries: the amplifier supply's +20 V to the box fan's regulator, and the regulator's 12 V to the box fan. The box fan runs on 12 V made from the amplifier supply's +20 V by an LM7812 regulator (chapter 5 (p. 27), section 5.6 (p. 31)).

Shape: two 2-pin connectors. `FAN_IN` carries +20 V and its return to the regulator's input. `FAN_OUT` carries the regulator's 12 V output and its return to the fan. Which board each connector sits on, and how the wires end at the supply and at the fan, are not recorded.



The box-fan harness. The supply end and the fan end are not recorded.

Wire by wire.

Connector, pin	Signal	What it is
FAN_IN 1	+20V	The regulator's input
FAN_IN 2	20V_GND	The amplifier supply's return
FAN_OUT 1	FAN_-	The fan's -, on 20V_GND
FAN_OUT 2	FAN_+	The fan's +, the regulator's 12 V output

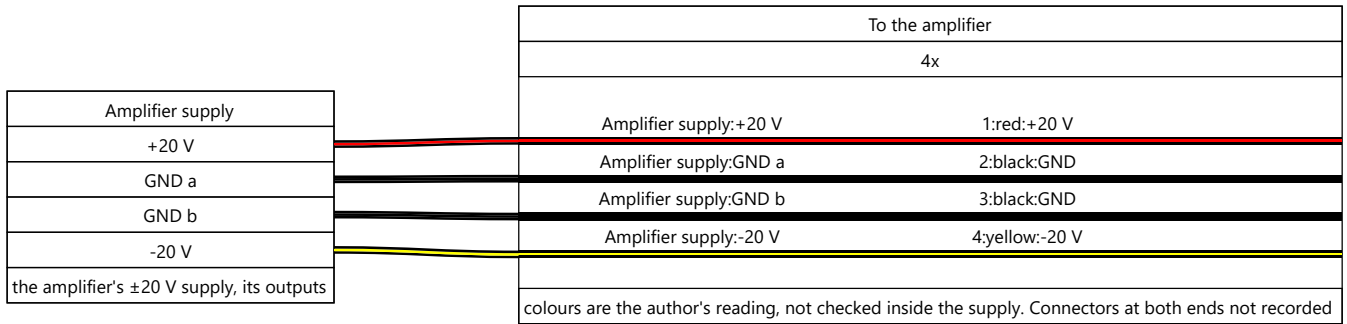
The fan runs only while the front power switch is on.

If it fails: the box fan stops while the radio plays (chapter 5 (p. 27), section 5.7 (p. 32)).

10.30 Amplifier supply wires — amplifier supply to the amplifier

Carries: the amplifier's ± 20 V supply, from the supply to the amplifier.

Shape: four wires. The connectors at both ends are not recorded.



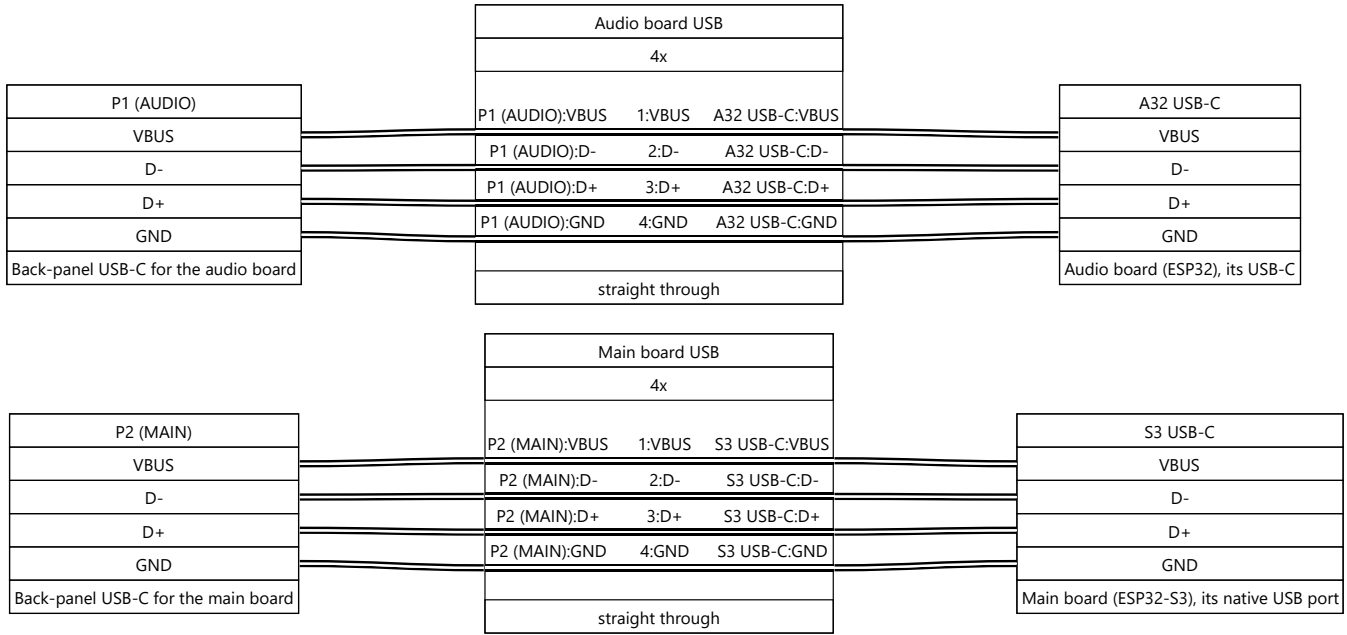
The amplifier supply's four wires.

Wire by wire. The drawing gives each wire's colour and what it carries; chapter 5 (p. 27), section 5.6 (p. 31), has the same as a table. These are the only wires in the machine whose colours are recorded. The colours are the author's reading, not checked inside the supply.

10.31 USB — back panel to each board

Carries: USB, from the back panel to each board.

Shape: two USB-C connectors on the back panel lead to the two boards. P2 (MAIN) goes to the main board's native USB port. P1 (AUDIO) goes to the audio board's USB-C. The cable type and length are not recorded.



The two USB connections, straight through.

Wire by wire. Each connection carries all four lines straight through: VBUS, D-, D+ and GND. On each back-panel connector the other three ground pins (A12, B1 and B12) are joined together. Nothing is recorded on the connectors' CC pins (the pins a USB-C host uses to detect a device).

Caution — USB power reaches the boards. VBUS is carried through, so a computer plugged into the back panel feeds a board's USB power input while the 5 V bus feeds its 5V pin. Chapter 2, section 2.5 (p. 15), quotes Espressif's guides on this.

If it fails: that board can no longer be reached over USB from the back panel.

10.32 Needle motor lead — needle motor driver to the needle motor

Carries: the four coil drives of the needle motor, a 28BYJ-48 stepper, and the coils' common. The driver outputs sink current to ground; the coils' common sits on 5 V.

Shape: the motor plugs into a 5-pin header on the needle motor driver board, marked "To 28BYJ-48" (J9). It is the connector that comes with the common ULN2003A boards. The motor's own lead is not recorded.

TO STEPPER		To the needle motor	
5-pin		5x	
OUT1 = COIL 4	1	TO STEPPER:1:OUT1 = COIL 4	1:COIL 4
OUT2 = COIL 3	2	TO STEPPER:2:OUT2 = COIL 3	2:COIL 3
OUT3 = COIL 2	3	TO STEPPER:3:OUT3 = COIL 2	3:COIL 2
OUT4 = COIL 1	4	TO STEPPER:4:OUT4 = COIL 1	4:COIL 1
COM = 5VDC	5	TO STEPPER:5:COM = 5VDC	5:COM
Needle motor driver board (ULN2003A), header J9, marked To 28BYJ-48		to the 28BYJ-48 stepper motor; its lead is not recorded	

The needle motor header. The motor end is not recorded.

Wire by wire.

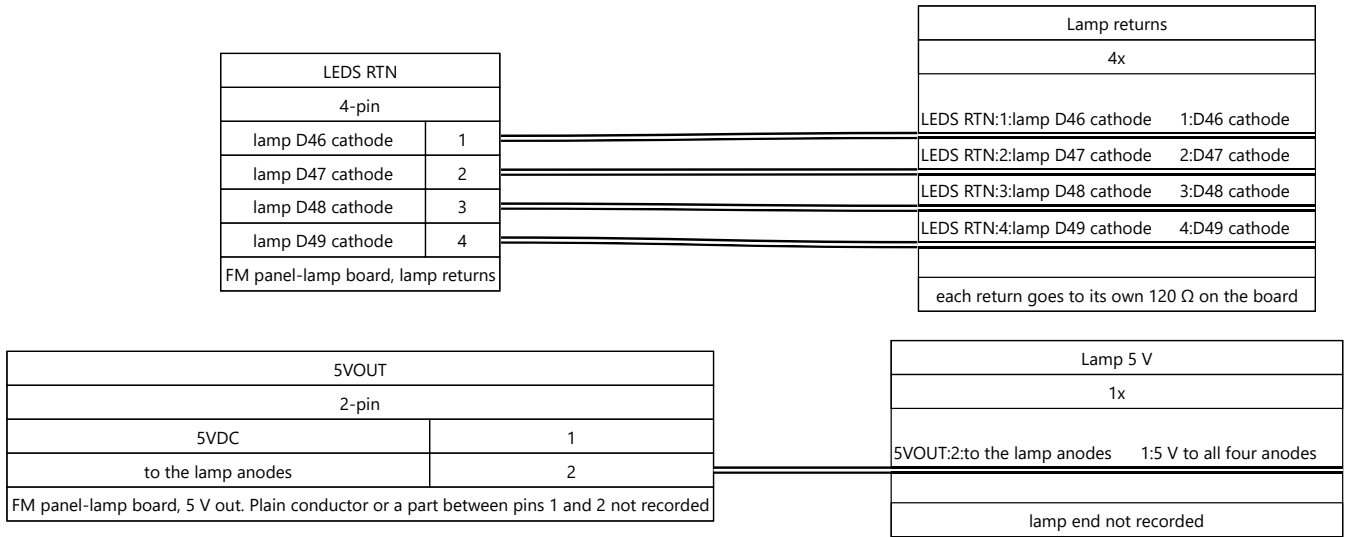
Header pin	Signal	What it is
1	OUT1	Driver output 1, to coil 4
2	OUT2	Driver output 2, to coil 3
3	OUT3	Driver output 3, to coil 2
4	OUT4	Driver output 4, to coil 1
5	COM	The coils' common, on 5 V

If it fails: the needle does not move, or moves erratically (chapter 8 (p. 45), section 8.7 (p. 48)).

10.33 Panel-lamp leads — panel-lamp board to the four FM lamps

Carries: 5 V to the four FM panel lamps (LEDs), and each lamp's return to the panel-lamp board.

Shape: two connectors on the panel-lamp board. 5VOUT (2 pins) carries 5 V to the four lamp anodes. LEDS RTN (4 pins) brings each lamp's cathode back to its own 120 Ω resistor on the board. The four resistors join on the MOSFET's drain (section 10.11 (p. 66)). The wiring at the lamps is not recorded.



The panel-lamp leads. The lamp end is not recorded.

Wire by wire. On **5VOUT**, as drawn, pin 1 is on the board's 5 V and pin 2 goes to the four anodes. Whether pin 1 and pin 2 are joined by a plain conductor or through a part is not recorded. On **LEDs RTN**, each pin takes one lamp:

LEDs RTN pin	What it is
1	Return from lamp D46's cathode
2	Return from lamp D47's cathode
3	Return from lamp D48's cathode
4	Return from lamp D49's cathode

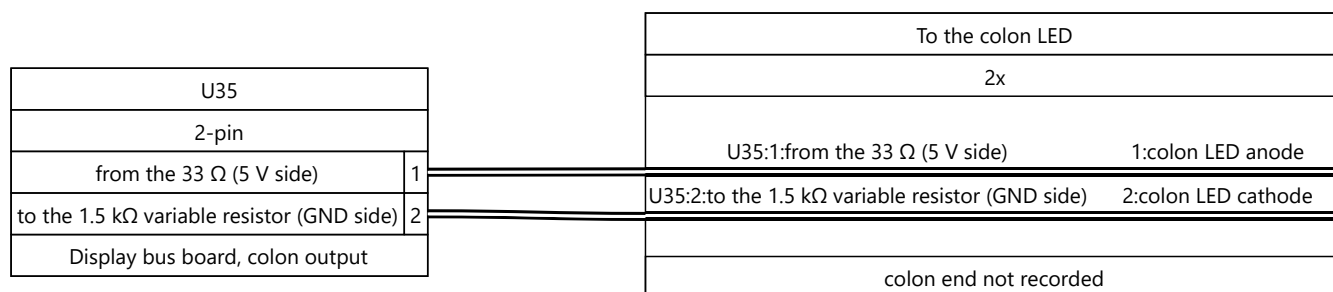
If it fails: a broken return leaves its one lamp dark; a broken 5 V lead leaves all four dark (chapter 9 (p. 50), section 9.6 (p. 55)).

10.34 Inside the clock display — colon and digits

Carries: the colon LED's two leads, and each digit's seven segment lines and its common anode. The colon's chain is 5 V → 33 Ω → colon LED → variable resistor (glued at 1.5 kΩ) → ground; the variable resistor sits at the colon module. Each digit is built of seven segment LEDs whose anodes join on pin 8 of its 8-pin header, with each segment's cathode on pins 1 to 7.

Shape — the colon. On the display bus board, the colon output (U35) has two pins. How the wires run to the colon is not recorded.

Shape — the digits. Each digit's header is drawn plugged into its own 8-pin socket, pin for pin; no cable is drawn. The display bus board has four 8-pin digit outputs, which carry the same signals on the same pin numbers. Whether those outputs are the sockets the digits plug into is not recorded.



The colon output. The colon end is not recorded.

Wire by wire — the colon. Pin 1 of the colon output comes from the 33 Ω, and pin 2 goes to the variable resistor. The colon LED sits between them, anode on pin 1.

Wire by wire — the digits. Pins 1 to 7 carry the same signal on every socket and every output; pin 8 is each digit's own common:

Pin	Signal on the socket and on the output	At the digit
1	SEG_A_BUS	Segment A cathode
2	SEG_B_BUS	Segment B cathode
3	SEG_C_BUS	Segment C cathode
4	SEG_D_BUS	Segment D cathode
5	SEG_E_BUS	Segment E cathode
6	SEG_F_BUS	Segment F cathode
7	SEG_G_BUS	Segment G cathode
8	D1, D2, D3 or D4	Common anode

Pin 8 is **D1** on output 1 (U18), **D2** on output 2 (U17), **D3** on output 3 (U19) and **D4** on output 4 (U32). Digit 1 is the rightmost.

If it fails: a broken colon lead leaves the colon dark. A bad contact on one pin of a digit's header leaves that segment dark in that digit only; on pin 8, the whole digit goes dark (chapter 7 (p. 40), section 7.8 (p. 44)).

10.35 FM receiver antenna and the antenna coax

Neither of these is drawn as a cable, and no pins are recorded for them. So this page has no drawing and no wire table.

Carries: two antenna signals: the FM calibration receiver's antenna wire, and the FM antenna's coax into the machine.

Shape — the receiver's antenna. The FM calibration receiver's antenna is a 30 cm wire inside the tube radio's Faraday cage, bent in a U over the radio's FM oscillator section. It reaches the receiver board through a 2×3 pin grid (U71), so that you can unplug the receiver from its antenna during a repair. Which pins of the grid carry the wire is not recorded.

Shape — the antenna coax. The antenna's coax arrives at a BNC male connector (J10) on the back panel: its centre carries the antenna signal `ANT`, and its shell is on Earth. The tube radio's antenna input is a second coax connector (J53); chapter 13 (p. 93), section 13.11 (p. 128), gives the radio's side of it. **No cable joining J10 to J53 is drawn or recorded.** The antenna itself is in chapter 11 (p. 87); the earth path is in chapter 5 (p. 27), section 5.4 (p. 30).

If it fails: if the receiver's antenna wire or its pin grid fails, the receiver still answers the main board but finds no signal (chapter 8 (p. 45), section 8.7 (p. 48)).

10.36 Mains and earth wiring

The mains wiring (inlet, front power switch, 5 V supply, amplifier supply, line filter and tube radio) and the earth wiring (the inlet's earth, the amplifier's back plate, the three Faraday cages and the antenna's earth) are described in chapter 5 (p. 27), sections 5.2 (p. 28) to 5.4. They are not drawn here. The mains wires' gauges and colours are not recorded.

10.37 When something is wrong

Find what you see in the left column; the right column names the cable pages to check.

You see	Check
A cable needs replacing	Each wire is identified by its signal name, not by its position (section 10.1 (p. 56)). Each page says at which end the shield is joined to ground.
One segment dark on every digit	Segment drive (10.5), then segment buses (10.28)
One digit dark	Digit select (10.8), then digit rails (10.27)
Whole display dark, colon lit	Driver board 5 V (10.26)
Colon dark, digits lit	Colon 5 V (10.23), and its reversed polarity
Needle does not move	Needle motor control (10.6), motor driver 5 V (10.24), needle motor lead (10.32)
Needle never finds its reference point	Needle index sensor harness (10.9)
Main board reads "amplifier off" with the amplifier on	Amplifier power sensor cable (10.7)
Panel lamps dark	Panel-lamp control (10.11), then the lamp leads (10.33)
Radio silent, Bluetooth plays	Radio to ADC (10.22), ADC 5 V (10.25), the ADC branch of the audio harness (10.15)
Radio and Bluetooth silent, AUX plays	DAC to amplifier (10.21), the DAC branch of the audio harness and its power lead (10.15)
Source switch stuck on AUX in one or all positions	Source switch cables (10.20)
Volume knob does nothing	Front volume knob cables (10.13), audio board 3.3 V (10.19)
Bluetooth LED dark, or button dead	Bluetooth button and LED cable (10.14)
The two boards do not talk	Board link (10.12)

11. The FM antenna

The tube radio receives FM from an outdoor antenna that the author built for this machine: a three-element Yagi on the roof. Any FM or VHF antenna will work on the radio's antenna input. This chapter describes this one, as built and installed, so that you can check it, repair it or build your own. Its dimensions, mounting and aim suit this house and this location. Adapt them to yours.

Sections 11.1 and 11.2 are the author's own figures. Section 11.3 gives further details from the build notes; those details are not confirmed one by one, so they are weaker than 11.1 and 11.2.

11.1 The aerial

A **Yagi** is a directional antenna: one **driven element**, the only part wired to the cable, with unpowered **parasitic** elements beside it on a common support, the **boom**. The longer **reflector** sits behind the driven element and the shorter **director** in front, so the antenna listens best in the direction of the director.

This one is a **three-element Yagi**, mounted horizontally on the roof. The table gives each part's material and size.

Part	Material	Size
Director	plain aluminium U-channel, 3.2 mm thick, 12.6 mm high, 25.5 mm wide	133.8 cm long
Driven element	½ in type L copper tube, folded (below)	139 cm per element; about 1390 mm tip to tip
Reflector	the same aluminium U-channel as the director	150.9 cm long
Boom	3030 aluminium extrusion	1170 mm long

The driven element is a folded dipole: two parallel copper tubes, joined at both ends by soldered 90-degree tubing, with the two conductors **50 mm** apart, centre to centre. Only this element is wired to the coax.

Nothing touches the boom electrically. The director and the reflector touch neither the boom nor anything else. The driven element reaches the boom only through a non-conductive clamp.

11.2 The feedline and its earthing

The **feedline** is the coax cable that carries the signal from the antenna down to the radio. The table gives how the feedline, its earthing and the mast are installed.

What	As installed
Feedline	Coax. At present it is at least 100 ft long. At its final place it is to be about 50 ft in total, from the bulkhead (the F-81 connector at the feedpoint, section 11.3 (p. 88)) to the radio.
Grounding block	Fitted outside, out of the weather. Its exact position does not matter.
Earth wire	8 AWG stranded copper, ferruled , run from the grounding block to the electricity meter box (Hydro-Québec) and landed on a lug on the meter base , beside one an electrician made for the external wiring.

What	As installed
Mast	The same 3030 aluminium extrusion as the boom, 4 ft high , fixed to the roof with continuous aluminium extrusion. The mast is centred on the boom.
Height and aim	About 4 ft over its base and 1 ft over the tip of the roof , clear of reflections from the roof. The roof's slope faces away from Mount Royal; the aerial points as near Mount Royal as it can .
Guys	Dacron lines, two from each end of the boom . As fitted, the antenna cannot move short of a very big accident.

The coax shield is earthed at the grounding block. The grounding block is outside. An 8 AWG stranded copper run goes from the grounding block to a lug on the meter base.

The boom is earthed through the structure. The boom is continuous metal through the mast to the roof's metal. The metal roof is electrically connected to the electrical mast, and so to earth. The boom and the coax shield's 8 AWG run therefore reach the same earth electrode system.

What follows from this. The elements sit on insulating bushings (section 11.3 (p.88)), so they are isolated from that earthed structure. The coax shield is earthed by its own 8 AWG run from the grounding block to the meter-base lug, not by a separate wire from the boom.

At the radio end. The coax arrives at the machine's antenna connector, a BNC (J10) on the back panel. Its centre carries the antenna signal, ANT ; its shell, and so the coax shield, is on Earth. The tube radio's own antenna input is another coax connector (J53); no cable joining J10 to J53 is recorded. Chapter 5, section 5.4 (p. 30), gives the earth path inside the cabinet. Chapter 13, section 13.11 (p. 128), gives the radio's side, where the shield reaches the chassis only through one safety capacitor (C51).

Caution — the shield and the chassis. The coax shield is on Earth. The radio's chassis is on mains neutral and is never joined to Earth. The shield's only path to the radio's chassis is the safety capacitor (C51). Chapter 2 explains what the chassis is joined to and why it matters.

11.3 Build details from the notes

These details come from the build notes and are not confirmed one by one.

Element positions. The table gives where each element sits along the boom, with the reflector at 0 cm.

Element	Position
Reflector	0 cm
Driven element	55 cm
Director	110 cm

Elements. Both U-channels are mounted opening downwards. The centres of the three elements lie in one plane, within 5 mm.

Feed gap. A **20 mm** gap is cut in one tube of the folded dipole only. The feedpoint is there.

How each element is mounted. Each element sits in a stack of parts on the boom, listed here from the top down:

1. a nut;

2. a Belleville washer (a dished spring washer);
3. a neoprene washer, of the kind used with metal-roofing screws;
4. a G10 washer (glass-epoxy laminate);
5. the element, in an ASA bushing;
6. an ASA plate, continuous with that bushing;
7. the boom, through a 90-degree clamp.

The driven element is mounted the same way. **No static-bleed resistors are fitted.** These would be 1 MΩ resistors from the parasitic elements to the boom, to let static charge drain away.

Feedpoint. The feedpoint is where the coax joins the driven element. An **F-81 bulkhead** connector (an F-type coax connector made to pass through a panel) sits in the ASA clamp. Its shell is on one side of the 20 mm gap, its centre pin on the other. **Six salvaged HDMI ferrite sleeves** (17.4 mm outside, 9.7 mm inside, 28.5 mm long) sit on the feedpoint as a **common-mode choke**, to keep signal currents off the outside of the coax shield.

Feedline and grounding block. The feedline is **RG-6** coax. The grounding block is a **Perfect Vision PVGB1HFWS**: UL listed, rated to 3 GHz, F-type, single, with a weather boot. It is a plain block. It has **no gas discharge tube** (no surge arrester).

11.4 What is not measured

- **No performance figure is recorded:** no SWR (standing-wave ratio, a measure of how well the antenna matches its cable), no field strength, and no comparison of reception with another antenna.
- **No drawing shows the antenna.** The drawings show only the radio end: the antenna connector and the radio's antenna input.
- **The cable from the antenna connector to the radio's antenna input** is not recorded (chapter 10 (p.56), section 10.35 (p.85)). How the RG-6 feedline is fitted to the BNC antenna connector (J10) is not recorded either.

11.5 When something is wrong

The table lists what you may notice and where to look first.

You see	Check
FM weak or noisy on every station, while AM and SW work	The FM antenna path, which only FM uses: the antenna connector (J10), the coax inside the cabinet, the radio's antenna input (J53) and its coupling capacitor (C3) (chapter 13 (p.93), section 13.2 (p.98)). Then the coax outside, the grounding block and the feedpoint.
The aerial has been moved or re-aimed	Its aim (as near Mount Royal as it can, here) and its height clear of the roof (section 11.2 (p.87)).

12. The amplifier and speakers

Every sound the machine makes leaves through one amplifier and its two speakers. The DAC's output (the RADIO and BT sources) and the AUX input both arrive at the amplifier's input panel, which adds them together. This chapter says what the amplifier and the speakers are, what the input panel does, and what has been measured there. It describes this speaker set; adapt it to yours.

The amplifier's own circuit is not covered. No drawing shows its boards.

12.1 What the amplifier and speakers are

The amplifier and speakers came from a **Kramer Tavor 5-0** powered speaker pair, part number 60-00009010: active left, passive right, 2×30 W RMS, 4 Ω . The active speaker held the amplifier; the passive one did not. The table lists the parts.

Part	What it is
Power stage	a TDA7265
Input board	three op-amp chips: a TL074 , a TL072 and a C4558
Supply	the amplifier's own isolated mains switching supply, measured at ± 19.33 V with no load . Two TO-220 regulators (a common three-pin power package) make ± 12 V for the op-amps. It reaches the amplifier on four wires, the only wires in the machine whose colours are recorded (chapter 5 (p. 27), section 5.6 (p. 31)). It also feeds the box fan and the amplifier power sensor (chapter 5 (p. 27), section 5.6 (p. 31); chapter 9 (p. 50), section 9.5 (p. 54)).
Auto mode	a mode of the original amplifier; it has been removed. Nothing more about it is recorded.
Volume control	a B10k dual-gang potentiometer (section 12.2 (p. 90))
Speakers	on each side, a 5.25 in polypropylene woofer and a 0.5 in Mylar dome tweeter , with a first-order passive crossover at 8 kHz

Passive radiators, not ports. The original speaker cabinets were ported. In this machine the speakers work with **passive radiators**, added by the author, and are not ported. A passive radiator is a cone with no magnet or coil of its own. The air in the enclosure moves it, where a port would let that air out through a tube. Its printed parts (a holder, two protectors, a membrane weight and a TPU membrane) are listed in Appendix D (p. 143).

The earth plate. Of the original cabinets, only the active speaker's aluminium back plate remains. It is at the back of the cabinet. The machine's mains inlet came on this plate. The plate carries the earth star point, and the amplifier's ground is joined to it. Chapter 5, section 5.3 (p. 28), explains that joint and what it means for the DC ground.

12.2 The volume control

The amplifier's original dual-gang volume potentiometer has been replaced with a **B10k dual-gang potentiometer**: two potentiometers on one shaft, one per channel. Its two gangs are separate, as the original's were. It has a power switch, which is not used.

- It sits **on the amplifier's circuit board** and is reached **only from the back**.
- It **does not reach silence** at its minimum.
- It is **not** the front volume knob. The front knob is read digitally by the audio board and acts on RADIO and BT only (chapter 9 (p. 50), section 9.4 (p. 53)). Chapter 6, section 6.8 (p. 38), sets out all three volume controls in the chain.

12.3 The input panel

The input panel is a small circuit board carrying the amplifier's inputs. The cable sheet's `AMP_` signals (chapter 10 (p. 56)) land on its pins. Where the panel sits in the cabinet is not recorded.

It adds up its inputs; it does not choose between them. If two or three inputs carry sound at the same time, the amplifier plays their sum.

A **balanced** input takes each channel on two wires, a plus and a minus, and amplifies the difference between them. The table lists the panel's three inputs, their contacts, and whether each is grounded with nothing plugged in.

Input	Contacts	With nothing plugged in
Balanced	L+, L-, ground, R-, R+	not grounded
AUX	a TRS jack (tip, ring, sleeve)	grounded
RCA pair	two RCA jacks	grounded

The DAC reaches the panel through the balanced input. Its cable puts a 470 Ω resistor in each of its four signal legs and carries six ferrite cores (chapter 6 (p. 33), section 6.6 (p. 37); chapter 10 (p. 56), section 10.21 (p. 75)).

The AUX jack reaches the amplifier's input panel, which sums it with the other inputs; the run is not drawn. Where the AUX jack sits in the cabinet is not recorded; it is not among the front-panel controls (chapter 3 (p. 16), section 3.2 (p. 16)). AUX never passes through the audio board. Nothing is recorded on the RCA pair (chapter 6 (p. 33), section 6.7 (p. 38)).

12.4 What was measured at the panel and on the cable

These readings show how the amplifier's balanced input behaves at DC, which matters if you chase a noise or a fault on the DAC-to-amplifier link. All the readings below are resistance, taken with a multimeter, one reading each.

At the amplifier's balanced input, with the DAC-to-amplifier cable unplugged at the amplifier:

Between	Reading
L- and <code>AMP_G</code> (ground)	open: 3 M Ω and rising, as a capacitor charged
R- and <code>AMP_G</code>	open: 3 M Ω and rising, as a capacitor charged

On the DAC-to-amplifier cable, unplugged. These readings were taken before the 470 Ω resistors on the two plus legs (L+ and R+) were fitted:

Between	Reading
L– and ground	470 Ω
R– and ground	470 Ω
L– and R–	a little under 1 k Ω
L+ and ground	4.2 M Ω
R+ and ground	4.2 M Ω

What this supports, and no more. At the amplifier's connector, neither minus leg has a DC path to the amplifier's ground. That is what a differential, AC-coupled input looks like on a meter. Such an input amplifies the difference between its two legs, through capacitors that pass audio but block DC. It is one meter reading, taken once; nothing inside the amplifier was examined.

The frying noise once heard at the amplifier, and the parts later added to the DAC-to-amplifier cable, are in chapter 6 (p. 33), section 6.6 (p. 37).

12.5 When something is wrong

The table lists what you may notice and where to look first.

You see	Check
No sound from any source, with the front power switch on	The amplifier supply: +20 V and –20 V on its four wires (chapter 5 (p. 27), section 5.6 (p. 31)). Then the amplifier's own volume control, at the back.
The amplifier is never fully silent	Normal: its volume control does not reach silence at its minimum.
Two sources heard at once	Normal: the input panel adds its inputs. Unplug whatever is playing into AUX (chapter 6 (p. 33), section 6.7 (p. 38)).
Radio and Bluetooth silent, AUX plays	Not the amplifier: the DAC side (chapter 6 (p. 33), section 6.9 (p. 39)).
A frying noise at the amplifier	The DAC-to-amplifier cable, its four 470 Ω resistors and its ferrites (chapter 6 (p. 33), section 6.9 (p. 39)).

13. The tube radio (LLOYDS TM-838N)

The tube radio is the part of Ambersong that receives. It is a LLOYDS TM-838N, an AM, short-wave (SW) and FM table set with seven tubes, kept on its own chassis inside the cabinet. When you listen to the radio, every station is found, amplified and detected by these tubes; the rest of the machine only digitises the result. This chapter describes the set stage by stage, as fitted in this machine, so that you can check or repair it without its schematic.

This is one TM-838N as fitted and repaired. Another set of the same model will differ in details.

Danger — a radio with no mains transformer. The radio's chassis sits on mains neutral, and its high-voltage supply (B+) is made straight from the mains. Read chapter 2 (p. 13) before you work on it, and unplug the cord first.

13.1 Overview

13.1.1 What the set is

The set receives three bands, chosen with its four-position **mode switch**: **SW**, **AM**, **FM** and **FM-AFC**. FM-AFC is FM with **automatic frequency control (AFC)** switched on: a voltage from the FM detector keeps the FM oscillator on the station (section 13.3 (p. 101)).

It is a **superheterodyne**. A **mixer** combines the incoming station with the signal of a **local oscillator** inside the set, and passes on the difference, the **intermediate frequency (IF)**. The IF is the same for every station, so the stages after the mixer stay tuned to it and never move.

FM has its own **front end**, the stages that take the station at its own radio frequency (**RF**) and mix it down to the IF. AM and SW share another front end, and use a different IF:

Band	Front end	IF
FM	the FM RF amplifier and mixer (V1), with the FM local oscillator (V2)	10.6 MHz
AM and SW	the converter (V3), a mixer tube with its own oscillator	455 kHz

The two paths meet at the **first IF amplifier (V4)**, which amplifies both IFs. After it, the FM IF goes to the **FM limiter (V5)** and the **ratio detector**. The AM IF goes to the **AM detector**, the two diodes inside V6. The audio unit of the mode switch picks the audio of one of the two detectors. From there the sound passes through the radio's **volume control**, the **first audio amplifier** (V6's triode), the **output stage (V7)** and the **output transformer**.

The FM IF is 10.6 MHz. It was set when a professional, Radio Hosvep on Park Ave., tuned the FM path. The readings of the FM calibration receiver, the single-chip FM receiver the main board uses to calibrate the dial (chapter 8 (p. 45)), agree: they find the FM local oscillator one IF below each station. 10.7 MHz, the figure the SAMS service data gives (section 13.1.3 (p. 94)), is the design value. Only the FM path was tuned: the three FM IF transformers (L10, L12 and L14) and what touches them. The AM and SW path was not.

The set is built the old "AC/DC" way. It has no mains transformer: its seven tube heaters are wired in series across the mains (section 13.9 (p. 121)), and a single rectifier makes its B+ straight from the mains (section 13.10 (p. 125)).

13.1.2 How it sits in Ambersong

The radio is a unit of its own inside the cabinet. The table lists every place where it meets the rest of the machine:

Where the radio meets the machine	What happens there	More in
Mains	The radio is fed from the front power switch through the mains line filter. It has its own power switch and a 0.75 A fuse.	chapter 5 (p. 27); 13.10
Chassis	The radio's circuit common, CHASSIS, is on mains neutral. It is never joined to Earth or to the Faraday cages.	chapter 2 (p. 13)
FM antenna	The antenna's coax arrives at the radio's coax input. Its shield is on Earth and reaches the chassis only through one safety capacitor.	13.2; chapter 11 (p. 87)
Sound out	The output transformer's secondary, loaded by a 7.5 Ω 5 W resistor, feeds the ADC (the analogue-to-digital converter board) through a shielded cable. The feed is mono.	13.11; chapter 6 (p. 33); chapter 10 (p. 56), section 10.22 (p. 76)
Speakers	SP1 and SP2 are the radio's original speakers as drawn by the factory. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.	13.8
Volume	The radio's own volume control sits at about one eighth of its travel in normal use. Above that the sound is too loud further down the chain; it is not clipped.	13.8; chapter 6 (p. 33)
Tuning	A printed front knob, on a printed shaft extension, turns the radio's original tuning knob, which drives the dial cord and the four-section tuning capacitor. A magnetic angle sensor reads its shaft; a motor, not the dial cord, moves the needle.	chapter 8 (p. 45)
FM calibration receiver	Its antenna is a 30 cm wire inside the radio's cage, bent in a U over the FM oscillator section.	13.3; chapter 8 (p. 45)
Cage and fan	The radio sits under its own earthed Faraday cage, with a 5 V fan blowing air into it.	chapters 3 and 5

13.1.3 Fitted values and original specifications

Each stage has a parts table with two value columns:

- **Fitted** is the part in this radio now.
- **Original** is what the two original documents give: the **factory drawing** and the **SAMS** service data. "Same" means both give the fitted value. Where they differ, the table gives both; the fitted value is what is in the radio. Where neither gives a value, the table says so.

Unless a table says otherwise, neither original gives a resistor's wattage or tolerance or a capacitor's voltage rating, and none is recorded for the fitted parts.

Substitutes and selection notes. The original specifications come with notes on catalogue substitutes and on choosing a replacement, from a component list the author compiled and has not published. That list is not in this book. A "listed substitute" in a parts table is one named in that list, and a note on what to match when you replace a part is that list's selection note. They describe the original part, though a few notes mention the fitted one.

Three-lead capacitors. Four ceramic capacitors have three leads: the FM supply bypass (C5), the AFC shunt (C11), the AFC supply bypass (C14) and one heater-string bypass (C48). The middle lead is wired to the chassis, and the two outer leads have 0 Ω between them.

Terminal numbers are logical. The numbers and letters given here for coil terminals and mode-switch contacts identify connections on the drawing and in the SAMS data. They are not the numbers on the parts' lugs, and which lug is which has not been recorded.

IF transformers. Each IF transformer sits in a shielded metal **can**, together with its own tuning capacitors. Neither original gives the values of those capacitors.

Trimmers and padders. Two kinds of small capacitor recur in the tuned circuits. A **trimmer** is a small adjustable capacitor set once, during alignment. A **padder** is a fixed capacitor in series with an oscillator coil; it makes the oscillator track the station across the dial.

13.1.4 Tube line-up

The set has seven tubes. "New old stock" means a tube that was made in the period and never used. The three new-old-stock tubes (V1, V2 and V6) were fitted by Radio Hosvep on Park Ave.; their types and connections are the original ones.

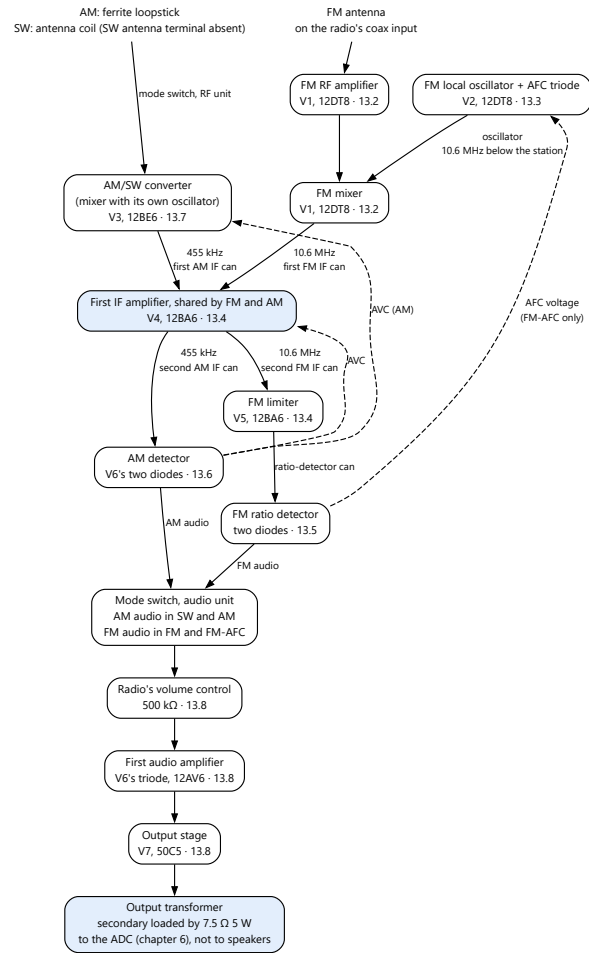
Tube	Type	What it does	Section
V1	12DT8 twin triode, new old stock	first triode: FM RF amplifier; second triode: FM mixer	13.2
V2	12DT8 twin triode, new old stock	first triode: FM local oscillator; second triode: AFC triode	13.3
V3	12BE6 pentagrid converter	AM and SW mixer with its own oscillator	13.7
V4	12BA6 pentode	first IF amplifier, for FM and AM	13.4, 13.6
V5	12BA6 pentode	FM limiter	13.4
V6	12AV6 duplex-diode triode, new old stock	its two diodes: AM detector; its triode: first audio amplifier	13.6, 13.8
V7	50C5 beam power tube	audio output	13.8

The tube pin assignments and the order of the heater string given in this chapter were checked on the chassis. The tubes' published ratings:

Type	Base	Heater
12DT8	9-pin miniature (noval, B9A), EIA basing 9DE	12.6 V, 0.15 A
12BE6	7-pin miniature (B7G), EIA basing 7CH	12.6 V, 0.15 A
12BA6	7-pin miniature (B7G), basing 7BK	12.6 V, 0.15 A
12AV6	7-pin miniature (B7G)	12.6 V, 0.15 A
50C5	7-pin miniature (B7G), EIA basing 7CV	50 V, 0.15 A, for series heater strings

Caution — a 12DT8 is not a 12AT7. The two are alike except for their heater and heater-to-cathode ratings, their capacitances and their **basing**. Do not assume a 12AT7 pinout when you wire or test a 12DT8 socket. The 12DT8's two triodes are separated by an internal shield on its own pin (pin 9).

13.1.5 Block diagram



The tube radio's stages, from the antennas to the output transformer.

Dashed lines are control voltages, not signal. **AVC** (automatic volume control) is a voltage from the AM detector that sets the gain of the first IF amplifier, and in AM of the converter, so that strong and weak stations come out at similar levels (section 13.6 (p. 111)).

13.1.6 The mode switch

The mode switch is one rotary switch with four positions. The factory drawing calls it a set of mechanically linked band, supply, audio, AFC and neon contacts; SAMS (its part M5) gives four positions and three physical sections. The drawing splits it into seven units, S1A to S1G. They are seven functions of one switch, not seven switches, and they all turn together.

The table gives, for each unit, where its common goes and what the common is joined to in each position:

Unit, and its job	Its common goes to	SW	AM	FM	FM-AFC	Contacts (common; SW, AM, FM, FM-AFC)
S1A , AFC off	chassis	not connected	not connected	the AFC control line: AFC off	not connected: AFC on	13; 1, 2, 3, 4

Unit, and its job	Its common goes to	SW	AM	FM	FM-AFC	Contacts (common; SW, AM, FM, FM-AFC)
S1B , audio	top of the radio's volume control	AM audio	AM audio	FM audio	FM audio	14; 5, 6, 7, 8
S1C , RF selection	the converter's signal grid (V3 pin 7) and the AM/SW RF tuning section (VC3)	SW antenna coil and its trimmer	AM loopstick and its trimmer	not connected	not connected	15; 9, 10, 11, 12
S1D , B+	the last point of the B+ filter (B115_A)	AM supply, AM_B	AM_B	FM supply, FM_B	FM_B	14; 5, 6, 7, 8
S1E , mode lamps	the neon feed, from the fused live through a 50 kΩ resistor (R32)	SW lamp (NE1)	AM lamp (NE2)	FM lamp (NE3)	FM lamp (NE3)	15; 9, 10, 11, 12
S1F , oscillator tank	the AM/SW oscillator tuning section (VC4) and feedback capacitor (C25)	SW padder (C26, 0.003 μF)	AM padder (C27, 350 pF) and AM oscillator trimmer (C23B)	not connected	not connected	15; 9, 10, 11, 12
S1G , oscillator feedback	the converter's cathode (V3 pin 2)	tap of the SW oscillator coil (L9)	tap of the AM oscillator coil (L8)	not connected	not connected	13; 1, 2, 3, 4

The contact numbers are logical (section 13.1.3 (p. 94)), and **the same number on two units is not the same contact**. SAMS describes the switch as three physical sections, or wafers, and the numbers repeat from one to another. S1A's common 13 is on the chassis, for example, while S1G's common 13 is on the converter's cathode. S1C's numbers are SAMS's markings for the switch's second section; S1F's and S1G's are its markings for the third.

What the B+ unit decides. AM_B feeds the converter (V3), whose plate supply passes through the primary of the first AM IF transformer. FM_B feeds the FM RF amplifier and mixer (V1), the FM oscillator and AFC triode (V2), the limiter (V5) and the first FM IF transformer's primary. So V1, V2 and V5 run only in FM and FM-AFC, and V3 only in SW and AM. The first IF amplifier (V4), the first audio amplifier (V6) and the output stage (V7) are fed from the B+ filter directly, in every position. Their supply is B115_A, and, through the output transformer, B130, the first filtered B+ point.

The mode lamps are not fitted. The neon bulbs NE1, NE2 and NE3 are absent from this radio. Their resistor (R32) and the S1E contacts are fitted.

Caution — replacing the mode switch. No commercial substitute is listed. A replacement must match this whole truth table, the shaft and detents, and the insulation and current ratings.

13.1.7 How parts are named in this chapter

Every part is named by what it does, with its reference from the radio's drawing in brackets: "the FM RF trimmer (C8)". **Always read the two together.** Many of the radio's references are also used for different parts elsewhere in the machine: C13 to C30, J2 to J4 and R17 exist on the main drawing too, and M1 is both the radio's fuse and the box fan. Where a reference stands alone in this chapter, it is the radio's part.

The first letters of the radio's references mean:

Letter	In the radio
V	a tube
R, C, L	resistor, capacitor, coil or transformer
VC	a section of the four-section tuning capacitor (VC1 to VC4, one assembly, SAMS part M2)
S	a switch: S1 is the mode switch, S2 the power and tone switch (SAMS part M6)
M	a part numbered from the SAMS list: the fuse (M1), the FM detector diodes (M3, M4)
J	a terminal or connector
NE, SP, T, X	neon lamp, speaker, transformer, rectifier

The output transformer (T1) carries the same reference in the radio and in the rest of the machine: it is one part.

Signal names in `code` font are the drawing's names for a connection, such as `FM_B` for the FM B+ supply. They are used where a connection reaches beyond one stage.

13.2 FM RF amplifier and mixer (V1, 12DT8)

V1 holds two triodes. The first is the **FM RF amplifier**: it lifts the weak signal from the antenna. Its grid is held at the chassis and the signal enters at its cathode, a **grounded-grid** amplifier. The second triode is the **FM mixer**. It takes the amplified station and the FM local oscillator's signal from V2. It sends their difference, the 10.6 MHz IF, to the first FM IF transformer. Both triodes take their B+ from `FM_B`, so they work only in FM and FM-AFC.

13.2.1 How it is joined

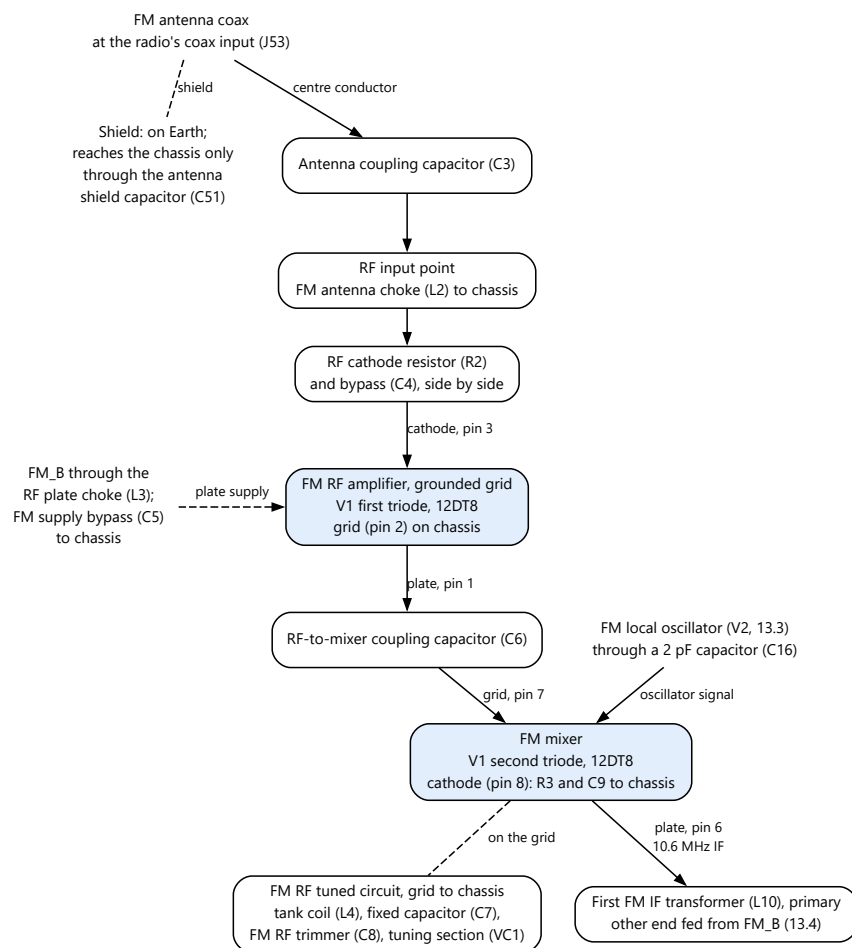
Antenna input. The FM antenna's coax arrives at the radio's coax input (J53). Its centre conductor goes through the antenna coupling capacitor (C3) to the **RF input point**. From that point the FM antenna choke (L2) goes to the chassis. The RF cathode resistor (R2) and the RF cathode bypass capacitor (C4), side by side, go from the same point to the RF triode's cathode. The coax shield is on Earth and reaches the chassis only through the antenna shield capacitor (C51). The radio first had a simple wire antenna; the coax input replaced it.

RF triode. The grid (pin 2) is on the chassis. The plate (pin 1) is fed from `FM_B` through the RF plate choke (L3), and `FM_B` is bypassed to the chassis there by the FM supply bypass capacitor (C5). The plate's signal goes to the mixer grid through the RF-to-mixer coupling capacitor (C6).

Mixer grid. The mixer grid (pin 7) sits on the **FM RF tuned circuit**, or **tank**. It is four parts, each joined from the grid to the chassis: the FM RF tank coil (L4), the tank's fixed capacitor (C7), the FM RF trimmer (C8) and the FM RF tuning section (VC1). The FM oscillator's signal arrives on the same grid through a 2 pF capacitor (C16, section 13.3 (p. 101)).

Mixer cathode and plate. The cathode (pin 8) goes to the chassis through the mixer cathode resistor (R3) and its bypass capacitor (C9), side by side. The plate (pin 6) goes to the primary of the first FM IF transformer (L10). The plate's B+ comes through that primary, whose other end is fed from `FM_B` (section 13.4 (p. 104)).

Heater and shield. The heater is on pins 4 and 5, in the series string (section 13.9 (p. 121)). The internal shield, pin 9, is on the chassis.



How the stage is joined (simplified).

13.2.2 V1's pins

The table gives each of V1's pins and what it is joined to.

Pin	What it is	Joined to
1	RF triode plate	RF plate choke (L3) to FM_B; coupling capacitor (C6) to the mixer grid
2	RF triode grid	chassis
3	RF triode cathode	RF cathode resistor (R2) and bypass (C4), to the RF input point
4	heater	heater string, toward the FM oscillator tube's heater through a heater choke (L15)
5	heater	heater string, from the converter's heater through a heater choke (L16)
6	mixer plate	primary of the first FM IF transformer (L10)
7	mixer grid	FM RF tuned circuit (L4, C7, C8, VC1); oscillator injection (C16)
8	mixer cathode	mixer cathode resistor (R3) and bypass (C9), to chassis
9	internal shield	chassis

13.2.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
FM RF amplifier and mixer tube (V1)	12DT8, new old stock	12DT8	
Radio's FM coax input (J53)	coax connector, RG179 coax	a simple wire antenna	centre to the antenna coupling capacitor (C3); shield to Earth
Antenna coupling capacitor (C3)	100 pF, safety class X1Y1	100 pF (factory and SAMS)	A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)); no voltage rating recorded. The substitutes listed for the original are DC-rated and do not establish a modern safety class or insulation design.
FM antenna choke (L2)	1 μ H, 23 turns	factory 1 μ H; SAMS adds 23 turns	
RF cathode resistor (R2)	70 Ω	factory 70 Ω ; SAMS 68 Ω , with 70 Ω as an alternate	
RF cathode bypass capacitor (C4)	0.002 μ F	same	
Antenna shield capacitor (C51)	0.001 μ F, safety class X1Y2	not recorded	The coax shield's only path to the chassis. A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)); no voltage rating recorded. Where it came from is not recorded.
RF plate choke (L3)	3 μ H	same	SAMS: DCR about 0.9 Ω .
FM supply bypass capacitor (C5)	0.002 μ F, three-lead ceramic	factory 0.002 μ F; SAMS schematic 0.01 μ F (0.002 μ F alternate); SAMS parts table 0.002 μ F	Middle lead to chassis (section 13.1.3 (p. 94)).
RF-to-mixer coupling capacitor (C6)	100 pF	same	
FM RF tank coil (L4)	tuned coil	no inductance in either original	
Tank's fixed capacitor (C7)	5 pF	same	
FM RF trimmer (C8)	trimmer	no value in either original	Range and dielectric not specified. Belongs with the tuning capacitor assembly.
FM RF tuning section (VC1)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Turns with the other three sections of the tuning capacitor. A replacement must match the minimum and maximum capacitance, tracking, shaft and gang geometry, and trimmers.
Mixer cathode resistor (R3)	1 k Ω	same	
Mixer cathode bypass capacitor (C9)	0.01 μ F	same	

13.2.4 How to check it

- **Coil resistance.** SAMS gives a DCR (DC resistance) of about $0.9\ \Omega$ for the RF plate choke (L3). A DCR is a check for an open winding, not a measure of inductance.
- **The three-lead bypass (C5).** Its two outer leads read $0\ \Omega$ to each other, and the middle lead goes to the chassis.
- **The antenna side.** The coax shield reaches the chassis only through the antenna shield capacitor (C51). Nothing else joins them.
- **Tube pins.** Use the table in section 13.2.2 (p. 99), not a 12AT7 pinout.

13.3 FM local oscillator and AFC (V2, 12DT8)

V2 also holds two triodes. The first is the **FM local oscillator**. It runs **one IF below the station**: oscillator frequency = station – 10.6 MHz. For a station at 100.0 MHz it runs at 89.4 MHz. Its tuning section (VC2) turns with the FM RF tuning section (VC1), so the oscillator follows the dial. The second triode is the **AFC triode**. Its plate is coupled into the oscillator's tuned circuit, and its grid takes the AFC control voltage from the ratio detector. In FM-AFC that voltage nudges the oscillator so the station stays tuned. In FM the mode switch holds it at the chassis, and AFC is off. Both triodes work only in FM and FM-AFC.

13.3.1 How it is joined

Supply. `FM_B` reaches this stage only through the AFC plate feed resistor (R8). After it, a local supply point is bypassed to the chassis by the AFC supply bypass capacitor (C14). This point feeds both triodes: the AFC plate through an RF choke (L1), and the oscillator plate through the oscillator plate feed resistor (R10).

Oscillator tuned circuit. The oscillator's tank is four parts, each joined from the tank to the chassis: the tank winding of the FM oscillator coil (L5), the tank's fixed capacitor (C17), the FM oscillator trimmer (C18) and the FM oscillator tuning section (VC2). Three capacitors meet the tank: the oscillator plate's (C20, 20 pF), the AFC plate's (C15, 10 pF), and the one that carries the oscillator's signal out to the FM mixer grid, V1 pin 7 (C16, 2 pF).

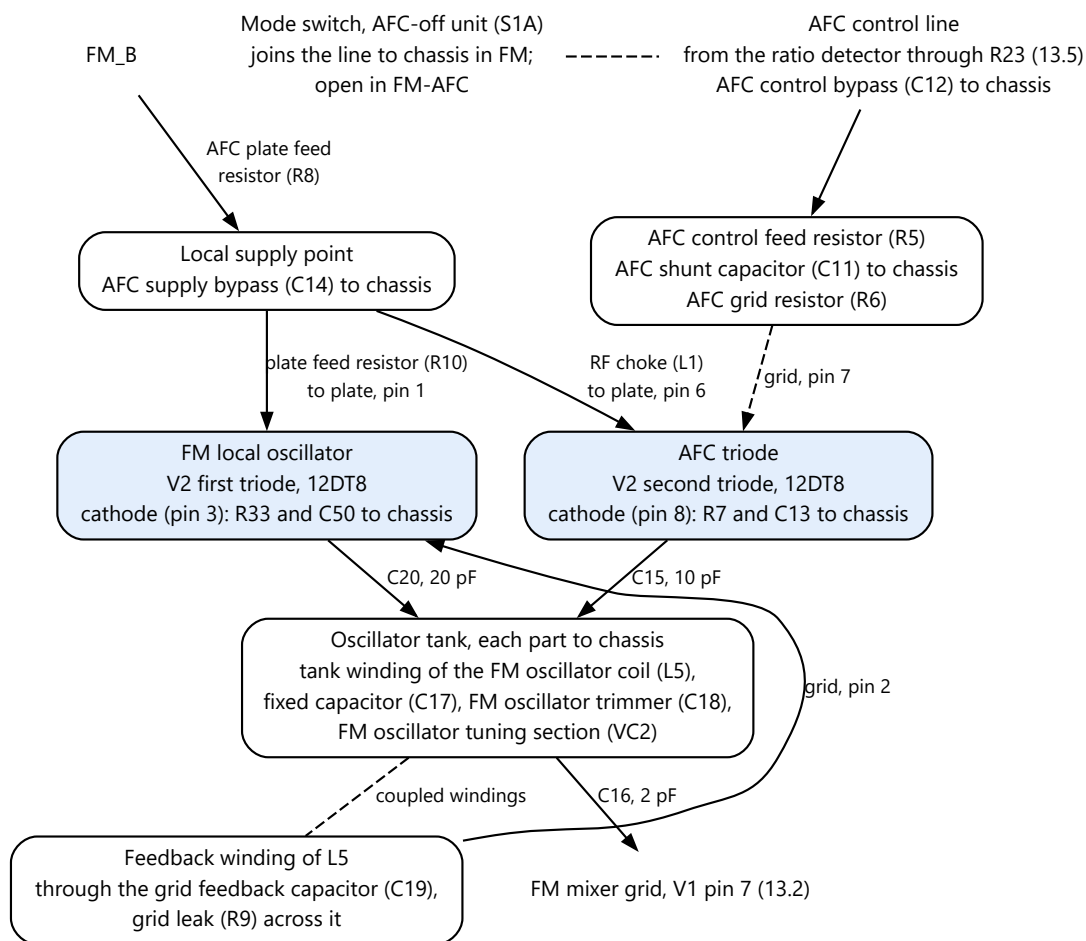
Oscillator feedback. The FM oscillator coil (L5) has two coupled windings: the tank winding and a **feedback winding**. One end of each is on the chassis. The feedback winding's other end goes to the oscillator grid (pin 2) through the grid feedback capacitor (C19), with the grid leak resistor (R9) across it. A **grid leak** is the resistor that gives a grid its DC path to the chassis. The oscillator cathode (pin 3) goes to the chassis through the oscillator cathode resistor (R33) and its bypass capacitor (C50), side by side.

AFC triode. The AFC control line arrives from the ratio detector through a $3\ \text{M}\Omega$ resistor (R23, section 13.5 (p. 108)) and is bypassed to the chassis by the AFC control bypass capacitor (C12). From there the AFC control feed resistor (R5) leads to a point that the AFC shunt capacitor (C11) joins to the chassis. From that point the AFC grid resistor (R6) leads on to the AFC grid (pin 7). The grid has no other connection. The AFC cathode (pin 8) goes to the chassis through the AFC cathode resistor (R7) and its bypass capacitor (C13), side by side. The AFC plate (pin 6) is fed through the RF choke (L1) and coupled to the oscillator tank through its coupling capacitor (C15).

In FM, the AFC-off unit of the mode switch (S1A) joins the AFC control line to the chassis. In FM-AFC that contact is open (section 13.1.6 (p. 96)).

Heater and shield. The heater is on pins 5 and 4. Pin 5 comes from the FM RF tube's heater through a heater choke (L15). **Pin 4 is the chassis end of the whole heater string:** V2 is the last tube in it (section 13.9 (p. 121)). The internal shield, pin 9, is on the chassis.

Near this stage. The FM calibration receiver's antenna, a 30 cm wire, lies over this section, bent in a U, inside the radio's cage. It is not connected to the radio (chapter 8 (p. 45)).



How the stage is joined (simplified).

13.3.2 V2's pins

The table gives each of V2's pins and what it is joined to.

Pin	What it is	Joined to
1	oscillator plate	plate feed resistor (R10) to the local supply; plate coupling capacitor (C20) to the tank
2	oscillator grid	grid feedback capacitor (C19) with grid leak (R9) across it, to the feedback winding
3	oscillator cathode	oscillator cathode resistor (R33) and bypass (C50), to chassis
4	heater	chassis: the end of the heater string
5	heater	heater string, from the FM RF tube through a heater choke (L15)
6	AFC plate	RF choke (L1) to the local supply; coupling capacitor (C15) to the tank
7	AFC grid	AFC grid resistor (R6) only

Pin	What it is	Joined to
8	AFC cathode	AFC cathode resistor (R7) and bypass (C13), to chassis
9	internal shield	chassis

13.3.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
FM oscillator and AFC tube (V2)	12DT8, new old stock	12DT8	
AFC control feed resistor (R5)	250 k Ω	same	
AFC shunt capacitor (C11)	0.002 μ F, three-lead ceramic, from the AFC grid network to chassis	factory 0.002 μ F; SAMS 0.001 μ F	The fitted C11 is a shunt from the AFC line (AFC_C, the point between the AFC control feed resistor R5 and the AFC grid resistor R6) to chassis. SAMS's substitute list is for its 0.001 μ F part. Middle lead to chassis (section 13.1.3 (p. 94)).
AFC grid resistor (R6)	100 Ω	same	
AFC control bypass capacitor (C12)	0.01 μ F	factory 0.01 μ F; SAMS 0.001 μ F	
Oscillator cathode resistor (R33)	600 Ω	not recorded	
Oscillator cathode bypass capacitor (C50)	0.01 μ F	not recorded	
AFC cathode resistor (R7)	600 Ω	same	
AFC cathode bypass capacitor (C13)	0.01 μ F	factory 0.01 μ F; SAMS schematic 0.002 μ F (0.01 μ F alternate); SAMS parts table 0.01 μ F	
AFC RF choke (L1)	3 μ H	same	SAMS: DCR about 0.9 Ω .
AFC plate feed resistor (R8)	1 k Ω	same	
AFC supply bypass capacitor (C14)	0.002 μ F, three-lead ceramic	same	Middle lead to chassis (section 13.1.3 (p. 94)).
AFC-to-oscillator coupling capacitor (C15)	10 pF	same	

Part	Fitted	Original	Notes
Oscillator-to-mixer coupling capacitor (C16)	2 pF	same	Several listed substitutes, Sprague among them, are 2.2 pF. Prefer a low-loss NP0 part of exactly 2 pF.
Oscillator tank's fixed capacitor (C17)	5 pF	same	
FM oscillator trimmer (C18)	trimmer	no value in either original	Range and dielectric not specified. Belongs with the tuning capacitor assembly.
FM oscillator tuning section (VC2)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Turns with the other three sections of the tuning capacitor.
FM oscillator coil (L5)	two coupled windings, tank and feedback	factory: two coupled windings; SAMS: one tapped winding; no inductance in either	Do not replace it with a tapped oscillator coil on the strength of its name.
Grid feedback capacitor (C19)	50 pF	same	
Grid leak resistor (R9)	20 k Ω	same	
Oscillator plate feed resistor (R10)	30 k Ω	factory 30 k Ω , with a different supply connection; SAMS 10 k Ω	
Oscillator plate coupling capacitor (C20)	20 pF, from the oscillator plate to the tank	factory 20 pF	SAMS's C20 is a different part, a 0.002 μ F bypass; its substitutes do not apply here. The factory part's dielectric, tolerance and voltage rating are unknown.

13.3.4 How to check it

- **The oscillator's frequency.** The FM calibration receiver hears this oscillator: its readings put the oscillator one IF, 10.6 MHz, below each station (chapter 8 (p. 45)).
- **AFC off in FM.** With the cord out, the mode switch's AFC-off unit (S1A) joins the AFC control line to the chassis on FM, and that contact is open on FM-AFC.
- **Coil resistance.** SAMS gives a DCR of about 0.9 Ω for the AFC RF choke (L1).
- **The three-lead capacitors (C11, C14).** The two outer leads read 0 Ω to each other, and the middle lead goes to the chassis.

13.4 The FM IF path (V4, V5, 12BA6)

The **IF amplifiers** raise the 10.6 MHz IF from the mixer to a level the detector can use. The first, V4, is shared with AM. Its grid circuit holds the secondaries of the first FM and the first AM IF transformers in series. Its plate circuit holds the primaries of the second FM and the second AM IF transformers in series. So the one tube amplifies both IFs. The second, V5, is the **FM limiter**: it clips the IF to a steady level, so that the detector answers to changes of frequency rather than of strength. V5 feeds the ratio-detector transformer.

13.4.1 How it is joined

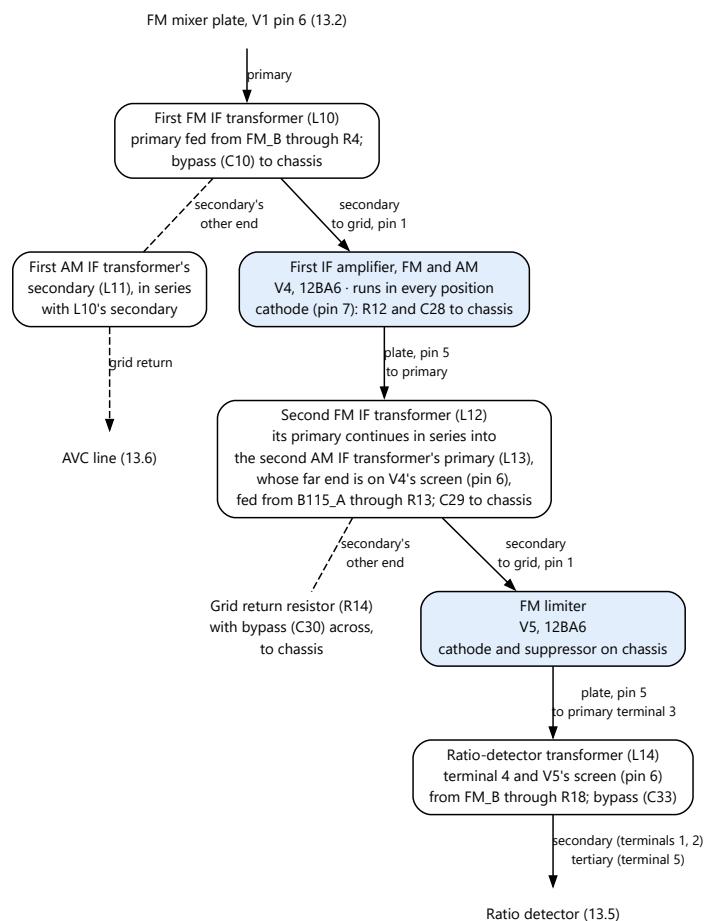
First FM IF transformer (L10). Its primary runs from terminal P, on the FM mixer plate (V1 pin 6), to terminal B, on a supply point. The first FM IF supply resistor (R4) feeds that point from FM_B, and the first FM IF supply bypass capacitor (C10) joins it to the chassis. Terminal G of L10's secondary drives V4's control grid (pin 1). Terminal E, the other end, is in series with the secondary of the first AM IF transformer (L11), joining L11's terminal G; L11's far end is on the AVC line. The letters are logical (section 13.1.3 (p. 94)). **V4's grid therefore returns to the AVC line through both secondaries.**

V4. The cathode (pin 7) goes to the chassis through the first IF cathode resistor (R12) and its bypass capacitor (C28), side by side. The suppressor grid (pin 2) is on the chassis. The plate (pin 5) feeds the primary of the second FM IF transformer (L12), and that primary continues in series into the primary of the second AM IF transformer (L13). The far end of that AM primary shares a point with V4's screen grid (pin 6). The AM IF screen supply resistor (R13, 1 k Ω) feeds that point from B115_A, and its bypass capacitor (C29, 0.01 μ F) joins it to the chassis (section 13.6 (p. 111)). **None of V4's supply comes from FM_B**, so V4 runs in every position.

V5. One end of the second FM IF transformer's secondary (L12) drives V5's control grid (pin 1). The other end goes to the chassis through the limiter grid return resistor (R14), with the limiter grid bypass capacitor (C30) across it. V5's cathode (pin 7) and suppressor grid (pin 2) are both straight on the chassis. The plate (pin 5) feeds terminal 3 of the ratio-detector transformer's primary (L14). The primary's other end, terminal 4, and V5's screen grid (pin 6) share a supply point. The ratio-detector supply resistor (R18) feeds it from FM_B, and the ratio-detector supply bypass capacitor (C33) joins it to the chassis.

Ratio-detector transformer (L14). Besides its primary it has a secondary (terminals 1 and 2) and a third winding, the **tertiary** (terminal 5). They feed the ratio detector (section 13.5 (p. 108)). Where the tertiary's other end goes is not recorded.

Cans and heaters. The cans of L10, L12 and L14 are on the chassis. V4's and V5's heaters are in the series string (section 13.9 (p. 121)).



How the stage is joined (simplified).

13.4.2 V4's and V5's pins

Both tubes are 12BA6 pentodes with the same basing. The table gives each pin for both tubes.

Pin	What it is	V4, first IF	V5, limiter
1	control grid	first FM IF transformer's secondary (L10), then first AM IF transformer's secondary (L11), to the AVC line	second FM IF transformer's secondary (L12), then the grid return resistor (R14) with its bypass (C30), to chassis
2	suppressor grid and internal shield	chassis	chassis
3, 4	heater	heater string	heater string
5	plate	second FM IF transformer's primary (L12), then second AM IF transformer's primary (L13)	ratio-detector transformer's primary (L14), terminal 3
6	screen grid	far end of the second AM IF transformer's primary (L13); screen supply resistor (R13) from B115_A; bypass (C29)	ratio-detector transformer's primary (L14), terminal 4; supply resistor (R18) from FM_B; bypass (C33)
7	cathode	cathode resistor (R12) with its bypass (C28), to chassis	chassis

13.4.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
First IF tube (V4)	12BA6	12BA6	
Limiter tube (V5)	12BA6	12BA6	
First FM IF transformer (L10)	IF can, tuned to 10.6 MHz	factory: FM IF (A); SAMS: 10.7 MHz	SAMS DCR: primary 0.8 Ω , secondary 0.8 Ω .
Second FM IF transformer (L12)	IF can, tuned to 10.6 MHz	factory: FM IF (B); SAMS: 10.7 MHz	SAMS DCR: primary 0.6 Ω , secondary 0.4 Ω .
Ratio-detector transformer (L14)	IF can with tertiary winding, tuned to 10.6 MHz	factory: ratio detector with tertiary winding; SAMS: 10.7 MHz	SAMS DCR: primary 1.2 Ω , secondary halves 0.4 Ω each, tertiary 0.5 Ω .
First FM IF supply resistor (R4)	1 k Ω	same	
First FM IF supply bypass capacitor (C10)	0.01 μ F	same	
First IF cathode resistor (R12)	70 Ω	factory 70 Ω ; SAMS 68 Ω , with 70 Ω as an alternate	
First IF cathode bypass capacitor (C28)	0.002 μ F	factory 0.002 μ F; SAMS 0.01 μ F	
Limiter grid return resistor (R14)	250 k Ω	same	
Limiter grid bypass capacitor (C30)	50 pF	same	SAMS prints Centralab DD-501 as the replacement here, but DD-500 for the other 50 pF part (C19), so the DD-501 number is not approved.
Ratio-detector supply resistor (R18)	1 k Ω	same	
Ratio-detector supply bypass capacitor (C33)	0.01 μ F	same	

For a replacement IF transformer, neither original gives what you need to choose one: the tuned frequency as fitted is 10.6 MHz, but the windings' coupling and impedance, the pinout, the internal capacitor values, the bandwidth and the mechanical fit are all unrecorded.

13.4.4 How to check it

- **Coil resistance.** The parts table gives the SAMS DCR figures, which describe the original parts. A DCR is a check for an open winding, not a measure of inductance or tuning.
- **Shared with AM.** V4's grid returns through the secondaries of both first IF transformers, FM and AM, and its plate is fed through the primaries of both second IF transformers. An open winding in any of these four stops V4 on both bands.

- **Tuning.** The FM IF transformers (L10, L12 and L14) were tuned to 10.6 MHz by a professional, Radio Hosvep on Park Ave. The AM IF transformers were not retuned.

13.5 The FM ratio detector and AFC control

The **ratio detector** turns the frequency swings of the FM IF into audio. It uses two diodes fed from the ratio-detector transformer (L14), a stabilising capacitor across their outputs, and a capacitor divider whose midpoint is the output. That output gives two things. One is the FM audio, through a **de-emphasis** filter that takes back the treble boost FM stations transmit. The other, through a 3 M Ω resistor, is the AFC control voltage for the FM oscillator (section 13.3 (p. 101)).

13.5.1 How it is joined

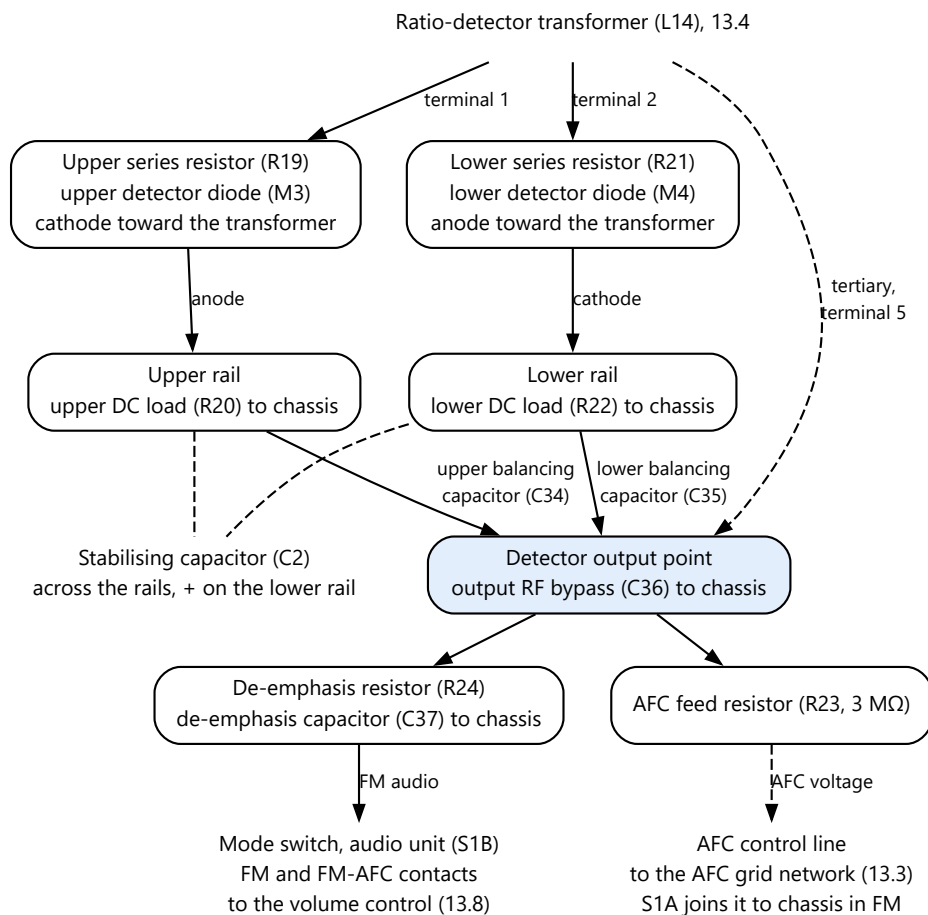
Diodes. One end of the ratio-detector transformer's secondary (L14, terminal 1) goes through the upper series resistor (R19) to the **cathode** of the upper detector diode (M3). M3's anode is on the **upper rail**. The other end (terminal 2) goes through the lower series resistor (R21) to the **anode** of the lower detector diode (M4). M4's cathode is on the **lower rail**.

Rails. The upper DC load resistor (R20) runs from the upper rail to the chassis, and the lower DC load resistor (R22) from the chassis to the lower rail, so the chassis is at their midpoint. The stabilising capacitor (C2), an electrolytic, sits across the two rails with its **+** on the **lower rail**.

Output. The upper balancing capacitor (C34) runs from the upper rail to the **detector output point**, and the lower balancing capacitor (C35) from that point to the lower rail. The transformer's tertiary winding (terminal 5) is on the same point. From it:

- the output RF bypass capacitor (C36) goes to the chassis;
- the de-emphasis resistor (R24) leads to the **FM audio**, with the de-emphasis capacitor (C37) from there to the chassis. The FM audio goes to the FM and FM-AFC contacts of the mode switch's audio unit (S1B), whose common feeds the top of the radio's volume control (section 13.8 (p. 117)). In SW and AM, that unit takes the AM audio instead;
- the AFC feed resistor (R23) leads to the **AFC control line**, which goes to the AFC triode's grid network (section 13.3 (p. 101)).

AFC control. The AFC-off unit of the mode switch (S1A) has its common on the chassis and its FM contact on the AFC control line. In **FM** it joins the line to the chassis, and AFC is off. In **FM-AFC**, and in SW and AM, the contact is open.



How the stage is joined (simplified).

13.5.2 Which way round the diodes and the stabilising capacitor go

The diodes are fitted in the **factory orientation, which is the opposite of the SAMS drawing's**. The table gives which way each one faces:

Part	Cathode	Anode
Upper detector diode (M3)	toward the transformer, through the upper series resistor (R19)	upper rail
Lower detector diode (M4)	lower rail	toward the transformer, through the lower series resistor (R21)

The stabilising capacitor (C2) has its **+ on the lower rail** and its – on the upper rail. SAMS shows its + on the upper rail; this radio does not follow SAMS here.

Caution — do not fit from SAMS. A detector diode or the stabilising capacitor fitted to the SAMS drawing is backwards in this radio. Note the orientation before you remove a part, and confirm it on the circuit before you fit the new one.

13.5.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
Stabilising capacitor (C2)	10 μ F electrolytic, + on the lower rail	10 μ F (factory and SAMS); factory polarity not resolved; SAMS: at least 50 V, + on the upper rail	No fitted voltage rating recorded.
Upper balancing capacitor (C34)	200 pF	same	Fitted dielectric not recorded.
Lower balancing capacitor (C35)	200 pF	same	Fitted dielectric not recorded.
Output RF bypass capacitor (C36)	200 pF	same	Fitted dielectric not recorded.
De-emphasis capacitor (C37)	0.001 μ F	same	1 nF, not 10 nF . Fitted dielectric not recorded.
Upper detector diode (M3)	a diode, type not recorded	a diode, type unmarked (factory); no part number (SAMS)	No evidence for any particular replacement type, such as a 1N34 or 1N60.
Lower detector diode (M4)	a diode, type not recorded	a diode, type unmarked (factory); no part number (SAMS)	As the upper diode.
Upper series resistor (R19)	1 k Ω	same	
Upper DC load resistor (R20)	10 k Ω	same	
Lower series resistor (R21)	1 k Ω	same	
Lower DC load resistor (R22)	10 k Ω	same	
AFC feed resistor (R23)	3 M Ω	same	
De-emphasis resistor (R24)	50 k Ω	same	
Mode switch, AFC-off unit (S1A)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
Mode switch, audio unit (S1B)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

13.5.4 How to check it

- **Orientation.** Check both detector diodes and the stabilising capacitor against section 13.5.2 (p. 109).
- **The rails.** Each DC load resistor (R20, R22) runs from one rail to the chassis.
- **AFC off in FM.** See section 13.3.4 (p. 104).
- **Transformer.** The SAMS DCR figures for the ratio-detector transformer (L14) are in section 13.4.3 (p. 107).

13.6 The 455 kHz AM IF and AM detector

In SW and AM, the converter (V3, section 13.7 (p. 113)) sends its 455 kHz IF through the **first AM IF transformer** (L11) to the first IF amplifier (V4), the tube the FM path also uses (section 13.4 (p. 104)). From V4 the IF goes through the **second AM IF transformer** (L13) to the **AM detector**: the two diodes inside V6, a 12AV6. The detector recovers the audio from the IF and also makes the **AVC** voltage, which sets the gain of V4 and, in AM, of the converter (section 13.1.5 (p. 96)).

13.6.1 How it is joined

First AM IF transformer (L11). One end of its primary (terminal B) is on the converter's plate (V3 pin 5). The other end (terminal P) is on the AM supply, **AM_B**, together with the converter's screen grids (V3 pin 6). The mode switch's B+ unit (S1D) puts B+ on **AM_B** in SW and AM only (section 13.1.6 (p. 96)). **No capacitor is on AM_B**. SAMS shows a 0.002 μF bypass there; this radio has none.

Into V4. The secondary of L11 is in series with the secondary of the first FM IF transformer (L10): one end (terminal G) joins L10's secondary, and the other end (terminal E) is on the **AVC line**. V4's control grid (pin 1) is at the far end of L10's secondary, so it returns to the AVC line through both secondaries (section 13.4.1 (p. 105)).

Out of V4. V4's plate feeds the primary of the second FM IF transformer (L12), then, in series, the primary of the second AM IF transformer (L13). The far end of L13's primary shares a point with V4's screen grid (pin 6). The **AM IF screen supply resistor** (R13, 1 k Ω) feeds that point from the B+ filter (**B115_A**, section 13.10 (p. 125)), and its bypass capacitor (C29, 0.01 μF) joins it to the chassis. So L13 and V4 have B+ in every position of the mode switch.

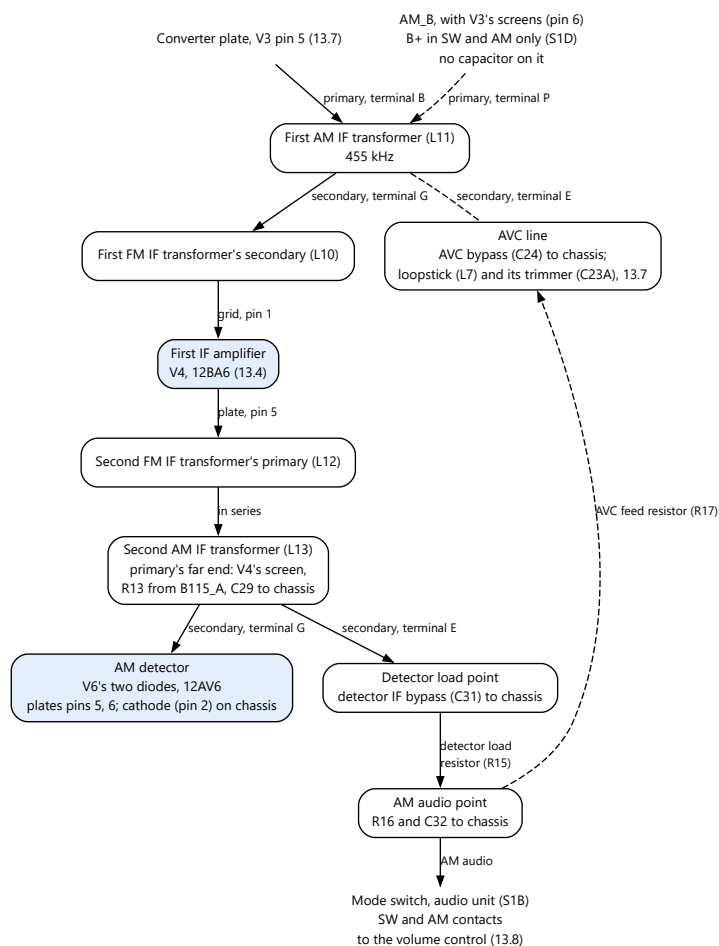
Detector. One end of L13's secondary (terminal G) goes to **both diode plates** of V6, pins 5 and 6. The diodes' cathode is pin 2, which V6's triode shares, and it is on the chassis. The other end of the secondary (terminal E) is the **detector load point**. From there:

- the detector IF bypass capacitor (C31, 100 pF) goes to the chassis;
- the detector load resistor (R15, 50 k Ω) leads to the **AM audio point**.

AM audio point. Three parts leave it: the detector DC return resistor (R16, 1 M Ω) and the audio IF bypass capacitor (C32, 100 pF), both to the chassis, and the **AVC feed resistor** (R17, 2 M Ω) to the AVC line. The AM audio goes to the SW and AM contacts of the mode switch's audio unit (S1B), whose common feeds the top of the radio's volume control (section 13.8 (p. 117)).

AVC line. It joins five things: the far end of the AVC feed resistor (R17), terminal E of L11's secondary, one end of the AM loopstick (L7), one side of the AM antenna trimmer (C23A), and the **AVC bypass capacitor** (C24, 0.05 μF), whose other side is on the chassis. In AM the converter's signal grid returns to this line through the loopstick (section 13.7 (p. 113)).

Cans. The cans of L11 and L13 are on the chassis.



How the stage is joined (simplified).

13.6.2 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
First AM IF transformer (L11)	IF can, 455 kHz	factory: AM IF (A); SAMS: 455 kHz	SAMS DCR: primary 22 Ω , secondary 22 Ω . Not returned.
Second AM IF transformer (L13)	IF can, 455 kHz	factory: AM IF (B); SAMS: 455 kHz	SAMS DCR: primary 22 Ω , secondary 20 Ω . Not returned. Its secondary feeds V6's diodes.
AM IF screen supply resistor (R13)	1 k Ω	same	Feeds the far end of L13's primary and V4's screen.
AM IF screen supply bypass capacitor (C29)	0.01 μ F	same	
Detector IF bypass capacitor (C31)	100 pF	same	
Detector load resistor (R15)	50 k Ω	same	
Detector DC return resistor (R16)	1 M Ω	same	

Part	Fitted	Original	Notes
Audio IF bypass capacitor (C32)	100 pF	same	
AVC feed resistor (R17)	2 MΩ	same	
AVC bypass capacitor (C24)	0.05 μF	0.05 μF (factory and SAMS); SAMS: 50 V	No fitted voltage rating recorded.
AM detector (V6's two diodes)	12AV6, new old stock	12AV6	The whole tube is in section 13.8 (p. 117).
Mode switch, audio unit (S1B)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

For a replacement AM IF transformer, neither original gives what you need: the windings' coupling and impedance, the pinout, the internal capacitor values, the bandwidth and the mechanical fit are all unrecorded. The tuned frequency is 455 kHz.

13.6.3 How to check it

- **Coil resistance.** The parts table gives the SAMS DCR figures for L11 and L13, which describe the original parts. A DCR is a check for an open winding, not a measure of inductance or tuning.
- **Shared with FM.** An open secondary in L11, or an open primary in L13, also stops V4 on FM (section 13.4.4 (p. 107)).
- **Tuning.** The AM IF transformers were not retuned when the FM path was tuned (section 13.1.1 (p. 93)).
- **Detector.** V6's diode plates are pins 5 and 6, both on the same end of L13's secondary; the cathode, pin 2, is on the chassis.

13.7 AM and SW input, converter (V3, 12BE6) and the SW and AM local oscillator

V3 is a **pentagrid converter**: one tube with five grids that is both the mixer and the local oscillator for AM and SW. Its oscillator section runs on its first grid and cathode. The station enters on its third grid, the **signal grid**. The IF, 455 kHz, leaves from its plate into the first AM IF transformer (L11). The mode switch picks the input coil and the oscillator coil for each band. The input coil is the ferrite **loopstick** (a ferrite rod carrying an antenna winding) in AM, and the SW antenna coil in SW. V3 takes its B+ from **AM_B**, so it works only in SW and AM.

The external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is. Every other input part is fitted too: the SW coupling capacitor (C21), the SW antenna coil (L6) and its trimmer (C22A), and the loopstick's trimmer (C23A).

SAMS gives the bands as 540 to 1620 kHz for AM (the loopstick) and 3.5 to 9 MHz for SW (the SW antenna coil).

The stage has its own **trimmers** and **padders**, the small capacitors that set its alignment and tracking (section 13.1.3 (p. 94)).

13.7.1 How it is joined

SW input. The SW antenna coupling capacitor (C21) has one lead where the SW antenna terminal (J2) would be; nothing else is on that lead. Its other lead goes to one end of the primary of the SW antenna coil (L6); the other end of that primary is on the chassis. One end of L6's secondary is the **SW RF point**, which the SW antenna trimmer (C22A) joins to the chassis, and which goes to the SW contact of the mode switch's RF unit (S1C). The other end of the secondary is on the chassis.

AM input. One end of the loopstick (L7) is the **AM RF point**, which goes to the AM contact of S1C. The other end is on the AVC line (section 13.6 (p. 111)). The AM antenna trimmer (C23A) sits across the loopstick, from the AVC line to the AM RF point.

Signal grid. The common of S1C, the AM/SW RF tuning section (VC3, whose other side is on the chassis) and V3's signal grid (pin 7) are one point. In SW, S1C puts the SW RF point on it; in AM, the AM RF point; in FM and FM-AFC, nothing. So in AM the grid returns to the AVC line through the loopstick, and in SW it returns to the chassis through the SW coil's secondary.

Plate and screens. The plate (pin 5) goes to the primary of the first AM IF transformer (L11). The screen grids (pin 6) are on **AM_B** with the other end of that primary. No capacitor is on **AM_B** (section 13.6.1 (p. 111)).

Oscillator grid. The oscillator grid (pin 1) goes to the chassis through the oscillator grid leak resistor (R11, 20 k Ω ; grid leak: section 13.3.1 (p. 101)). The grid also goes through the oscillator feedback capacitor (C25, 100 pF) to the **oscillator tuning point**.

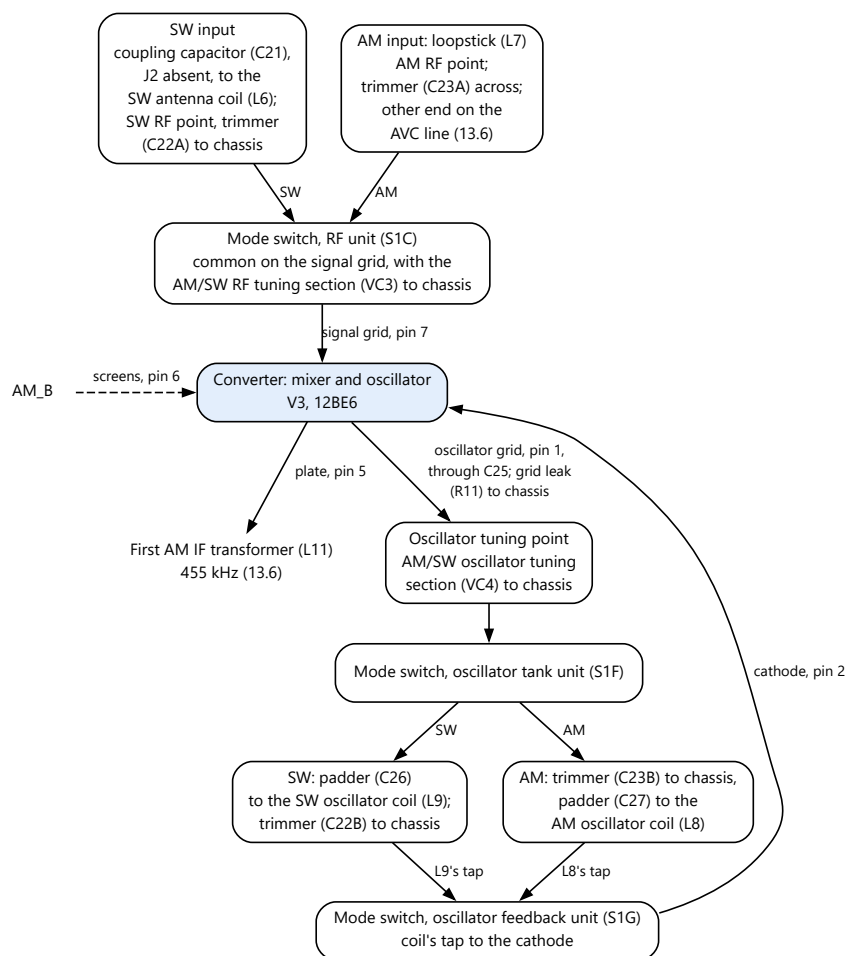
Oscillator tuning point. It joins C25, the AM/SW oscillator tuning section (VC4, whose other side is on the chassis) and the common of the mode switch's oscillator tank unit (S1F).

- **In SW**, S1F goes through the SW oscillator padder (C26, 0.003 μ F) to the top of the SW oscillator coil (L9). The SW oscillator trimmer (C22B) joins that top to the chassis.
- **In AM**, S1F goes to a point that the AM oscillator trimmer (C23B) joins to the chassis. From there the AM oscillator padder (C27, 350 pF) leads to the top of the AM oscillator coil (L8).

The bottoms of both oscillator coils are on the chassis.

Cathode. The cathode (pin 2) goes only to the common of the mode switch's oscillator feedback unit (S1G). In SW, S1G puts it on the tap of the SW oscillator coil (L9); in AM, on the tap of the AM oscillator coil (L8). In FM and FM-AFC the cathode is connected to nothing, and the tube has no B+.

Heater. The heater is on pins 3 and 4, in the series string (section 13.9 (p. 121)).



How the stage is joined (simplified).

13.7.2 V3's pins

The table gives each of V3's pins and what it is joined to.

Pin	What it is	Joined to
1	oscillator grid (grid 1)	grid leak (R11) to chassis; feedback capacitor (C25) to the oscillator tuning point
2	cathode	oscillator feedback unit (S1G) only: SW oscillator coil's tap in SW, AM oscillator coil's tap in AM
3	heater	heater string, toward the FM RF tube's heater through a heater choke (L16)
4	heater	heater string, from the first IF tube's heater; bypass capacitor (C47) to chassis
5	plate	primary of the first AM IF transformer (L11)
6	screen grids (grids 2 and 4)	AM_B, with the other end of L11's primary
7	signal grid (grid 3)	RF unit's common (S1C) and the AM/SW RF tuning section (VC3)

13.7.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
Converter tube (V3)	12BE6	12BE6	
SW antenna terminal (J2)	absent	factory: antenna terminal with a series coupling capacitor; SAMS: a terminal, no rating	The external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is.
SW antenna coupling capacitor (C21)	450 pF	same	Several listed substitutes, Sprague among them, are 470 pF. The listed substitutes are DC-rated and give no safety class for an antenna input on a mains-connected chassis.
SW antenna coil (L6)	two-winding coil	factory: two-winding SW matching coil; SAMS: DCR 0.3 Ω each winding, band 3.5–9 MHz	Inductance not given.
SW antenna trimmer (C22A)	trimmer	no value in either original	Range and dielectric not specified. Part of the band alignment: do not choose one by appearance.
AM loopstick (L7)	loopstick	factory: ferrite bar with antenna winding; SAMS: DCR 0.6 Ω , band 540–1620 kHz	Inductance not given.
AM antenna trimmer (C23A)	trimmer	no value in either original	As the SW antenna trimmer.
AM/SW RF tuning section (VC3)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Serves both SW and AM. Turns with the other three sections of the tuning capacitor.
Mode switch, RF unit (S1C)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
Oscillator grid leak resistor (R11)	20 k Ω	same	
Oscillator feedback capacitor (C25)	100 pF	same	
AM/SW oscillator tuning section (VC4)	section of the four-section tuning capacitor	part of SAMS's tuning assembly M2; no range given	Serves both SW and AM.
Mode switch, oscillator tank unit (S1F)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
SW oscillator padder (C26)	0.003 μ F	0.003 μ F (3000 pF), SAMS: ± 5 %	A stable, low-loss RF capacitor. The listed substitute is silvered mica, 500 V. The fitted part's type and rating are not recorded.
SW oscillator trimmer (C22B)	trimmer	no value in either original	As the SW antenna trimmer.
SW oscillator coil (L9)	tapped coil	factory: tapped oscillator coil; SAMS: DCR 0.3 Ω and 0.1 Ω sections	

Part	Fitted	Original	Notes
AM oscillator padder (C27)	350 pF	same	Do not round it to 330 pF: the padder sets the tracking. Sprague 5GA-T35 is 350 pF; Aerovox DI-330 and other listed substitutes are not.
AM oscillator trimmer (C23B)	trimmer	no value in either original	As the SW antenna trimmer.
AM oscillator coil (L8)	tapped coil	factory: tapped oscillator coil; SAMS: DCR 4.6 Ω and 0.9 Ω sections	SAMS: on the listed Merit, Miller and Stancor substitutes, ignore the extra tap.
Mode switch, oscillator feedback unit (S1G)	one unit of the mode switch	SAMS part M5	Section 13.1.6.

The AVC bypass capacitor (C24) is in section 13.6 (p. 111).

13.7.4 How to check it

- **Coil resistance.** SAMS gives these DCRs for the original coils: the SW antenna coil (L6) 0.3 Ω each winding; the loopstick (L7) 0.6 Ω ; the AM oscillator coil (L8) 4.6 Ω and 0.9 Ω for its two sections; the SW oscillator coil (L9) 0.3 Ω and 0.1 Ω . A DCR finds an open winding; it does not measure inductance.
- **The mode switch.** With the cord out, the mode switch's RF, oscillator tank and oscillator feedback units (S1C, S1F, S1G) each join their common to their SW contact in SW and their AM contact in AM. In FM and FM-AFC they join it to nothing (section 13.1.6 (p. 96)).
- **Alignment.** The AM and SW path was not retuned when the FM path was (section 13.1.1 (p. 93)). No alignment figures for the trimmers are recorded.

13.8 The audio amplifier (V6, 12AV6, and V7, 50C5), tone switch and output transformer

The audio picked by the mode switch's audio unit (S1B) reaches the top of the radio's **volume control** (R1). Its wiper drives the triode inside V6, the **first audio amplifier**, which raises the audio's voltage. V7, a 50C5 **beam power tube** (a tube built to deliver power to a load), is the **output stage**: it drives the primary of the **output transformer** (T1). A **tone switch** can join one or two small capacitors from V6's plate to the chassis, which takes some of the treble away. V6 and V7 have B+ in every position of the mode switch.

13.8.1 How it is joined

Volume control (R1). Its top (pin 1) comes from the common of the audio unit (S1B). Its bottom (pin 3) is on the chassis. Its wiper (pin 2) goes through the grid coupling capacitor (C38, 0.01 μ F) to V6's triode grid (pin 1). The audio grid leak resistor (R25, 2 M Ω) joins that grid to the chassis.

V6's triode. The cathode (pin 2), shared with the AM detector diodes, is on the chassis. The plate (pin 7) is fed from the B+ filter (B115_A) through the audio plate load resistor (R26, 250 k Ω). Three more parts leave the plate: the audio plate shunt capacitor (C39, 100 pF) to the chassis, the output coupling capacitor (C42, 0.01 μ F) to V7's grid, and the first tone capacitor (C40).

Tone switch. The first tone capacitor (C40, 0.002 μF) runs from V6's plate to a point the tone switch calls LOW. The second tone capacitor (C41, 0.002 μF) runs from LOW to a point called MID. The tone unit's common is on the chassis. The table shows what each tone position switches in:

Tone position	What it switches in, from V6's plate to the chassis
H (high)	nothing: the H contact is connected to nothing
M (medium)	C40 and C41 in series: 0.001 μF
L (low)	C40 alone: 0.002 μF

The factory drawing and SAMS agree on this. SAMS numbers the switch's lugs 13 for the common, 2 for OFF, 3 for H, 4 for M and 5 for L. The letters H, M and L above are the drawing's; no record maps them to the physical lugs.

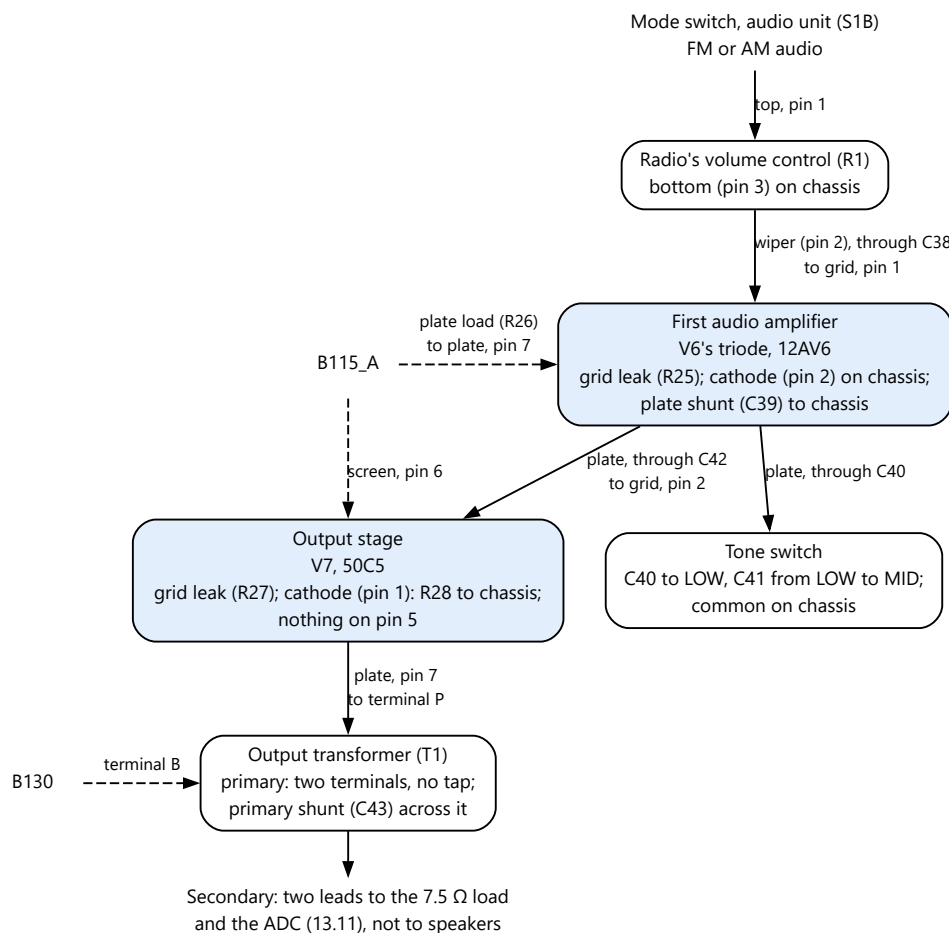
V7. The control grid (pin 2) takes the audio from C42 and returns to the chassis through the output grid leak resistor (R27, 500 k Ω). The cathode (pin 1) goes to the chassis through the output cathode resistor (R28, 150 Ω 1 W); nothing else is on the cathode. The screen grid (pin 6) is on the B+ filter (B115_A). The plate (pin 7) goes to one end of the output transformer's primary (terminal P) and to the primary shunt capacitor (C43, 0.0047 μF).

Nothing is wired to pin 5.

Output transformer (T1). Its primary has **two terminals and no tap**. Terminal P is on V7's plate. Terminal B is on B130, the first filtered point of the B+ supply (section 13.10 (p. 125)), and C43 sits across the two. The secondary's two leads leave the radio for the ADC (section 13.11 (p. 128)).

SP1 and SP2 are the radio's original speakers as drawn by the factory, in series across the secondary. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.

Heaters. V6's heater is on pins 3 and 4, V7's on pins 3 and 4, both in the series string (section 13.9 (p. 121)).



How the stage is joined (simplified).

13.8.2 V6's and V7's pins

The table gives each pin of both tubes and what it is joined to.

Pin	V6, 12AV6	Joined to	V7, 50C5	Joined to
1	triode grid	grid coupling capacitor (C38) from the volume wiper; grid leak (R25) to chassis	cathode	output cathode resistor (R28) to chassis
2	cathode, shared by the triode and both diodes	chassis	control grid	output coupling capacitor (C42); grid leak (R27) to chassis
3	heater	heater string, toward the first IF tube's heater	heater	heater string, toward the limiter's heater
4	heater	heater string, from the limiter's heater	heater	the fused live: the start of the heater string
5	diode plate	second AM IF transformer's secondary (L13), terminal G	second control-grid pin	nothing
6	diode plate	the same point as pin 5	screen grid	B+ filter (B115_A)
7	triode plate	plate load (R26) from B115_A; C39, C42, C40	plate	output transformer primary, terminal P; C43

The 50C5's published ratings: plate at most 150 V, screen at most 130 V, plate dissipation 7 W, screen dissipation 1.4 W.

13.8.3 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
AM detector and first audio tube (V6)	12AV6, new old stock	12AV6	
Output tube (V7)	50C5	50C5	
Radio's volume control (R1)	500 k Ω potentiometer	500 k Ω (factory and SAMS); SAMS: ½ W or less	Audio taper, as the original specification infers from its listed substitutes. No power switch is drawn on it. Sits at about one eighth of its travel in normal use (section 13.1.2 (p. 94)). For a replacement, measure the shaft, flat, bushing and rotation.
Grid coupling capacitor (C38)	0.01 μ F	0.01 μ F; SAMS: at least 450 V	A low-leakage coupling capacitor. The listed film substitute is 600 V, $\pm$ 10 %.
Audio grid leak resistor (R25)	2 M Ω	same	
Audio plate load resistor (R26)	250 k Ω	same	
Audio plate shunt capacitor (C39)	100 pF	same	
Output coupling capacitor (C42)	0.01 μ F	0.01 μ F; SAMS: at least 450 V	As the grid coupling capacitor.
Output grid leak resistor (R27)	500 k Ω	same	
Output cathode resistor (R28)	150 Ω , 1 W	same; SAMS: 1 W	Use at least 1 W, with room around it for the heat.
First tone capacitor (C40)	0.002 μ F	same	
Second tone capacitor (C41)	0.002 μ F	same	
Power and tone switch (S2)	one switch: a tone unit and a power unit	SAMS part M6; factory: OFF, H, M, L, with a linked power contact	No part number or contact rating is listed. A replacement needs mains-rated contacts and insulation, the same contact sequence, shaft and mounting. The power unit is in section 13.10 (p. 125).

Part	Fitted	Original	Notes
Output transformer (T1)	output transformer, two-terminal primary, no tap	factory: single-ended, two-terminal primary, two speakers in series; SAMS: part 38.11.1, 7000 Ω primary tapped at 2500 Ω , 6–8 Ω secondary	The fitted transformer has the factory form. Its impedances and DCRs are not recorded. SAMS's DCRs (active primary 176 Ω , unused section 146 Ω , secondary 0.9 Ω) describe SAMS's tapped version, not the fitted part. Do not apply SAMS's full-primary 7000 Ω to its active tap.
Primary shunt capacitor (C43)	0.0047 μ F	0.005 μ F (factory and SAMS); SAMS: at least 450 V	No fitted voltage rating recorded. For a replacement, also consider its pulse and transient rating.
Speakers (SP1, SP2)	not on the output: it feeds the 7.5 Ω load and the ADC	factory: two speakers in series; SAMS: 4 $\times$ 6 inch, permanent magnet, 3–4 Ω each, 6–8 Ω the pair	The radio's original speakers as drawn by the factory.

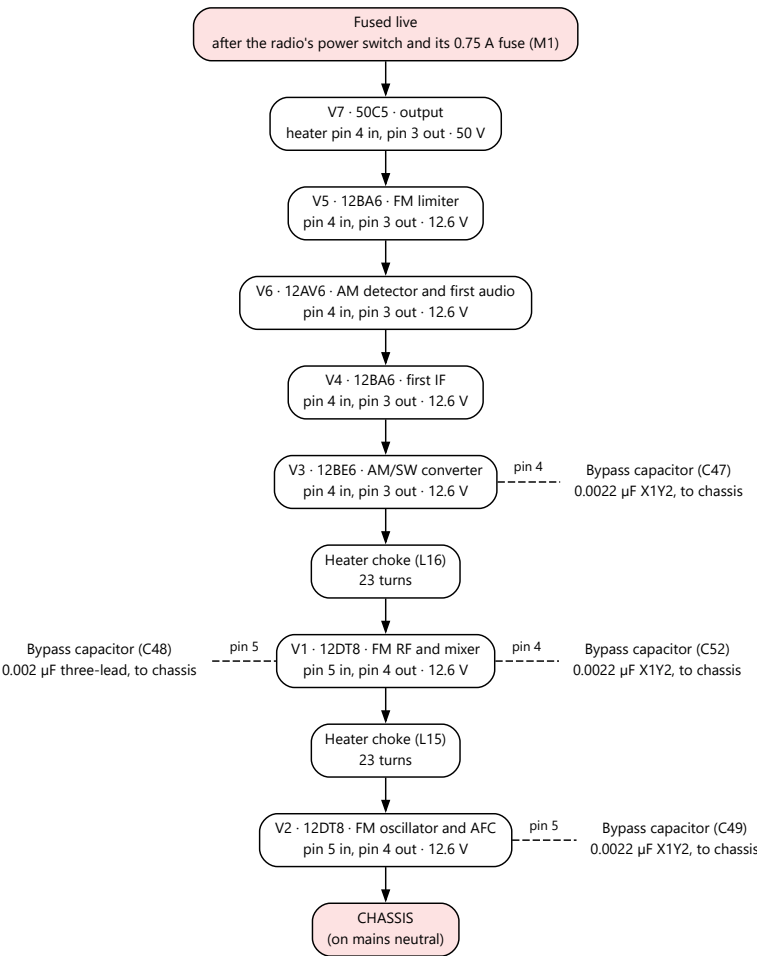
13.8.4 How to check it

- **Tone switch.** With the cord out: in L, 0.002 μ F (C40) is between V6's plate and the chassis; in M, 0.001 μ F (C40 and C41 in series); in H, neither.
- **The output transformer.** Its primary has two terminals only. SAMS's DCR figures do not describe it.
- **V7, pin 5.** Nothing is wired to it.
- **Volume.** The radio's own volume control belongs at about one eighth of its travel. Higher settings make the sound too loud further down the chain (chapter 6 (p. 33), section 6.8 (p. 38)).

13.9 The series heater string, the mode supply and the indicator lamps

The seven tube heaters are wired **in series**, one after the other, from the fused live to the chassis, so one current flows through all of them. Every tube here has a 0.15 A heater, the usual heater current of AC/DC sets. The 50C5 has a 50 V heater, the six others 12.6 V each (section 13.1.4 (p. 95)). Because the string is one chain, **one open heater or choke stops all seven heaters**. The same switch unit that sends B+ to the AM or the FM stages (S1D) and the unit that would light a mode lamp (S1E) are also described here.

13.9.1 The heater string



The series heater string, from the fused live to the chassis.

The order was checked on the chassis:

Order	Tube or part	In	Out	At the junction after it
1	output tube V7, 50C5	pin 4, on the fused live	pin 3	
2	limiter V5, 12BA6	pin 4	pin 3	
3	detector and first audio tube V6, 12AV6	pin 4	pin 3	
4	first IF tube V4, 12BA6	pin 4	pin 3	the heater-string bypass capacitor (C47) to chassis
5	converter V3, 12BE6	pin 4	pin 3	
6	heater choke (L16)			the three-lead heater bypass capacitor (C48) to chassis
7	FM RF and mixer tube V1, 12DT8	pin 5	pin 4	a heater bypass capacitor (C52) to chassis
8	heater choke (L15)			a heater bypass capacitor (C49) to chassis

Order	Tube or part	In	Out	At the junction after it
9	FM oscillator and AFC tube V2, 12DT8	pin 5	pin 4, on the chassis	

The two **heater chokes** (L15, L16) are RF chokes: coils that pass the heater current but block radio frequencies. The choke between V1 and V2 (L15) has a bypass capacitor to the chassis on each side (C52 and C49). The choke between V3 and V1 (L16) has one, on V1's side (C48); its V3 side has none. With the heater-string bypass capacitor at V3's pin 4 (C47), the string has four bypass capacitors. V1 sits between the two chokes, and V2 is the last tube: its pin 4 is the chassis end of the whole string. The internal shields of both 12DT8s (pin 9) are on the chassis.

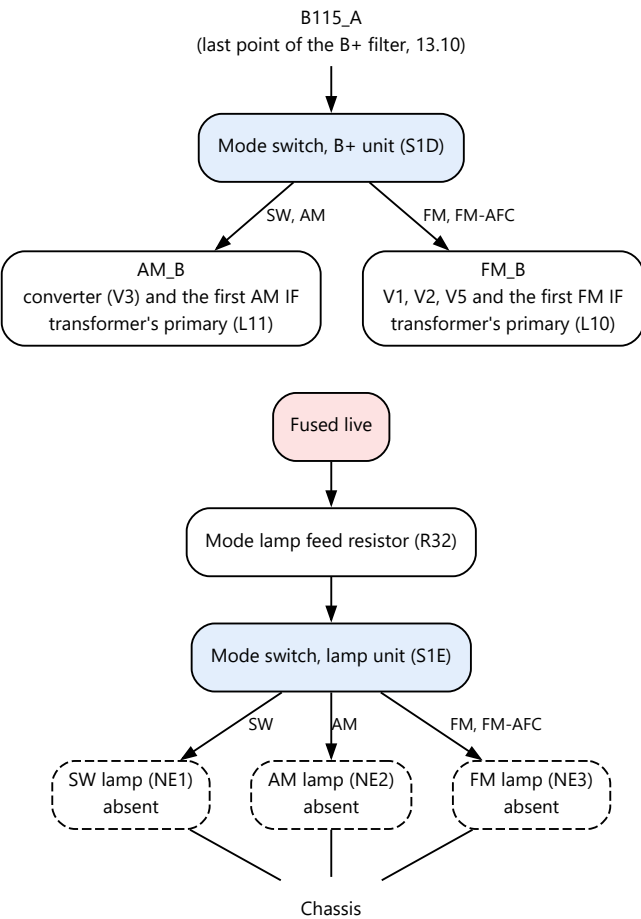
13.9.2 Mode supply

The mode switch's B+ unit (S1D) has its common on the B+ filter (B115_A). It puts B+ on **AM_B** in SW and AM, and on **FM_B** in FM and FM-AFC. Section 13.1.6 lists what each supply feeds.

13.9.3 Indicator lamps

The mode lamp feed resistor (R32, 50 kΩ) runs from the fused live to the common of the mode switch's lamp unit (S1E). S1E sends that feed to the SW lamp (NE1) in SW, the AM lamp (NE2) in AM, and the FM lamp (NE3) in FM and FM-AFC. Each lamp's other side goes to the chassis.

The neon bulbs NE1, NE2 and NE3 are absent from this radio. R32 and the S1E contacts are fitted.



How the stage is joined (simplified).

13.9.4 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
Heater choke between V3 and V1 (L16)	RF heater choke	factory: series heater RF choke; SAMS: 23 turns	23 turns alone does not define a replacement: the winding's diameter, length, wire gauge, core, current and insulation rating are not recorded.
Heater choke between V1 and V2 (L15)	RF heater choke	factory: series heater RF choke; SAMS: 23 turns	As the choke between V3 and V1.
Heater-string bypass capacitor (C47), at V3's pin 4	0.0022 μF, safety class X1Y2	0.002 μF (factory and SAMS)	Replaced the original. A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)); no voltage rating recorded.
Heater bypass capacitor (C48), at V1's pin 5	0.002 μF, three-lead ceramic, the original part	0.002 μF (factory and SAMS)	Middle lead to chassis (section 13.1.3 (p. 94)).
Heater bypass capacitor (C52), at V1's pin 4	0.0022 μF, safety class X1Y2	not recorded	Replaced the original part, whose value is not recorded. A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)); no voltage rating recorded.

Part	Fitted	Original	Notes
Heater bypass capacitor (C49), at V2's pin 5	0.0022 μ F, safety class X1Y2	0.002 μ F (factory and SAMS)	Replaced the original. A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)); no voltage rating recorded.
Mode switch, B+ unit (S1D)	one unit of the mode switch	SAMS part M5	Section 13.1.6.
Mode switch, lamp unit (S1E)	one unit of the mode switch, contacts fitted	SAMS part M5	Section 13.1.6.
Mode lamp feed resistor (R32)	50 k Ω	same	Original wattage not printed. Work out the lamp's voltage and current, and the resistor's dissipation, before you buy one.
SW lamp (NE1)	absent	a neon lamp, sharing the 50 k Ω feed resistor	Neither original gives a lamp model, striking or running voltage, current, size or brightness.
AM lamp (NE2)	absent	as the SW lamp	
FM lamp (NE3)	absent	as the SW lamp	Lit in FM and FM-AFC.

A safety marking alone does not make a part equivalent at RF: match its construction and its place on the chassis.

13.9.5 How to check it

- **All heaters cold.** With the cord out, check the string end to end, in the order of the table in section 13.9.1 (p. 122): each heater, then each choke. One open part breaks the whole chain.
- **The string's ends.** V7's pin 4 is on the fused live; V2's pin 4 is on the chassis.
- **The two 12DT8s.** V1 is the tube between the two chokes; V2 is the tube whose heater returns to the chassis.
- **The three-lead capacitor (C48).** Its two outer leads read 0 Ω to each other, and the middle lead goes to the chassis.

13.10 The original transformerless AC supply and B+

"Original" here means the radio's own supply, not part of Ambersong; this section describes it as fitted now. The set has no mains transformer. Its live terminal goes through the radio's power switch and fuse to the **fused live**, which feeds the heater string (section 13.9 (p. 121)) and the B+ supply. The neutral terminal **is** the chassis.

The **B+** is made by one silicon diode used as a **half-wave rectifier**: it passes only one half of each mains cycle. Three electrolytic capacitors then smooth it, with two **dropping resistors** between them (series resistors that lower and filter the supply).

Everything in this section is connected to the mains: read chapter 2 (p. 13), section 2.1 (p. 13), first.

13.10.1 How it is joined

Mains in. The radio's live terminal (J3) goes to the input of the power unit of the power and tone switch (S2). That unit's output goes straight into the mains fuse (M1, 0.75 A), with no wire between them. The fuse's output is the **fused live**. The radio's neutral terminal (J4) is the chassis itself. Which lug of S2 carries the power contact is not recorded.

From the fused live, four things leave:

- the heater string, at V7's pin 4 (section 13.9 (p. 121));
- the mode lamp feed resistor (R32, section 13.9 (p. 121));
- the **mains bypass capacitor** (C44, 0.01 μ F, safety class X1Y2), to the chassis;
- the **surge resistor** (R29, 28 Ω 3 W), to the rectifier.

Rectifier. R29 goes to the anode of the silicon rectifier (X1). Its cathode is the raw B+ point, B_RAW. The rectifier shunt capacitor (C46, 0.0047 μ F) sits across the rectifier.

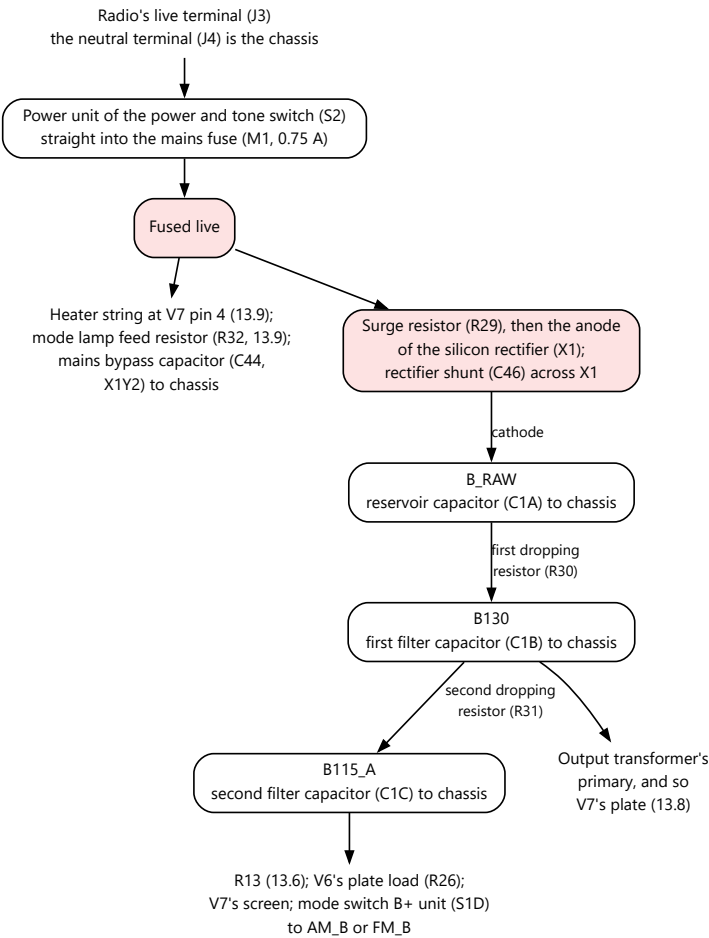
Filter. Each B+ point has its own 47 μ F capacitor, + to the point and – to the chassis:

Point	Its capacitor	Reached through	Feeds
B_RAW	reservoir capacitor (C1A)	the rectifier (X1)	the first dropping resistor only
B130	first filter capacitor (C1B)	first dropping resistor (R30, 150 Ω 5 W)	the output transformer's primary, and so V7's plate (section 13.8 (p. 117)); the second dropping resistor
B115_A	second filter capacitor (C1C)	second dropping resistor (R31, 300 Ω 1 W)	the AM IF screen supply resistor (R13, section 13.6 (p. 111)); V6's plate load (R26); V7's screen grid; the mode switch's B+ unit (S1D), which feeds AM_B or FM_B (section 13.1.6 (p. 96))

The **reservoir capacitor** is the first one after the rectifier: it holds the B+ up between the rectifier's pulses.

The names are not voltages. The numbers in B130 and B115_A are the SAMS data's nominal names. No B+ voltage of this radio is recorded.

Two coils are gone. Two of the radio's original coils, L17 and L18, were removed by accident. The external mains line filter took their place (chapter 5 (p. 27), section 5.2 (p. 28)). What they were is not recorded.



How the stage is joined (simplified).

13.10.2 Parts

The stage's parts, with their fitted and original values (section 13.1.3 (p. 94)):

Part	Fitted	Original	Notes
Radio's live terminal (J3)	fitted	no original specification recorded	Takes the switched live through the line filter (section 13.11 (p. 128)).
Radio's neutral terminal (J4)	fitted	no original specification recorded	It is the chassis. Takes neutral through the line filter.
Power and tone switch, power unit (S2)	one switch with the tone unit (section 13.8 (p. 117))	SAMS part M6: a separate AC power pole and tone contacts	Which lug carries the power contact is not recorded.
Mains fuse (M1)	0.75 A; type and voltage not recorded	factory: no fuse drawn; SAMS: 0.75 A, type GJV, in a different position, omitted in some versions	Here it sits between the power switch and the fused live. The original specification's selection note: use a matching speed, voltage, breaking capacity and package.
Mains bypass capacitor (C44)	0.01 μ F, safety class X1Y2	factory: 0.01 μ F, no voltage, tolerance or safety class given	SAMS numbers its version of this bypass C45; SAMS's 400 V for it does not set this part's rating. This radio has no C45. A safety-rated part (chapter 2 (p. 13), section 2.4 (p. 14)).

Part	Fitted	Original	Notes
Surge resistor (R29)	28 Ω , 3 W	same; SAMS: 3 W	Use at least 3 W, with room for the heat. SAMS's listed substitute, IRC MR 2, is a configurable 10 W wirewound assembly, not a fixed 28 Ω part.
Rectifier (X1)	silicon rectifier; part number and rating not recorded	factory: silicon rectifier; SAMS: rating not printed, output current measured at 0.090 A	Choose at least the 400 V, 0.5 A class of the period, with suitable surge, heat and leakage ratings. A higher rating alone does not give the same B+. One listed substitute, the RCA 1N2861, carries a conflicting 105 V AC capacitor-input rating.
Rectifier shunt capacitor (C46)	0.0047 μ F	0.005 μ F (factory and SAMS); SAMS: at least 450 V	No fitted voltage rating recorded. Consider its pulse and transient rating.
Reservoir capacitor (C1A)	47 μ F, Rubycon, 315 V, a separate capacitor	40 μ F, 150 V, one section of a common-negative can (factory and SAMS)	Match polarity, ripple current, temperature and start-up voltage.
First dropping resistor (R30)	150 Ω , 5 W	150 Ω ; SAMS: 2 W	
First filter capacitor (C1B)	47 μ F, Rubycon, 315 V, a separate capacitor	40 μ F, 150 V, one section of the same can	As the reservoir capacitor.
Second dropping resistor (R31)	300 Ω , 1 W	same; SAMS: 1 W	
Second filter capacitor (C1C)	47 μ F, Rubycon, 315 V, a separate capacitor	40 μ F, 150 V, one section of the same can	As the reservoir capacitor.

13.10.3 How to check it

All with the cord out.

- **Fuse.** 0.75 A, between the power switch and the fused live.
- **Rectifier direction.** Anode toward the surge resistor (R29), cathode toward the reservoir capacitor (C1A).
- **Capacitor polarity.** The reservoir and the two filter capacitors (C1A, C1B, C1C) each have + on their B+ point and – on the chassis.
- **Safety parts.** The mains bypass capacitor (C44) is X1Y2 rated (chapter 2 (p. 13), section 2.4 (p. 14)).

13.11 Where the tube radio meets Ambersong

Section 13.1.2 lists the meeting points. This section gives the radio's side of each.

Mains. The radio's live terminal (J3) takes the **switched live**, after the front power switch, through the external mains line filter; its neutral terminal (J4) takes **neutral** through the same filter. The filter sits under its own earthed Faraday cage. There is no isolation transformer. So the radio needs both the front power switch and its own power switch (S2) on to run (chapter 5 (p. 27), section 5.2 (p. 28)). S2 is not on the front panel; where it sits, and which of its positions have power on, are not recorded. No fuse is recorded on the machine's mains side; the tube radio has its own 0.75 A fuse (M1, section 13.10 (p. 125)).

Chassis. Because J4 is the chassis, CHASSIS is on mains neutral. It is never joined to Earth, to the Faraday cages or to the DC ground (chapter 2 (p. 13), section 2.3 (p. 14); chapter 5 (p. 27), section 5.3 (p. 28)).

Antenna. The radio's FM input is a coax connector (J53, RG179 coax). It replaced the radio's original simple wire antenna. Its centre goes through the antenna coupling capacitor (C3, X1Y1) to the FM RF stage (section 13.2 (p. 98)). Its shield is on Earth and reaches the chassis only through the antenna shield capacitor (C51, 0.001 μ F, X1Y2). The machine's antenna connector is J10, a BNC on the back panel. No cable joining J10 to J53 is recorded (chapter 10 (p. 56), section 10.35 (p. 85)). The antenna itself is in chapter 11 (p. 87); the earth path is in chapter 5 (p. 27), section 5.4 (p. 30). On the AM and SW side, the external SW antenna terminal (J2) is not fitted; the internal loopstick antenna (L7) is (section 13.7 (p. 113)).

Sound out. The output transformer (T1) is one part: its primary is the radio's (section 13.8 (p. 117)), its secondary feeds Ambersong. The secondary's two leads carry RAD_L and RAD_R. A **7.5 Ω 5 W resistor** (marked 5W7 Ω 5J) sits across them as the transformer's load. A shielded cable takes them to the radio input jack at the ADC (J51). Which lead goes to which contact does not matter. On the cable side of that jack the shield and foil are on the sleeve, RAD_L on the ring and RAD_R on the tip. On its output side the sleeve and ring are on ground, and the tip goes into the 4.7 k Ω resistor of the input attenuator. So the feed is mono: RAD_L goes to ground with the cable's shield and foil, and RAD_R goes through the 4.7 k Ω (chapter 6 (p. 33), section 6.4 (p. 36); chapter 10 (p. 56), section 10.22 (p. 76)).

Volume. Of the three volume controls in the chain, the radio's own (R1) is the first. It is not on the front panel, and sits at about one eighth of its travel in normal use (chapter 6 (p. 33), section 6.8 (p. 38)).

FM calibration receiver. Its antenna, a 30 cm wire bent in a U, lies over the FM oscillator section inside the radio's cage. It is not connected to the radio (section 13.3 (p. 101); chapter 8 (p. 45), section 8.5 (p. 47)).

13.12 When something is wrong

Find what you see in the left column; the right column says where to look first.

You see	Where to look
Radio silent on every band; no heater warms up	The radio's own power switch (S2) and its fuse (M1). Then the heater string, end to end (section 13.9.5 (p. 125)): one open heater or choke stops all seven.
Radio silent on every band; heaters warm	Parts every band shares: the B+ chain (section 13.10 (p. 125)), the first IF tube (V4), V6 and V7, the volume control and the output transformer.
FM silent, SW and AM work	What runs only in FM and FM-AFC: the mode switch's B+ unit on FM_B, V1, V2, V5, and the FM windings of the IF transformers (sections 13.2 (p. 98) to 13.5).
SW and AM silent, FM works	What runs only in SW and AM: AM_B, the converter (V3), the AM windings of the IF transformers and V6's diodes (sections 13.6 (p. 111), 13.7).
FM works in FM but not in FM-AFC	The only difference between the two positions is the AFC-off unit (S1A). Look at the AFC path: the AFC feed from the ratio detector (section 13.5 (p. 108)), the AFC triode and its grid network (section 13.3 (p. 101)).
Radio too loud further down the chain	The radio's own volume control: it belongs at about one eighth of its travel (section 13.8 (p. 117)).
Treble missing	The tone switch: L and M shunt treble to the chassis; H switches nothing in (section 13.8 (p. 117)).

Appendix A. Datasheet facts used in this machine

This appendix gathers the makers' figures that matter for this machine: pinouts, limits, addresses and ratings, part by part. Only the facts the machine uses are here. You need it when you replace a part, check a reading against a limit, or adapt the design to other parts. The chapters tell you what each part does; abbreviations are explained in the glossary.

Every figure is the **maker's**, for the catalogue part. It is not a measurement of the part fitted here. Sometimes the maker's document describes a related part rather than the fitted one: a module the boards may not carry, or a reference development board the fitted board may not match. The entry then says so. There are no entries for the real-time clock module: its datasheet is not included in this book (Appendix C (p. 139)).

A.1 Main board: ESP32-S3

The fitted module is marked **ESP32-S3-N16R8**. The chip read itself as an ESP32-S3 (QFN56), revision v0.2, with 8 MB of PSRAM inside (chapter 4 (p. 19), section 4.1 (p. 19)). The module figures below are from Espressif's **ESP32-S3-WROOM-1** module datasheet. That the fitted module is a WROOM-1 is not established.

The table gives each fact, the maker's figure, and why it matters on this machine.

Fact	Maker's figure	Why it matters here
Memory, variant N16R8	16 MB flash (quad SPI) and 8 MB PSRAM (octal SPI); operating temperature −40 to 65 °C	the fitted module's marking
GPIO35, GPIO36, GPIO37	On modules with octal PSRAM, connected to the PSRAM and not available for other uses . Espressif's ESP32-S3-DevKitC-1 guide says the same for its boards.	not used here (chapter 4 (p. 19), section 4.2 (p. 20))
GPIO47, GPIO48 voltage	1.8 V only on modules whose chip is an ESP32-S3R16V. The chip datasheet gives the ESP32-S3R8 a 3.3 V flash supply (VDD_SPI).	They carry the FM calibration receiver's bus. That they work at the normal 3.3 V on this board is an inference from these two figures; the datasheet does not state it directly.
Supply (VDD33)	3.0 to 3.6 V, 3.3 V typical; absolute maximum 3.6 V	
Strapping pins	GPIO0 (weak pull-up) and GPIO46 (weak pull-down) set the boot mode; GPIO45 (weak pull-down) sets the flash supply voltage; GPIO3 (floating) sets the JTAG signal source. They are read at reset. GPIO1 is not a strapping pin.	none of the four is used; GPIO1 reads the amplifier power sensor
GPIO19, GPIO20	USB D− and D+, connected by default to the chip's USB Serial/JTAG controller. Reconfiguring them as ordinary GPIOs disables USB-JTAG.	the main board's USB port
GPIO39 to GPIO42	the JTAG interface: MTCK (GPIO39), MTDO (GPIO40), MTDI (GPIO41), MTMS (GPIO42)	used here for the angle sensor's clock and three digit selects
UART0	TXD0 on GPIO43, RXD0 on GPIO44	not used
RGB LED (DevKitC-1 guide)	on GPIO48 on the first version of the DevKitC-1 board, on GPIO38 on version 1.1	The fitted board's RGB LED is on GPIO48, which also carries the calibration receiver's clock.

Fact	Maker's figure	Why it matters here
Power inputs (DevKitC-1 guide)	the USB ports, the 5V pin and the 3V3 pin are "three mutually exclusive ways to provide power to the board"	see the caution in chapter 2 (p. 13), section 2.5 (p. 15)

A.2 Audio board: ESP32

The audio board is meant to carry an **ESP32-WROOM-32E** module; whether it does is not established. Its chip read itself as an ESP32-D0WD-V3. The module figures are from Espressif's **ESP32-WROOM-32** datasheet. The board figures are from Espressif's **ESP32-DevKitC V4** guide. Whether the fitted board matches that board is not established.

The table is laid out as in section A.1.

Fact	Maker's figure	Why it matters here
GPIO6 to GPIO11	connected to the module's SPI flash; not recommended for other uses	not connected here
Strapping pins	GPIO0 (pull-up) and GPIO2 (pull-down) set the boot mode; GPIO12 (MTDI, pull-down) selects the flash supply voltage at reset; GPIO15 (MTDO, pull-up); GPIO5 (pull-up)	GPIO0 carries the converters' master clock
GPIO34 to GPIO39	input only : no output driver and no internal pull-up or pull-down (ESP32 chip datasheet). On the module: GPIO36 is ADC1 channel 0, GPIO39 channel 3, GPIO34 channel 6, GPIO35 channel 7.	the volume knob (GPIO35) and the source switch (GPIO36)
Supply (VDD33)	3.0 to 3.6 V, 3.3 V typical; absolute maximum −0.3 to 3.6 V	
GPIO16, GPIO17 (DevKitC V4)	available only on boards with ESP32-WROOM or ESP32-SOLO-1 modules; boards with ESP32-WROVER modules reserve them	GPIO17 is the word clock here
Capacitor C15 (DevKitC V4)	On earlier DevKitC V4 boards, C15 may make the board boot into download mode, and may disturb a clock output on GPIO0.	GPIO0 is the master clock output here
Power inputs (DevKitC V4)	Micro USB, the 5V pin and the 3V3 pin are three mutually exclusive ways to power the board; power must come from one and only one of them, "otherwise the board and/or the power supply source can be damaged"	see chapter 2 (p. 13), section 2.5 (p. 15)

A.3 The ADC: PCM1802 (Texas Instruments)

The ADC turns the tube radio's audio into digital samples for the audio board (chapter 6 (p. 33)). The table gives the maker's figures for its supplies and setting pins.

Fact	Maker's figure
Supplies	VCC (pin 5): analogue, 5 V. VDD (pin 14): digital, 3.3 V.
Setting pins	PDWN (pin 7), BYPAS (pin 8), OSR (pin 16), FMT0, FMT1, MODE0 and MODE1 are Schmitt-trigger inputs with an internal pull-down of about 50 kΩ, and are 5 V tolerant.
PDWN	power-down, active low: the chip runs only while PDWN is high
Format	FMT1 low and FMT0 high: I2S, 24-bit
Interface mode	MODE1 and MODE0 both low: slave mode (system clock 256, 384, 512 or 768 times the sample rate)

Fact	Maker's figure
Slave mode	BCK, LRCK and FSYNC are inputs. The chip sends its data only while FSYNC is high.
OSR	low: oversampling $\times 64$; high: $\times 128$
BYPAS	low: normal mode, with the high-pass filter cutting DC; high: filter bypassed
SCKI (pin 15)	the system clock input, 256, 384, 512 or 768 times the sample rate; Schmitt-trigger, 5 V tolerant

How each setting pin is tied on this machine's module is in chapter 6 (p. 33), section 6.3 (p. 34).

A.4 The DAC: PCM5102A (Texas Instruments)

The DAC turns the audio board's digital audio back into analogue audio for the amplifier (chapter 6 (p. 33)). The table gives the maker's figures for its pins, supplies and modes.

Fact	Maker's figure
Pins	OUTL 6, OUTR 7, AVDD 8, AGND 9, DEMP 10, FLT 11, SCK 12, BCK 13, DIN 14, LRCK 15, FMT 16, XSMT 17, DVDD 20
Digital inputs	failsafe LVCMOS Schmitt-trigger inputs
Supplies	AVDD: 3.3 V. DVDD: 1.8 V or 3.3 V. Absolute maximum 3.9 V on AVDD, CPVDD and DVDD.
XSMT	soft mute when low, un-mute when high. Where it is not needed, it can be tied straight to AVDD.
FMT	low: I2S; high: left-justified
FLT	low: normal-latency filter; high: low-latency filter
DEMP	de-emphasis for 44.1 kHz audio: off when low, on when high
Zero-data mute	After 1024 word-clock periods of zero data (21 ms at 48 kHz), on both channels by default, the chip puts its output into full analogue mute.
Master clock and PLL	The internal PLL is turned off as soon as an external system clock (SCK) is supplied. If BCK and LRCK start correctly while SCK stays at ground for 16 word-clock periods, the PLL starts and makes its own system clock from BCK.

A.5 Amplifier power sensor and box-fan regulator

PC817 optocoupler (Sharp). An optocoupler passes a signal across by light: an LED on one side lights a phototransistor on the other, with no wire between them.

Fact	Maker's figure
Pins (DIP-4)	1 anode, 2 cathode, 3 emitter, 4 collector
Absolute maximum, input (LED)	forward current 50 mA; peak forward current 1 A; reverse voltage 6 V; power dissipation 70 mW
Absolute maximum, output	collector-emitter 80 V; emitter-collector 6 V; collector current 50 mA
Current transfer ratio (CTR)	50 to 400 % at 5 mA LED current and 5 V collector-emitter, 25 °C; 50 % minimum. The CTR is the output current as a share of the LED current.

LM7812 12 V regulator (Texas Instruments). It makes the box fan's 12 V from the amplifier supply's +20 V (chapter 5 (p. 27), section 5.6 (p. 31)).

Fact	Maker's figure
Pins (TO-220)	1 input, 2 ground, 3 output
Output	11.5 to 12.5 V at 25 °C; 11.4 to 12.6 V over an input of 14.5 to 27 V, 5 mA to 1 A output, at most 15 W dissipated. The maker specifies it at 19 V in.

The amplifier supply that feeds it measured 19.33 V with no load (chapter 5 (p. 27), section 5.6 (p. 31)).

A.6 Display and motor drivers

ULN2003A (Texas Instruments). The chip holds seven Darlington transistor pairs, each able to pull a load to ground. It is used on the display driver board and on the needle motor driver.

Fact	Maker's figure
Inputs	1B to 7B on pins 1 to 7
Outputs	1C to 7C on pins 16 to 10 (open collector: each can only pull low)
E (pin 8)	common emitter of all channels, usually tied to ground
COM (pin 9)	common cathode of the built-in flyback diodes; required for inductive loads
Input resistor	2.7 kΩ in series with each base, so the inputs work straight from 5 V or 3.3 V logic
Absolute maximum	collector–emitter 50 V; input 30 V; peak collector current 500 mA; all referred to E

FQP27P06 P-channel MOSFET (onsemi). These are the digit switches of the clock display (chapter 7 (p. 40)).

Fact	Maker's figure
Drain–source voltage	–60 V
Drain current, continuous	–27 A (case at 25 °C)
Gate–source voltage	±25 V maximum
Gate threshold	–2.0 to –4.0 V
On-resistance	70 mΩ maximum at –10 V gate–source, –13.5 A
Package and pins	TO-220, pins G, D, S

A.7 Connectors

JST XH. These are the 2-pin sockets of the 5 V and 3.3 V buses. They are identified only by the drawing's text "JST XH".

Fact	Maker's figure
Pitch	2.5 mm
Current	3 A AC/DC, with 22 AWG wire
Voltage	250 V AC/DC
Temperature	–25 to +85 °C
Wire	30 to 22 AWG (0.05 to 0.33 mm ²); insulation 0.9 to 1.9 mm outside diameter

IEC C14 mains inlet. No datasheet names the fitted inlet. One maker's C14 inlet, the Schurter 6100-4, is rated 10 A, protection class I.

A.8 Needle, tuning and lamp parts

AS5600 magnetic angle sensor (the tuning shaft).

Fact	Maker's figure
I2C address	0x36 (7-bit)
DIR pin	on ground: the reading increases clockwise; on the supply: counter-clockwise
SDA, SCL	"consider external pull-up"
Supply	VDD3V3 absolute maximum 4.0 V. For 3.3 V operation, the VDD5V and VDD3V3 pins must be tied together. Whether the fitted sensor's board ties them is not recorded.

RDA5807M FM receiver (the FM calibration receiver).

Fact	Maker's figure
Pins	ground 1, 3 and 8; FM_IN 2; SCLK 4 (serial clock); SDIO 5; RCLK 6 (32.768 kHz crystal or reference clock input); VDD 7; ROUT and LOUT 9 and 10
Supply	1.8 to 3.3 V, 3.0 V typical, through an internal regulator
Control	I2C only
Address	0010000 binary (0x10). This revision of the datasheet documents no other address. The fitted chip answers at 0x11 (observed on the running machine, not a datasheet figure).
Bypass	the maker's reference design puts a 22 nF supply bypass capacitor close to pin 7

The decoupling fitted on this machine's module is in chapter 8 (p. 45), section 8.5 (p. 47).

IRL540N MOSFET (the panel-lamp switch).

Fact	Maker's figure
Gate	logic-level gate drive
Gate threshold	1.0 to 2.0 V
On-resistance	0.044 Ω at 10 V gate, 18 A; 0.053 Ω at 5.0 V, 18 A; 0.063 Ω at 4.0 V, 15 A

A3144 Hall-effect switch (the needle index).

Fact	Maker's figure
Supply	4.5 to 24 V, with a built-in regulator
Output	open collector, sinks up to 25 mA; needs a pull-up for logic circuits
Pins	viewed from the branded side: 1 supply, 2 ground, 3 output
Switching	the output (pin 3) goes low when the field at the sensor rises above the operate point, and high again when it falls below the release point
Availability	discontinued ; the maker points A3144 users to the A1104

A.9 Tubes

The tubes' bases, heater ratings and pins, and the 50C5's maximum ratings, are in chapter 13 (p. 93) (sections 13.1.4 (p. 95) and 13.8.2). These figures are published tube data. Some of it comes from reference sites rather than from the tubes' makers.

50C5 typical class-A operation. The 50C5 is the radio's output tube (chapter 13 (p. 93), section 13.8 (p. 117)). The table gives two typical operating points: the voltages on its plate, control grid and screen grid, the plate current, the load and the output power.

Plate voltage	Grid voltage	Screen voltage	Plate current	Load	Output power
110 V	−7.5 V	110 V	50 mA	2.5 kΩ	1.9 W
120 V	−8 V	110 V	49 mA	2.5 kΩ	2.3 W

Appendix B. Where the schematics differ from this book

You do not need the KiCad drawings to use this book. This appendix is for readers who also open them: read it before you compare a drawing with the machine or with this book. The author drew the machine to show a person what was built, not to make a circuit board. So a drawing sometimes places a part where it reads best rather than where it sits. **Where a drawing and this book differ, this book is right.**

B.1 Drawing errors still in the published sheets

Two known errors are left in the drawings on purpose. Both are wrong names on library symbols, and neither changes any wiring. The table gives what the drawing shows and what is in the machine.

On the drawing	In the machine
The main board's module is drawn with Espressif's ESP32-S3-WROOM-2 library symbol, with the value "ESP32-S3-WROOM-1_N16R8".	The fitted module is marked ESP32-S3-N16R8 (chapter 4 (p. 19), section 4.1 (p. 19)).
The tube radio's mode-switch units (S1A to S1G) use a library symbol named TM-838M_UNIT1 .	The set is a TM-838N .

Do not take the WROOM-2's datasheet for this board. It describes the module named by the symbol, not the fitted one. That module:

- carries only the ESP32-S3R8V or ESP32-S3R16V chip, with octal flash on every variant;
- runs GPIO47 and GPIO48 at 1.8 V;
- has its pins 28 to 30 not connected, so it has no IO35 to IO37 pads, while the drawn symbol has pins 35, 36 and 37.

Appendix A (p. 130) gives the figures that do apply.

B.2 Parts drawn where they read best

On the main sheet, these parts are drawn for clarity, not where they sit. The sub-sheet or the cable sheet shows the real place. The table gives where each is drawn and where it really is.

On the main sheet	In the machine
The source switch's 22 k Ω and 68 k Ω resistors, drawn in the front switch's block	on the audio board (chapter 9 (p. 50), section 9.3 (p. 52))
The 0.1 μ F filter capacitor on GPIO36, drawn at the ladder side of the 1 k Ω	on the pin side, at the audio board
The colon's 33 Ω and its 1.5 k Ω variable resistor, both drawn on the 5 V side of the colon LED	5 V $\rightarrow$ 33 Ω $\rightarrow$ colon $\rightarrow$ variable resistor $\rightarrow$ ground, as the display sub-sheet draws it. The variable resistor is glued at 1.5 k Ω (chapter 7 (p. 40), section 7.6 (p. 43)).

On the main sheet	In the machine
The four 470 Ω resistors of the DAC-to-amplifier link, drawn on the DAC's output pins	in the cable, as the cable sheet draws them: the two plus legs' at the DAC end, the two minus legs' in the cable, at an end not recorded (chapter 6 (p. 33), section 6.6 (p. 37); chapter 10 (p. 56), section 10.21 (p. 75))

B.3 Reading notes

Names and symbols

- **Rails with two names** are one conductor: `5VDC_S` and `5VDC`; `PSU_GND` and `GND`; `A_VDC` and `A_3V3` (About this book, *Notation*).
- `A_I00A` and `A_I00D` are both the audio board's GPIO0, named twice so each converter's wiring reads clearly. The two clock contacts of the audio harness both carry GPIO0; neither belongs to one converter.
- **One label on several wires.** On the Bluetooth and source-switch sheet, "GPIO" names both the H and the L leg. On the ADC sheet, "ATTENUATOR" names both the ring and the tip of the radio input. On the USB-C sheet, one "TO ... USB-C" name sits on all four wires of each port (VBUS, D-, D+ and ground). These are visual aids, not shorts. Each is a separate wire, and this book describes each one separately.
- **GND twice on one connector** (the digit-rail cable's 2-pin connector) is the cable's shield and foil, ending on the driver board's common ground (chapter 10 (p. 56), section 10.27 (p. 79)).
- `20V_GND` is drawn on the main sheet for completeness only. The sheet's note beside it reads: "The AMP PSU '20V_GND' is shown for completeness only; the actual path goes to the AMP's -20V and +20V inputs." The amplifier supply's four wires, as recorded, are in chapter 5 (p. 27), section 5.6 (p. 31), and chapter 10 (p. 56), section 10.30 (p. 81).
- **The real-time clock** is drawn with a DS3231M symbol. The fitted clock is a DS3231 module of the ZS-042 style (chapter 4 (p. 19), section 4.7 (p. 25)). The symbol's eight ground pins are hidden on the drawing, and lie on the main sheet's `A_VDC` wire. They belong to the drawn symbol only and are disregarded: they are not a connection between 3.3 V and ground. The module's ground comes on its power cable (chapter 10 (p. 56), section 10.16 (p. 71)).
- **The clock module's I2C connector** is drawn with pins 1 and 2 not connected, pin 3 on SDA and pin 4 on SCL. That is right: a 4-pin socket with two pins used.

References

- **The radio input jack at the ADC** (`RADIO_TO_ADC`) is J51, as on the main sheet; the ADC's own sheet numbers it J42. This book uses J51.
- **That jack's output side.** The ADC's sheet labels both its ring and its tip ATTENUATOR. As fitted, the ring is on ground with the sleeve, and only the tip feeds the 4.7 k Ω (chapter 6 (p. 33), section 6.4 (p. 36)).
- **The DAC's output plug**, named `DAC_OUT` on both, is J49 on the cable sheet and J40 on the DAC sheet.
- **The same reference on two sheets** can mean two different parts: C13 to C30, J2 to J4, R17 and M1, for example, are used both on the main sheet and in the tube radio. Chapter 13, section 13.1.7 (p. 97), explains how this book keeps them apart.

What to ignore

- **The TL074 circuit off the page** of the main sheet is a draft of the amplifier's input board, drawn to help visualise it. Disregard it.

- **Unlabelled rectangles over modules** are the printed shells that hold or cover them.

What a drawing does and does not tell you

- **Parts drawn only on the main sheet are fitted.**
- **Capacitors drawn as polarised** are fitted the right way round as drawn; the others are film or ceramic. In the tube radio's polarised capacitors (C1A to C1C and C2), pin 1 is the + terminal.
- **Pin numbers on cable connectors** count along the drawn symbol. They are not contact numbers, and a sub-sheet sometimes orders a connector's pins differently from the cable sheet. This book prints the cable sheet's order. Neither order says anything about the real connector (chapter 10 (p. 56), section 10.1 (p. 56)).

The tube radio sheet

- **The FM IF is 10.6 MHz.** The sheet's "10.7 MHz FM IF Path" section title and the fields of the first two FM IF transformers (L10, L12) read 10.7 MHz, the design value; the third, the ratio-detector transformer (L14), is drawn in its own section but its field reads 10.7 MHz too. The fitted IF is 10.6 MHz for all three (L10, L12 and L14), set when Radio Hosvep on Park Ave. professionally tuned the FM path (chapter 13 (p. 93), section 13.1.1 (p. 93)).
- **Terminal numbers are logical.** The identifiers on the coils L5, L6 and L8 to L14 and on the mode switch's units S1A to S1G name connections, not physical lugs.
- **One part, several units.** A tube is drawn as separate units: its signal sections in their circuits, its heater on the heater string. S1A to S1G are seven drawn units of one mode switch.
- **Three-lead capacitors** (C5, C11, C14, C48) are drawn as two-terminal capacitors. On the part, the middle lead goes to the chassis and the two outer leads read 0 Ω to each other (chapter 13 (p. 93), section 13.1.3 (p. 94)).
- **"50C5" on the output transformer's primary** means the radio's output stage.
- **The speakers SP1 and SP2** are the radio's original speakers as the factory drew them. In this machine the radio's output feeds the 7.5 Ω load and the ADC, not speakers.
- **Drawn but absent:** the neon lamps NE1 to NE3 and the external SW antenna terminal (J2) are not in this radio.
- **Values and specifications.** A part's value on the sheet is its fitted value. Its hidden specification fields hold the original factory and SAMS values; chapter 13 (p. 93) prints both.
- **Notes that point elsewhere.** The sheet's notes that send you to "the audio sheet" or "the Main Sheet" are left as written, because the sheet is usually one of a larger set.

Appendix C. What this book does not cover, and open points

This book states only what is known about the fitted machine. This appendix gathers, in one place, what it leaves out on purpose and what is not known. Read it before you plan a repair or a change, or before you rely on a detail the book does not give. Each open point is stated where it matters, in the chapter named. It is listed here so that you can see the gaps at a glance.

C.1 Outside the book's scope

The table lists what the book leaves out on purpose.

Not covered	Notes
The firmware	What the two boards run is in the companion book, the Ambersong Firmware Gospel.
The amplifier's own circuit	Chapter 12 says what the amplifier is and how it connects. No drawing shows its boards.
Wire colours	Except the amplifier supply's four wires (chapter 5 (p. 27), section 5.6 (p. 31)).
Connector details	Which side of a connector is pin 1, the crimp and connector housings, and the order of the contacts inside a connector. Pin numbers in this book are the cable sheet's (chapter 10 (p. 56), section 10.1 (p. 56)).
Cable lengths and routing	The only length facts: every cable inside the cabinet is under 14 in, and the DAC-to-amplifier cable is under 12 in (chapter 10 (p. 56), section 10.21 (p. 75)).
Screws of the printed plates	Mentioned, not detailed (chapter 3 (p. 16)).
Where each module sits in the cabinet	Beyond "each board in its own printed case, or on a printed carrying plate, screwed to the cabinet" (chapter 3 (p. 16), section 3.1 (p. 16)). The amplifier's aluminium back plate, with the mains inlet, is at the back of the cabinet, and the antenna connector is on the back panel; where the amplifier's input panel sits is not recorded.
The tube radio's original documents	The factory schematic and the SAMS service data are scans, and the SAMS data is not included. Their values reach this book only through the radio drawing's specification fields and notes: the "Original" columns of chapter 13 (p. 93).
The real-time clock's datasheet	The fitted clock is a DS3231 module of the ZS-042 style. Its datasheet limits are not included in this book, so Appendix A (p. 130) has no entry for it (chapter 4 (p. 19), section 4.7 (p. 25)).

C.2 Open points

The open points are grouped by chapter. In each table, a row names a topic and says what is not known about it.

The boards (chapter 4)

Topic	What is not known
Module makers	Neither module's maker is identified. The main board module's FCC grantee code (2AD6X: the part of its FCC radio-approval ID that names the maker) differs from Espressif's (2AC7Z). The audio board's code (2BB77) is not identified with a maker. Whether either module matches any Espressif module datasheet is not established.

Topic	What is not known
Flash chips	The makers behind the flash manufacturer IDs 0x68 (main board) and 0x5e (audio board) are not identified.
Main board against the DevKitC-1	Compatible in firmware and pin-compatible for every pin the machine uses. The pins it does not use have not been checked.
Audio board against the DevKitC V4	Whether it matches, its header order included, is not established.
Firmware text that disagrees with this book	Four lines of the firmware's pin definitions disagree with the hardware; the firmware is the one to change. A comment reads "680 ohm on the 5 V rail": the panel-lamp resistors are 120 Ω (chapter 9 (p. 50), section 9.1 (p. 50)). A comment reads "4.7k pull-ups to S3_3V3": the angle sensor's pull-ups are 10 k Ω (chapter 4 (p. 19), section 4.4 (p. 23)). The define <code>A32_PCM5102_XSMT 16</code> makes the audio board drive GPIO16 high as the DAC's soft mute, but GPIO16 is not wired and XSMT is pulled up on the DAC module itself, so it drives nothing (chapter 6 (p. 33), section 6.5 (p. 37)). The define <code>S3_LIMIT_RIGHT 4</code> is used nowhere, and GPIO4 goes nowhere beyond its own pull-up and capacitor (chapter 4 (p. 19), section 4.4 (p. 23)).

Power and grounds (chapter 5)

Topic	What is not known
DC ground to Earth	The DC ground reaches Earth through the amplifier's ground (a washer on its Bass pot, chapter 5 (p. 27), section 5.3 (p. 28)). By which path the <code>GND</code> buses reach the amplifier's ground is not recorded. "Almost 0 Ω " is the author's figure; the instrument and when it was read are not recorded.
Bus positions	Which 5 V bus socket each power cable uses; which of the clock module, the volume knob and the source switch plug into 3.3 V bus sockets; which of the 3.3 V jacks are bus positions and which sit on the audio board (chapter 10 (p. 56)).
Amplifier supply wires	Their colours are the author's reading, not checked inside the supply.
Box-fan harness	Which board its two plugs sit on, and how the wires end at the supply and at the fan.
Mains wiring	The wires' gauges and colours.
Mains fuse	No fuse is recorded on the machine's mains side; the tube radio has its own 0.75 A fuse (chapter 13 (p. 93), section 13.10 (p. 125)). Whether the mains inlet or the supply modules carry fuses inside is not recorded.

Audio (chapters 6, 9 and 12)

Topic	What is not known
ADC's PDWN and FSYNC	As drawn, they are joined on a <code>P_3V3</code> label that goes nowhere else. The converter runs, so PDWN is not held low; the module's own wiring is not recorded (chapter 6 (p. 33), section 6.3 (p. 34)).
DAC data pull-down	Where the 100 k Ω to ground sits along the DAC data line (chapter 4 (p. 19), section 4.5 (p. 24)).
AUX and RCA	The AUX jack reaches the amplifier's input panel, but the run is not drawn. Nothing is recorded on the panel's RCA pair (chapter 6 (p. 33), section 6.7 (p. 38)).
DAC-to-amplifier cable	Where along the cable the two minus legs' 470 Ω resistors sit (chapter 10 (p. 56), section 10.21 (p. 75)).
Radio-to-ADC cable	Whether it is soldered or plugged at the output transformer's secondary (chapter 10 (p. 56), section 10.22 (p. 76)).

Topic	What is not known
Amplifier input	What is known rests on one meter reading, taken once; nothing inside the amplifier was examined (chapter 12 (p. 90), section 12.4 (p. 91)).

Clock display (chapter 7)

Topic	What is not known
Digit order	The order (GPIO42, GPIO2, GPIO40, GPIO41 for digits 1 to 4) is the one drawn in chapter 7 (p. 40), section 7.4 (p. 42), and chapter 10 (p. 56), section 10.8 (p. 63), and the one the display shows. Which physical board or cable puts the digits in this order is not established; the author believes it is the digit driver board.
Digit sockets	Whether the display sub-sheet's digit outputs are the sockets the digits plug into (chapter 10 (p. 56), section 10.34 (p. 84)).
Colon cable	Where its reversed polarity physically sits (chapter 10 (p. 56), section 10.23 (p. 76)).
Digit-rail cable	What its shield and foil do at the display end (chapter 10 (p. 56), section 10.27 (p. 79)).

Dial and sensors (chapter 8)

Topic	What is not known
Index magnet	Which pole faces the Hall switch.
Angle sensor cable	The connector at the sensor end, and its pin order (chapter 10 (p. 56), section 10.10 (p. 65)).
Calibration receiver's antenna plug	Which pins of the 2 × 3 grid carry the antenna wire (chapter 10 (p. 56), section 10.35 (p. 85)).
Needle motor lead	Anything beyond the socket's five pins (chapter 10 (p. 56), section 10.32 (p. 83)).

FM antenna (chapter 11)

Topic	What is not known
Performance	No SWR, no field strength and no reception comparison has been measured.
Antenna coax inside	The cable from the antenna connector, on the back panel, to the radio's antenna input (chapter 10 (p. 56), section 10.35 (p. 85)).
Build details	The details of section 11.3 (p. 88) come from the build notes and were not confirmed one by one.

The tube radio (chapter 13)

Topic	What is not known
Lugs	Which physical lug carries each logical terminal number of the coils and the mode switch (section 13.1.3 (p. 94)).
The ratio detector's tertiary winding (L14)	Where its other end goes (section 13.4 (p. 104)).
Original specifications	Not recorded for the oscillator cathode resistor (R33) and its bypass capacitor (C50) (section 13.3 (p. 101)).
The antenna shield capacitor (C51)	Where it came from (section 13.2 (p. 98)).

Topic	What is not known
Fitted part types and ratings	Not recorded for several parts: among them the FM detector diodes (M3, M4), the mains fuse's type and voltage (M1), the rectifier's part number and rating (X1), and some capacitors' dielectrics. Each stage's parts table says which (sections 13.2 (p. 98) to 13.10).
Power switch	Which lug of the power and tone switch (S2) carries the power contact (section 13.10 (p. 125)), which of its positions have power on, and where the switch sits (it is not on the front panel).

Appendix D. Printed parts

This appendix lists every printed part of the build. You need it when you reprint a broken or lost part, or when you build a cabinet of your own.

Every printed part is an STL file (the file a 3D printer works from) in the `hardware/STLs/` folder of the published Ambersong project files, named as in the tables below: 76 files. **All are PLA, except the passive radiator's membrane, which is TPU** (a flexible filament). They are the author's parts for this cabinet and these boards; adapt them to yours. In the file names, MCU means a microcontroller board, PSU a power supply, and LEDITRON the clock display.

Not printed: the source switch and the power switch (fixed in the wooden front panel), the mains inlet (on the amplifier's aluminium back plate), and the standoffs of the amplifier's ± 20 V supply, which are plastic screws and standoffs.

D.1 Speakers

These are the passive radiators that replace the speakers' original ports, and the speaker holder (chapter 12 (p. 90), section 12.1 (p. 90)).

File	What it is
AUDIO - Passive Radiator - Holder.stl	passive radiator: holder
AUDIO - Passive Radiator - Inside Protector.stl	passive radiator: inside protector
AUDIO - Passive Radiator - Membrane Weight.stl	passive radiator: membrane weight
AUDIO - Passive Radiator - Outside Protector.stl	passive radiator: outside protector
AUDIO - Passive Radiator - TPU Membrane.stl	passive radiator: membrane, printed in TPU
AUDIO - Speakers Holder and Seal.stl	speaker holder and seal

D.2 Back

The files whose names begin with BACK are the back panel's anchors and vents, and the parts for the USB-C ports and the antenna connector (chapter 3 (p. 16), sections 3.1 (p. 16) and 3.4).

File	What it is
BACK - Dual USB-C Ports.stl	the two back-panel USB-C ports (chapter 4 (p. 19), section 4.6 (p. 24))
BACK - Electronics - BNC Antenna Interface.stl	the antenna connector's interface: the connector (J10) is a BNC on the back panel (chapter 11 (p. 87))
BACK - Fan Exhaust.stl	the box fan's exhaust
BACK - Fan Intake.stl	the louvred air intakes
BACK - Panel Anchors Bottom.stl	back panel anchors, bottom
BACK - Panel Anchors Inner Corners.stl	back panel anchors, inner corners
BACK - Panel Anchors Outer Corners.stl	back panel anchors, outer corners

File	What it is
BACK - Panel Anchors.stl	back panel anchors

D.3 Front panel

These are the parts of the front panel, the FM dial and the tuning knob (chapter 3 (p. 16), section 3.2 (p. 16); chapter 8 (p. 45), section 8.1 (p. 45)).

File	What it is
FRONT - FM Dial - Back.stl	FM dial back; it is also the stepper motor's mount
FRONT - FM Dial - Cover.stl	FM dial cover
FRONT - FM Dial - Spacer.stl	FM dial spacer
FRONT - FM Dial - Stepper Needle and Magnet Holder.stl	the mount on the stepper motor's shaft that carries the needle and the index magnet
FRONT - Panel Anchors - Insidemost.stl	front panel anchors, insidemost
FRONT - Panel Anchors - Outsidemost.stl	front panel anchors, outsidemost
FRONT - Tuner Pot Knob.stl	the printed front tuning knob. On the printed shaft extension, it turns the radio's original tuning knob, which drives the dial cord and the tuning capacitor (chapter 3 (p. 16), section 3.2 (p. 16)). "Pot" is the file's word: tuning is by the tuning capacitor.
FRONT - Tuning Pot Extender PLA Shaft.stl	the tuning shaft extension
FRONT - Tuning Shaft Bearing Seat.stl	the tuning shaft's bearing seat
FRONT - Volume Pot Knob.stl	the front volume knob (chapter 9 (p. 50), section 9.4 (p. 53))

D.4 Board cases and holders, inside the cabinet

Each board or module sits in its own printed case, or on a printed carrying plate, screwed to the cabinet (chapter 3 (p. 16), section 3.1 (p. 16)).

File	What it is
IN - Electronics - 5V BUS case Bottom.stl	5 V bus case, bottom (chapter 5 (p. 27), section 5.5 (p. 30))
IN - Electronics - 5V BUS case Top.stl	5 V bus case, top
IN - Electronics - 5V PSU Case Bottom.stl	5 V power supply case, bottom
IN - Electronics - 5V PSU Case Top.stl	5 V power supply case, top
IN - Electronics - 5V+3V3 BUS case Bottom.stl	5 V and 3.3 V bus case, bottom
IN - Electronics - 5V+3V3 BUS case Top.stl	5 V and 3.3 V bus case, top
IN - Electronics - AC Line Filter Case.stl	mains line filter case (chapter 5 (p. 27), section 5.4 (p. 30))
IN - Electronics - AS5600 Case - Bottom.stl	the angle sensor's cage, bottom (chapter 5 (p. 27), section 5.4 (p. 30))

File	What it is
IN - Electronics - AS5600 Case - Connecting Rod.stl	the printed shaft that carries the angle sensor's magnet (chapter 8 (p. 45), section 8.1 (p. 45))
IN - Electronics - AS5600 Case - Top.stl	the angle sensor's cage, top
IN - Electronics - Audio Listener Case BOTTOM.stl	the source-switch board's case, bottom (chapter 9 (p. 50), section 9.3 (p. 52))
IN - Electronics - Audio Listener Case TOP.stl	the source-switch board's case, top
IN - Electronics - Audio MCU Case BOTTOM.stl	audio board case, bottom
IN - Electronics - Audio MCU Case TOP.stl	audio board case, top
IN - Electronics - Backing Panel 1.stl	a panel that holds several of the cases
IN - Electronics - Bluetooth LED Holder.stl	Bluetooth LED holder
IN - Electronics - Bluetooth LED Pusher.stl	Bluetooth LED pusher (the LED is also the button)
IN - Electronics - Box Fan LM7812 Case Bottom.stl	the box fan's regulator case, bottom (chapter 5 (p. 27), section 5.6 (p. 31))
IN - Electronics - Box Fan LM7812 Case Top.stl	the box fan's regulator case, top
IN - Electronics - DS3231 RTC Case BOTTOM.stl	real-time clock module case, bottom
IN - Electronics - DS3231 RTC Case TOP.stl	real-time clock module case, top
IN - Electronics - Digit Drivers Case BOTTOM.stl	digit driver case, bottom (chapter 7 (p. 40))
IN - Electronics - Digit Drivers Case TOP.stl	digit driver case, top
IN - Electronics - Front Panel FM LEDs Case BOTTOM.stl	FM panel lamp board case, bottom (chapter 9 (p. 50), section 9.1 (p. 50))
IN - Electronics - Front Panel FM LEDs Case TOP.stl	FM panel lamp board case, top
IN - Electronics - Hall Sensor Stepper Limit Case.stl	the needle index sensor's case (chapter 8 (p. 45), section 8.3 (p. 46))
IN - Electronics - LEDITRON Back.stl	clock display back
IN - Electronics - LEDITRON Bus Case Bottom.stl	clock display bus case, bottom
IN - Electronics - LEDITRON Bus Case TOP.stl	clock display bus case, top
IN - Electronics - PCM1802 ADC Case BOTTOM.stl	ADC case, bottom
IN - Electronics - PCM1802 ADC Case TOP.stl	ADC case, top
IN - Electronics - PCM5102A DAC Case Bottom.stl	DAC case, bottom
IN - Electronics - PCM5102A DAC Case TOP.stl	DAC case, top
IN - Electronics - Power Detect Case BOTTOM.stl	amplifier power sensor case, bottom (chapter 9 (p. 50), section 9.5 (p. 54))
IN - Electronics - Power Detect Case TOP.stl	amplifier power sensor case, top
IN - Electronics - RDA5807M - Bottom.stl	FM calibration receiver case, bottom (chapter 8 (p. 45), section 8.5 (p. 47))

File	What it is
IN - Electronics - RDA5807M - Cage Holder.stl	FM calibration receiver cage holder
IN - Electronics - RDA5807M - Middle Bottom.stl	FM calibration receiver case, middle bottom
IN - Electronics - RDA5807M - Middle TOP.stl	FM calibration receiver case, middle top
IN - Electronics - RDA5807M - TOP.stl	FM calibration receiver case, top
IN - Electronics - S3 Main MCU Backing.stl	main board backing
IN - Electronics - S3 Main MCU Case BOTTOM.stl	main board case, bottom
IN - Electronics - S3 Main MCU Case TOP.stl	main board case, top
IN - Electronics - Stepper Driver ULN2003A Case - Bottom.stl	needle motor driver case, bottom (chapter 8 (p. 45), section 8.2 (p. 46))
IN - Electronics - Stepper Driver ULN2003A Case - Top.stl	needle motor driver case, top

D.5 The tube radio's Faraday cage

These two parts make the cage over the tube radio (chapter 3 (p. 16), section 3.5 (p. 17); chapter 5 (p. 27), section 5.4 (p. 30)).

File	What it is
IN - Tube Radio - Faraday Cage 1.stl	the tube radio's Faraday cage, part 1; the cage fan's mount is part of it
IN - Tube Radio - Faraday Cage 2.stl	the tube radio's Faraday cage, part 2

D.6 Clock display

These are the clock display's housing and digits (chapter 7 (p. 40)).

File	What it is
LEDITRON - Colon - Blanket.stl	colon blanket
LEDITRON - Colon.stl	colon
LEDITRON - Digit.stl	digit
LEDITRON - Front Case.stl	front case
LEDITRON - Glass Spacer.stl	glass spacer

Glossary

The terms and names you meet in this book, in alphabetical order. Look a term up here whenever a chapter uses it without explaining it, or when you open an appendix. Signal names are in `code font`, as on the drawings. Where a term has its own section, the chapter is given.

`20V_GND` — The amplifier supply return: the middle (0 V) output of the amplifier's ± 20 V supply. One of the machine's four grounds. Its two middle black wires go to the amplifier's ground, so it is joined to Earth and to the DC ground (chapter 5 (p. 27), section 5.3 (p. 28)).

`5VDC` — The 5 V bus, from the 5 V power supply. `5VDC_S` is the same conductor, named before the bus (chapter 5 (p. 27), section 5.5 (p. 30)).

A32 — The audio board: the ESP32 development board that handles sound, Bluetooth, the front controls and the real-time clock (chapter 4 (p. 19)).

`A_IOn` — A signal on pin GPIO_n of the audio board: `A_I022` is its GPIO22.

`A_VDC`, `A_3V3` — Two names for one rail: the audio side's 3.3 V, from the audio board's 3.3 V output, shared out by the 3.3 V bus.

Absolute maximum — A maker's limit that must not be exceeded, even briefly. It is not a working value (Appendix A (p. 130)).

AC/DC set — A tube radio with no mains transformer. Its tube heaters are wired in series across the mains, and its B+ is rectified straight from the mains, so its chassis is tied to the mains. The tube radio here is one (chapters 2 and 13).

ADC — Analogue-to-digital converter: here a PCM1802 board that turns the tube radio's audio into digital samples for the audio board (chapter 6 (p. 33)).

AFC — Automatic frequency control: a voltage from the FM detector that keeps the FM oscillator on the station. Switched on in the radio's FM-AFC position (chapter 13 (p. 93)).

Alignment — Adjusting a radio's trimmers, padders and IF transformers so that its tuned circuits track together across the band. Only the tube radio's FM path was professionally tuned; its AM and SW path was not retuned (chapter 13 (p. 93), sections 13.1.1 (p. 93) and 13.7).

AM — Amplitude modulation; on this radio, the medium-wave band. See also SW and FM.

AMP signals — The cable sheet's names for the pins of the amplifier's input panel: `AMP_L+`, `AMP_L-`, `AMP_R+`, `AMP_R-` and `AMP_G`, the ground (chapters 6 and 12).

ASA — A weather-resistant plastic. The antenna's element bushings, their plates and the feedpoint clamp are ASA (chapter 11 (p. 87), section 11.3 (p. 88)).

Attenuator — A resistor network that lowers a signal's level. The radio's output passes one on its way into the ADC, starting with a 4.7 k Ω resistor (chapter 6 (p. 33), section 6.4 (p. 36)).

Audio board — The ESP32 development board that handles the sound, Bluetooth, the front controls and the real-time clock. Also written A32 (chapter 4 (p. 19)).

AUX — The auxiliary analogue input. The AUX jack reaches the amp's input panel, which sums it with the other inputs; the run is not drawn. It never passes through the audio board (chapter 6 (p. 33), section 6.7 (p. 38)).

AVC — Automatic volume control: a voltage from the AM detector that lowers the gain of earlier stages on strong stations, so that strong and weak stations come out at similar levels (chapter 13 (p. 93)).

AWG — American Wire Gauge: a wire-size scale in which a larger number means a thinner wire.

B+ — The tube radio's high-voltage DC supply for its tubes, made by rectifying the mains. The numbers in its point names (**B130**, **B115_A**) are the original service data's nominal figures, not measured voltages (chapter 2 (p. 13); chapter 13 (p. 93), section 13.10 (p. 125)).

Balanced input — An input that takes each channel on two wires, a plus and a minus, and amplifies the difference between them. The DAC reaches the amplifier through one (chapter 12 (p. 90), section 12.3 (p. 91)).

BCK, LRCK, SCK — The three clocks of the I2S audio bus: the bit clock, the word clock (left and right samples) and the master or system clock (chapter 6 (p. 33), section 6.2 (p. 34)).

BNC — A coaxial connector with a bayonet lock. The machine's antenna connector, on the back panel, is one (chapter 11 (p. 87)).

Boom — The metal bar that carries an antenna's elements (chapter 11 (p. 87)).

Bus — Here, a star-point bus: a row of 2-pin power sockets in which every power pin is one wire and every ground pin is one wire. The machine has a 5 V bus and a 3.3 V bus (chapter 5 (p. 27), section 5.5 (p. 30)).

Bypass capacitor — A capacitor that gives AC a short path to ground or chassis around a part, while the part keeps its DC level. Many of the tube radio's stages use them (chapter 13 (p. 93)).

CABLE S/FTP — On a connector pin of the drawings: the cable's shield and foil end there, on ground, and are left unconnected at the other end unless the cable's page says otherwise (chapter 10 (p. 56), section 10.1 (p. 56)).

CBB22 — A common type of metallised polypropylene film capacitor. One sits at the ADC's input (chapter 6 (p. 33), section 6.4 (p. 36)).

CHASSIS — The tube radio's circuit common: its metal chassis and everything joined to it. **It is on mains neutral** and is joined to none of the other three grounds: Earth, the DC ground and the amplifier supply return (chapter 2 (p. 13), section 2.3 (p. 14); chapter 5 (p. 27), section 5.3 (p. 28)).

Class A — An amplifier stage whose tube conducts through the whole signal cycle. Appendix A (p. 130) gives the 50C5's typical class-A operating points.

Coax — Coaxial cable: a centre conductor inside a tubular shield, made for radio signals. The FM antenna's feedline is RG-6 coax; the radio's FM input takes RG179 coax (chapters 11 and 13).

Common anode — A display digit whose segment LEDs share one anode. The digit lights when its anode is switched to the supply and a segment's cathode is pulled low (chapter 7 (p. 40)).

Common-mode choke — Here, ferrite cores on a cable or a feedpoint. They resist currents that flow the same way on all the conductors, such as noise picked up along a shield, while passing the signal.

Coupling capacitor — A capacitor that passes a signal from one point to the next while blocking DC. The radio's FM antenna input reaches its first stage through one (C3) (chapter 13 (p. 93), section 13.2 (p. 98)).

CP210x — A Silicon Labs USB-to-serial bridge chip. The audio board's USB port uses one (chapter 4 (p. 19)).

- CTR** — Current transfer ratio: an optocoupler's output current as a share of its LED current (Appendix A (p. 130)).
- DAC** — Digital-to-analogue converter: here a PCM5102A board that turns the audio board's digital audio back into analogue audio for the amplifier (chapter 6 (p. 33)).
- Darlington** — Two transistors in tandem that act as one with a very high gain. The ULN2003A holds seven (Appendix A (p. 130)).
- DC ground (GND)** — The common return of everything on 5 V and 3.3 V. It reaches the amplifier's ground and so is on Earth at almost 0 Ω ; by which path it reaches the amplifier's ground is not recorded (chapter 5 (p. 27), section 5.3 (p. 28)).
- DCR** — The DC resistance of a coil, measured with an ohmmeter (chapter 13 (p. 93)).
- De-emphasis** — A treble cut that undoes the treble boost added to FM broadcasts before they are sent. The DAC's de-emphasis is switched off; the tube radio has its own (chapters 6 and 13).
- Decoupling capacitor** — A capacitor from a supply pin to ground, close to a part, that keeps the supply steady when the part draws current in bursts.
- Detector** — The stage that recovers the audio from the IF signal. The tube radio has an AM detector and, for FM, a ratio detector (chapter 13 (p. 93), sections 13.5 (p. 108) and 13.6).
- Development board** — A module on a small board with pin headers and a USB port. Both microcontroller boards are development boards (chapter 4 (p. 19)).
- Dielectric** — The insulating material inside a capacitor, which gives the capacitor its type: film, ceramic, mica or electrolytic. Chapter 13 gives it where it is recorded.
- Director, reflector, driven element** — The elements of a Yagi antenna. Only the driven element is wired to the cable; the reflector sits behind it and the director in front (chapter 11 (p. 87)).
- Download mode** — The boot mode in which an ESP32 waits to receive new firmware instead of running its program (Appendix A (p. 130)).
- Dual-gang, gang** — A gang is one section of a potentiometer or capacitor; a dual-gang part has two sections turned by one shaft. The amplifier's volume control is a dual-gang potentiometer (chapter 12 (p. 90), section 12.2 (p. 90)).
- Earth** — The mains protective earth, from the mains inlet's earth pin. The Faraday cages, the antenna connector's shell, the amplifier's ground, the DC ground and the amplifier supply return are on it (chapter 5 (p. 27), section 5.3 (p. 28)).
- Earth star point** — The earth point on the amplifier's aluminium back plate. The amplifier's ground sits on it, through a washer on the Bass pot (chapter 5 (p. 27), section 5.3 (p. 28)). Not the same as the DC ground's *star point*.
- eFuse** — A one-time setting burned into a chip. The main board's flash voltage is set by eFuse (chapter 4 (p. 19), section 4.1 (p. 19)).
- EMI filter** — See *mains line filter*.
- ESP32, ESP32-S3** — Espressif's microcontroller families. The main board is built on an ESP32-S3, the audio board on an ESP32 (chapter 4 (p. 19)).

F-type — The screw-on coax connector of TV and satellite cables. The antenna's feedpoint connector (an F-81 bulkhead) and its grounding block are F-type (chapter 11 (p. 87)).

Faraday cage — An earthed metal enclosure that shields what is inside from electrical noise, and the outside from it. The machine has three, made of aluminium tape (chapter 5 (p. 27), section 5.4 (p. 30)).

FCC ID — The US radio-approval code printed on a module. Its first part, the grantee code, names the company that obtained the approval (chapter 4 (p. 19); Appendix C (p. 139)).

Feedline, feedpoint — The feedline is the coax from the antenna to the radio. The feedpoint is where the feedline joins the driven element, at a 20 mm gap in the folded dipole (chapter 11 (p. 87)).

Ferrite core — A ring or sleeve of ferrite slipped over a cable to damp high-frequency noise on it (chapter 10 (p. 56)).

Fitted, original — In chapter 13 (p. 93), *fitted* is the part in this radio now; *original* is what the factory drawing and the SAMS service data give.

Flash — Memory that keeps the program with the power off. The main board has 16 MB, the audio board 4 MB (chapter 4 (p. 19), section 4.1 (p. 19)).

Flyback diode — A diode across a coil that absorbs the voltage spike when the coil is switched off. The ULN2003A has them built in (Appendix A (p. 130)).

FM, FM-AFC — Frequency modulation, the VHF broadcast band. FM-AFC is the radio's FM position with automatic frequency control switched on (chapter 13 (p. 93), section 13.1 (p. 93)).

Folded dipole — A driven element made of two parallel conductors joined at both ends, fed at a gap in one of them. The antenna's driven element is one (chapter 11 (p. 87)).

Gas discharge tube (GDT) — A surge arrester often fitted in coax grounding blocks. The antenna's grounding block has none (chapter 11 (p. 87)).

GPIO — General-purpose input/output: a pin of a microcontroller that its program can read or drive.

Grid leak — A high-value resistor that sets a tube grid's DC level by letting charge leak away (chapter 13 (p. 93)).

Grounding block — A block that joins two lengths of coax. The antenna's is fitted outside, out of the weather. An 8 AWG stranded copper run goes from the grounding block to a lug on the meter base, earthing the coax shield (chapter 11 (p. 87), section 11.2 (p. 87)).

Hall-effect switch — A sensor whose output switches when a magnet comes close enough. The needle's index sensor, an A3144, is one (chapter 8 (p. 45), section 8.3 (p. 46)).

Header — A row of pins on a board, onto which a connector plugs.

Heater string — The tube radio's seven tube heaters wired in series, one after the other, so one current flows through all of them. One open heater stops them all (chapter 13 (p. 93), section 13.9 (p. 121)).

I2C — A two-wire bus, data and clock, that a board uses to talk to several chips by address. The angle sensor, the FM calibration receiver and the real-time clock use it.

I2S — A bus that carries digital audio between chips, with a bit clock, a word clock, a master clock and a data line per direction (chapter 6 (p. 33), section 6.2 (p. 34)).

IEC C14 — The standard three-pin mains inlet, the same kind as on a computer. The machine's mains inlet is one (chapter 5 (p. 27)).

IF — Intermediate frequency: the fixed frequency to which a superheterodyne turns every station. The tube radio's FM IF is 10.6 MHz, its AM IF 455 kHz (chapter 13 (p. 93)).

IF transformer — A tuned transformer that passes the IF from one stage to the next, in a shielded metal can with its own tuning capacitors (chapter 13 (p. 93)).

IN/LISTENER/OUT board — The drawings' name for the source-switch board: the small board, in its own printed case, that passes the audio board's 3.3 V and the two source-switch legs on to the front switch. Its connector is named **LISTENER** (chapter 10 (p. 56), section 10.20 (p. 73)).

Index — The needle's one fixed reference point: a magnet on the needle mount passing over a Hall-effect switch (chapter 8 (p. 45), section 8.3 (p. 46)).

Isolation transformer — A transformer that separates a circuit from the mains. The tube radio has none, which is why its chassis is on mains neutral.

JST XH — A family of small connectors with a 2.5 mm pitch, used for the 5 V and 3.3 V buses (Appendix A (p. 130)).

JTAG — A debug interface for chips. On the main board, GPIO39 to GPIO42 are the chip's JTAG pins; the machine uses them for other jobs (chapter 4 (p. 19); Appendix A (p. 130)).

Kapton — A heat-resistant insulating tape, used to cover the aluminium tape of the Faraday cages.

Ladder — Here, the resistor divider that turns the three positions of the source switch into three voltages for the audio board (chapter 9 (p. 50), section 9.3 (p. 52)).

LED — Light-emitting diode.

Leditron — The name of the clock display: four seven-segment digits and a colon, made from single LEDs (chapter 7 (p. 40)).

Limiter — An FM stage that clips the IF signal to a steady level, so that changes in strength do not reach the detector (chapter 13 (p. 93)).

Listed substitute — A catalogue replacement part named for one of the tube radio's original parts in the author's unpublished component list. The list is not in this book; chapter 13 (p. 93) names a substitute only where it tells something about the part (chapter 13 (p. 93), section 13.1.3 (p. 94)).

Local oscillator — The oscillator inside a superheterodyne whose signal the mixer combines with the station. The tube radio's FM local oscillator runs one IF below the station (chapter 13 (p. 93)).

Logic-level MOSFET — A MOSFET that turns fully on with a gate voltage a logic pin can give. The panel-lamp switch, an IRL540N, is one (chapter 9 (p. 50), section 9.1 (p. 50)).

Loopstick — A coil wound on a ferrite rod that serves as an AM antenna. The tube radio's AM input uses its internal loopstick (chapter 13 (p. 93)).

LVC MOS — Low-voltage CMOS: a family of logic input and output levels (Appendix A (p. 130)).

Main board — The ESP32-S3 development board that runs the clock display, the dial needle and its sensors, the FM calibration receiver and the panel lamps. Also written S3 (chapter 4 (p. 19)).

Mains line filter — The EMI (electromagnetic interference) filter between the switched mains and the tube radio, under its own Faraday cage (chapter 5 (p. 27), section 5.2 (p. 28)).

Microcontroller, MCU — A small computer on one chip, with its own inputs and outputs. The machine has two, on the main board and the audio board (chapter 4 (p. 19)).

Mixer — The stage of a superheterodyne that combines the station with the local oscillator's signal to make the IF (chapter 13 (p. 93)).

Mode switch — The tube radio's four-position switch: SW, AM, FM and FM-AFC. The drawing splits it into seven units, S1A to S1G, that turn together (chapter 13 (p. 93), section 13.1.6 (p. 96)).

Module — The small shielded board that carries a microcontroller chip with its flash memory. Each development board carries one (chapter 4 (p. 19)).

MOSFET, P-channel MOSFET — A transistor switched by the voltage on its gate. A P-channel MOSFET switches a load to the supply; the clock digits are switched by four of them, FQP27P06 (chapter 7 (p. 40)).

Neon lamp — A small glow lamp that runs from mains voltage. The tube radio's mode lamps (NE1 to NE3) are drawn but not fitted (chapter 13 (p. 93), section 13.1.6 (p. 96)).

New old stock — A part made long ago and never used. Three of the radio's tubes are new old stock (chapter 13 (p. 93)).

NP0 — A ceramic capacitor dielectric whose value hardly changes with temperature, used where a value must stay exact (chapter 13 (p. 93)).

Op-amp — Operational amplifier: a small amplifier chip used for signal stages. The amplifier's input board holds three (chapter 12 (p. 90), section 12.1 (p. 90)).

Open collector — An output that can only pull its line low; a pull-up resistor sets the high level. The index sensor and the optocoupler have one.

Optocoupler — A part that passes a signal across by light, an LED lighting a phototransistor, with no wire between the two sides. The amplifier power sensor uses a PC817 (chapter 9 (p. 50), section 9.5 (p. 54)).

Output transformer — The transformer between the radio's output tube and its load. In this machine its secondary feeds a 7.5 Ω resistor and the ADC, not speakers (chapter 6 (p. 33), section 6.4 (p. 36)).

Package (TO-220, DIP-4, QFN56) — The body and pin layout of a chip or transistor. TO-220 is a power package with three legs and a metal tab; DIP-4 a four-pin through-hole package; QFN56 a flat chip package with 56 pads (Appendix A (p. 130)).

Padder, trimmer — Small capacitors in a tuned circuit. A trimmer is adjustable and sets the end of a tuning range; a padder, in series with the oscillator's tuning, keeps the oscillator in step with the signal circuits across the band (chapter 13 (p. 93)).

Passive radiator — A speaker cone with no magnet or coil of its own, moved by the air in the enclosure, used in place of a port (chapter 12 (p. 90), section 12.1 (p. 90)).

Pentagrid converter, pentode, triode, beam power tube — Tube types. A triode has one grid, a pentode three; a pentagrid converter has five grids and is mixer and oscillator in one tube; a beam power tube is built to deliver power to a load (chapter 13 (p. 93)).

Perfboard — A board with a grid of holes for hand-wired circuits. Several of the machine's small circuits are built on perfboard.

PLA, TPU, STL — PLA is the plastic of the printed parts; TPU is a flexible printing plastic, used only for the passive radiators' membranes; an STL is the file a printed part is made from (Appendix D (p. 143)).

Plate, cathode, control grid — Electrodes of a tube. The heated cathode gives off electrons, the plate (the anode) collects them, and the control grid between them sets how many get through (chapter 13 (p. 93)).

PLL — Phase-locked loop: a circuit that makes one clock from another. The DAC can make its own master clock this way from the bit clock (Appendix A (p. 130)).

Polarised capacitor — A capacitor with a + and a – terminal, such as an electrolytic, that must be fitted the right way round. In the tube radio's polarised capacitors (C1A to C1C and C2), pin 1 is the + terminal (Appendix B (p. 136)).

PSRAM — Extra memory beside a microcontroller's own. The main board's chip carries 8 MB (chapter 4 (p. 19)).

PSU — Power supply unit. Used in drawing names and printed-part file names.

Pull-up, pull-down — A resistor that holds a line high (to the supply) or low (to ground) when nothing else drives it.

PWM — Pulse-width modulation: switching a load on and off fast, with the on-time setting the average power. The FM panel lamps are dimmed by PWM (chapter 9 (p. 50), section 9.1 (p. 50)).

Ratio detector — The tube radio's FM detector: two diodes and a transformer that turn frequency changes into audio and largely ignore changes in strength (chapter 13 (p. 93), section 13.5 (p. 108)).

RCA — The common phono jack for line-level audio. The amplifier's input panel has an RCA pair; nothing is recorded on it (chapter 12 (p. 90)).

Rectifier — A part that turns AC into DC. The tube radio's rectifier (X1) makes its B+ from the mains (chapter 13 (p. 93)).

RF — Radio frequency: the station's signal as it arrives, before the mixer turns it into the IF. On FM, the RF amplifier (V1) raises it first (chapter 13 (p. 93), section 13.2 (p. 98)).

RMS — Root mean square. A power rating in W RMS is a continuous power, not a peak.

RTC — Real-time clock: a chip that keeps the time; a backup cell on its module keeps it running while the power is off. Here a DS3231 module (chapter 4 (p. 19), section 4.7 (p. 25)).

S3 — The main board, an ESP32-S3 development board (chapter 4 (p. 19)).

S3_3V3 — The main board's own 3.3 V output. It is a separate rail, not the 3.3 V bus.

S/FTP — See *shield and foil*.

Sample rate — The number of audio samples per second on each channel.

SAMS — SAMS Photofact, the published service data for the tube radio. With the factory drawing, it is one of the radio's two original documents.

Schmitt trigger — An input with two switching thresholds, one for a rising and one for a falling signal, so that a slow or noisy signal switches cleanly (Appendix A (p. 130)).

Screen grid — The grid of a pentode or beam tube between the control grid and the plate, held at a steady positive voltage (chapter 13 (p. 93)).

SDA, SCL — The data and clock lines of an I2C bus.

Shield and foil — The braided shield and the foil wrap of a shielded cable. The machine's shielded cables are S/FTP: a braided shield and a foil around twisted pairs (chapter 10 (p. 56), section 10.1 (p. 56)).

Slave mode — A converter mode in which the chip takes its clocks from outside instead of making them. The ADC runs in slave mode, clocked by the audio board (chapter 6 (p. 33)).

Source switch — The front three-position switch: RADIO, BT and AUX. It only tells the audio board which source was chosen; it switches no audio (chapter 9 (p. 50), section 9.3 (p. 52)).

SPI, quad SPI, octal SPI — A serial link between a chip and its memory, with four (quad) or eight (octal) data lines. On the main board's module variant (N16R8), the maker gives quad SPI flash and octal SPI PSRAM (Appendix A (p. 130)).

Standoff — A spacer post that holds a board or a supply off a surface. The amplifier's ± 20 V supply is held off the back plate by plastic screws and standoffs (chapter 5 (p. 27), section 5.3 (p. 28)).

Star point — The one place where all the DC ground wires meet: the ground pins of the 5 V and 3.3 V buses (chapter 5 (p. 27), section 5.5 (p. 30)). See also *earth star point*.

Stepper motor — A motor that turns in fixed steps, one for each change of its coil currents. The needle's motor is a 28BYJ-48 (chapter 8 (p. 45), section 8.2 (p. 46)).

Strapping pin — A microcontroller pin whose level is read at power-up to set how the chip starts (Appendix A (p. 130)).

Superheterodyne — A receiver that turns every station into one fixed intermediate frequency (IF) before amplifying and detecting it (chapter 13 (p. 93), section 13.1.1 (p. 93)).

Suppressor grid — The grid of a pentode nearest the plate. It stops electrons knocked off the plate from reaching the screen grid. In the radio's 12BA6 IF tubes (V4, V5) it is on the chassis (chapter 13 (p. 93), section 13.4 (p. 104)).

SW — Short wave, one of the tube radio's bands.

SWR — Standing-wave ratio: a measure of how well an antenna matches its cable. Not measured on this antenna (chapter 11 (p. 87)).

Taper — How a potentiometer's resistance changes along its travel. An audio taper changes slowly at first, to suit the ear. The radio's volume control is audio taper, as its original specification infers from its listed substitutes (chapter 13 (p. 93)).

Tertiary winding — A third winding on a transformer, besides the primary and the secondary. The ratio-detector transformer (L14) has one (chapter 13 (p. 93), section 13.4 (p. 104)).

TRS — Tip, ring, sleeve: a three-contact jack or plug, as on stereo headphones (chapter 6 (p. 33)).

Tube base — The pin layout of a tube. B7G is the 7-pin miniature base; B9A, or noval, the 9-pin miniature. An EIA basing code (such as 7CH or 9DE) says which electrode is on which pin (chapter 13 (p. 93), section 13.1.4 (p. 95)).

UART — A two-wire serial link, one wire each way, plus ground. The two boards talk over one (chapter 10 (p. 56), section 10.12 (p. 66)).

USB-Serial/JTAG — The ESP32-S3 chip's own USB port, which serves as both a serial link and a debug interface. The main board's USB port uses it (chapter 4 (p. 19)).

VBUS — The 5 V power wire of a USB cable (chapter 4 (p. 19), section 4.6 (p. 24)).

VHF — Very high frequency: the radio band that holds the FM broadcast band (chapters 11 and 13).

Wiper — The sliding contact of a potentiometer. Its voltage follows the knob's position (chapter 9 (p. 50), section 9.4 (p. 53)).

X1Y1, X1Y2 — Safety ratings for capacitors connected to the mains. An X rating is for a capacitor across the mains, a Y rating for one between the mains and a part a person may touch; X1Y1 and X1Y2 parts carry both. Replace such a part only with one of the same class (chapter 2 (p. 13), section 2.4 (p. 14)).

Yagi — A directional antenna: a driven element with unpowered elements beside it on a boom. The FM antenna is a three-element Yagi (chapter 11 (p. 87)).

Zero-data mute — The DAC's own mute: after a run of all-zero audio on both channels, it silences its output (Appendix A (p. 130)).

Index

#

5VDC_FM_LED 66
 5VDC_LIMIT 64
 5VDC_S 137, 147
 5VOUT 51, 83, 84
 20V_GND 14, 15, 29, 30, 54, 81, 137, 147

A

A32_3V3 23, 31, 73
 A32_5VIN 23, 26, 72
 A32_BT_LED 23, 68, 69
 A32_I2S 22, 70
 A32_PCM5102_XSMT 37, 140
 A32_RTC 22, 25, 72
 A32_SWITCH 23, 53, 55, 74
 A32_to_S3 21, 23, 66, 67
 A32_UART 21, 23, 67
 A32_VDC 25, 31, 71, 73
 A32_VDC_LISTENER 53, 55, 74
 A32_VDC_VOLUME 67, 68
 A32_VOL 23, 67, 68
 A1104 47, 134
 A.1 Main board: ESP32-S3 130
 A.2 Audio board: ESP32 131
 A.3 The ADC: PCM1802 (Texas Instruments) 131
 A.4 The DAC: PCM5102A (Texas Instruments) 132
 A.5 Amplifier power sensor and box-fan regulator 132
 A.6 Display and motor drivers 133
 A.7 Connectors 133
 A.8 Needle, tuning and lamp parts 134
 A.9 Tubes 135
 A_3V3 23, 31, 51, 55, 57, 69, 71, 73, 137, 147
 A_I00A 22, 23, 34, 70, 71, 137
 A_I00D 22, 23, 34, 70, 71, 137
 A_I04 22, 34, 70
 A_I017 22, 34, 70
 A_I018 22, 34, 70
 A_I019 22, 34, 70
 A_I021 22, 25, 72
 A_I022 22, 25, 72, 147
 A_I025 23, 51, 69
 A_I033 23, 51, 69
 A_I035 23, 53, 68
 A_I036 23
 A_I036_H 52, 53, 73, 74
 A_I036_L 52, 53, 73, 74
 A_I0n 56, 57, 147

ADC 5 V — 5 V bus to the ADC board 78
 ADC_5VIN 78
 AFC_C 103
 AM and SW input, converter (V3, 12BE6) and the SW and AM local oscillator 113
 AM_B 97, 111, 112, 113, 114, 115, 123, 124, 126, 127, 129
 AMP_ 91, 147
 AMP_G 38, 75, 91, 147
 AMP_IN 75
 AMP_L+ 38, 75, 147
 AMP_L- 38, 75, 147
 AMP_R+ 38, 75, 147
 AMP_R- 38, 75, 147
 Amplifier power sensor — main board to power sensor board 62
 Amplifier supply wires — amplifier supply to the amplifier 81
 ANT 86, 88
 AS5600 11, 17, 21, 30, 31, 45, 47, 65, 134, 144, 145
 Audio board 3.3 V — audio board to the 3.3 V supply 73
 Audio board 5 V — 5 V bus to audio board 72
 Audio board pin map 22
 Audio board support parts 24

B

B130 13, 97, 118, 119, 126, 127, 148
 B.1 Drawing errors still in the published sheets 136
 B.2 Parts drawn where they read best 136
 B.3 Reading notes 137
 B_RAW 126, 127
 Bluetooth button and LED — audio board to front panel 68
 Board link — main board to audio board 66
 Boards and cables 43
 Box-fan harness — amplifier supply to the box fan 80
 BT_LED 23, 68, 69
 Build details from the notes 88

C

C1A 126, 127, 128, 138, 153
 C1B 126, 127, 128
 C1C 126, 127, 128, 138, 153
 C2 108, 109, 110, 138, 153
 C3 14, 89, 98, 99, 100, 129, 148
 C4 98, 99, 100
 C5 95, 98, 99, 100, 101, 138

C6 98, 99, 100
 C7 98, 99, 100
 C8 97, 98, 99, 100
 C9 98, 99, 100
 C10 105, 106, 107
 C11 95, 101, 102, 103, 104, 138
 C12 101, 102, 103
 C13 97, 101, 102, 103, 137
 C15 101, 102, 103, 131
 C16 98, 99, 101, 102, 104
 C17 101, 102, 104
 C18 101, 102, 104
 C19 101, 102, 104, 107
 C20 101, 102, 104
 C21 113, 114, 115, 116
 C22A 113, 114, 115, 116
 C22B 114, 115, 116
 C23A 111, 112, 113, 114, 115, 116
 C23B 97, 114, 115, 117
 C24 111, 112, 113, 117
 C25 97, 114, 115, 116
 C26 97, 114, 115, 116
 C27 97, 114, 115, 117
 C28 105, 106, 107
 C29 105, 106, 111, 112
 C30 97, 105, 106, 107, 137
 C31 111, 112
 C32 111, 112, 113
 C33 105, 106, 107
 C34 108, 109, 110
 C35 108, 109, 110
 C36 108, 109, 110
 C37 108, 109, 110
 C38 117, 119, 120
 C39 117, 119, 120
 C40 117, 118, 119, 120, 121
 C41 118, 119, 120, 121
 C42 117, 118, 119, 120
 C43 118, 119, 121
 C44 14, 126, 127, 128
 C45 127
 C46 126, 127, 128
 C47 14, 115, 122, 123, 124
 C48 14, 95, 122, 123, 124, 125, 138
 C49 14, 122, 123, 125
 C50 101, 102, 103, 141
 C51 14, 88, 98, 99, 100, 101, 129, 141
 C52 14, 122, 123, 124
 C59 37
 C4558 90
 C.1 Outside the book's scope 139

C.2 Open points 139

Cables 48

CBB22 36, 37, 148

CHASSIS 13, 14, 27, 29, 30, 94, 122, 129, 148

Clock module control — audio board to the real-time clock module 72

Clock module power — 3.3 V jack to the real-time clock module 71

Colon 5 V — 5 V bus to the clock display's colon 76

COLON_IN 31, 43, 44, 58, 76, 77

COM 43, 46, 83, 133

Connectors and labels 17

Cooling 17

CR2032 25, 26

D

D1 41, 42, 44, 79, 85

D2 42, 79, 85

D3 42, 79, 85

D4 41, 42, 44, 79, 85

D5 42

D6 42

D7 42

D8 42

D9 42

D10 42

D11 42

D12 42

D13 42

D14 42

D15 42

D16 42

D17 42

D18 42

D19 42

D20 42

D21 42

D22 42

D23 42

D24 42

D25 42

D26 42

D27 42

D28 42

D46 84

D47 84

D48 84

D49 84

D.1 Speakers 143

D.2 Back 143

D.3 Front panel 144

D.4 Board cases and holders, inside the cabinet 144

D.5 The tube radio's Faraday cage 146

D.6 Clock display 146

DAC to amplifier — DAC board to the amplifier's balanced input 75

DAC_OUT 75, 137

Digit rails — display driver board to the display bus board 79

Digit select — main board to display driver board 63

DIGIT_IN 20, 21, 43, 63

DIGIT_OUT 44, 79

DIGIT_OUT1 79, 80

DISPLAY_IN 20, 43, 61

DISPLAY_OUT 44, 80

Driver board 5 V — 5 V bus to the display driver board 78

DRIVER_IN 44, 79

DRIVERS_5VIN 43, 44, 78, 79

E

Earth, cages and the antenna 30

ESP32-S3-N16R8 19, 130

ESP32-S3-WROOM-1 130

ESP32-S3-WROOM-2 136

ESP32-WROOM-32 131

ESP32-WROOM-32E 19, 131

F

FAN_+ 81

FAN_- 81

FAN_IN 32, 81

FAN_OUT 32, 81

FCC ID 19

FM local oscillator and AFC (V2, 12DT8) 101

FM receiver antenna and the antenna coax 85

FM receiver control — main board to FM calibration receiver 60

FM receiver power — main board to FM calibration receiver 59

FM RF amplifier and mixer (V1, 12DT8) 98

FM_B 97, 98, 99, 101, 102, 105, 106, 123, 124, 126, 127, 129

FM_LED_IN 21, 51, 66

From the DAC to the amplifier 37

Front volume knob — audio board and 3.3 V to the volume knob 67

G

GPI00 22, 23, 24, 34, 70, 71, 130, 131, 137

GPI01 20, 22, 23, 54, 55, 62, 63, 130

GPI02 20, 22, 23, 40, 41, 42, 44, 63, 64, 131, 141

GPI03 22, 23, 130

GPI04 20, 22, 24, 25, 33, 34, 37, 47, 65, 70, 140

GPI05 20, 23, 24, 25, 45, 46, 47, 48, 64, 65, 131

GPI06 20, 23, 41, 42, 43, 61, 131

GPI07 20, 42, 43, 61

GPI08 21, 42, 43, 61

GPI09 21, 45, 46, 48, 62

GPI010 21, 46, 62

GPI011 21, 23, 46, 62, 131

GPI012 21, 23, 46, 48, 62, 131

GPI013 21, 23, 26, 66, 67

GPI014 21, 23, 26, 66, 67

GPI015 21, 23, 42, 43, 61, 131

GPI016 21, 23, 37, 42, 43, 61, 131, 140

GPI017 21, 22, 34, 42, 43, 61, 70, 131

GPI018 21, 22, 34, 42, 43, 61, 70

GPI021 21, 22, 25, 50, 51, 55, 66

GPI025 23, 51, 55, 68

GPI026 21, 23, 26, 66, 67

GPI027 21, 23, 26, 66, 67

GPI033 23, 51, 55, 68

GPI035 22, 23, 53, 55, 67, 130, 131

GPI036 22, 23, 24, 52, 53, 55, 73, 130, 131, 136

GPI038 21, 24, 25, 45, 47, 65, 130

GPI039 21, 22, 23, 24, 25, 45, 47, 65, 130, 131, 151

GPI040 21, 22, 40, 41, 42, 44, 63, 64, 130, 141

GPI041 21, 22, 40, 41, 42, 44, 63, 130, 141

GPI042 21, 22, 40, 41, 42, 44, 63, 64, 130, 141, 151

GPI045 22, 130

GPI046 22, 130

GPI047 21, 24, 48, 60, 130, 136

GPI048 19, 21, 24, 25, 48, 60, 130, 136

GPI0n 56, 147

H

How it is driven 41

How to read the cable pages 56

How to use it 7

I

IEC C14 13, 17, 27, 28

Inside the clock display — colon and digits 84

IRL540N 50, 51, 66, 134, 151

IRL540N MOSFET 50

J

J2 97, 113, 114, 115, 116, 129, 137, 138

J3 125, 127, 128

J4 97, 125, 127, 128, 129, 137

J9 83

J10 18, 86, 88, 89, 129, 143

J40 137

J42 137

J49 137**J50** 76**J51** 36, 76, 129, 137**J53** 86, 88, 89, 98, 99, 100, 129**JST XH** 18, 30, 31, 57**L****L1** 101, 102, 103, 104**L2** 98, 99, 100**L3** 98, 99, 100, 101**L4** 98, 99, 100**L5** 101, 102, 104, 138**L6** 113, 114, 115, 116, 117, 138**L7** 111, 112, 113, 114, 115, 116, 117, 129**L8** 97, 114, 115, 117, 138**L9** 97, 114, 115, 116, 117**L10** 93, 98, 99, 105, 106, 107, 108, 111, 112, 124, 138**L11** 105, 106, 111, 112, 113, 114, 115, 124**L12** 93, 105, 106, 107, 108, 111, 112, 138**L13** 105, 106, 111, 112, 113, 119**L14** 93, 105, 106, 107, 108, 109, 110, 138, 141, 154**L15** 99, 102, 122, 123, 124**L16** 99, 115, 122, 123, 124**L17** 126**L18** 126**LEDITRON_IN** 44, 80**LIMIT** 20, 25, 47, 48, 64**LISTENER** 23, 74, 151**LVCMOS** 132, 151**M****M1** 97, 98, 122, 125, 127, 128, 129, 137, 142**M2** 98, 100, 104, 116**M3** 98, 108, 109, 110, 142**M4** 98, 108, 109, 110, 142**M5** 96, 110, 113, 116, 117, 125**M6** 98, 120, 127

Main board 5 V — 5 V bus to main board 60

Main board pin map 20

Main board support parts 23

Mains and earth wiring 86

Motor driver 5 V — 5 V bus to the needle motor driver 77

N

Needle index sensor — main board and 5 V bus to the index sensor 64

Needle motor control — main board to needle motor driver 62

Needle motor lead — needle motor driver to the needle motor 83

Notation 8

O**OUT1** 46, 83**OUT2** 46, 83**OUT3** 46, 83**OUT4** 46, 83

Overview 93

P**P_3V3** 34, 35, 140

Panel-lamp control — main board and 5 V bus to the panel-lamp board 66

Panel-lamp leads — panel-lamp board to the four FM lamps 83

PFET1 42**PFET4** 42**POWER_DET** 20, 62, 63

Printed parts 18

PSRAM 19, 22, 130, 153, 154**PSU_GND** 137**R****R1** 17, 117, 119, 120, 129**R2** 98, 99, 100**R3** 98, 99, 100**R4** 105, 106, 107**R5** 101, 102, 103**R6** 101, 102, 103**R7** 101, 102, 103**R8** 101, 102, 103**R9** 101, 102, 104**R10** 101, 102, 104**R11** 114, 115, 116**R12** 105, 106, 107**R13** 105, 106, 111, 112, 126, 127**R14** 105, 106, 107**R15** 111, 112**R16** 111, 112**R17** 97, 111, 112, 113, 137**R18** 105, 106, 107**R19** 108, 109, 110**R20** 108, 109, 110**R21** 108, 109, 110**R22** 108, 109, 110**R23** 101, 102, 108, 109, 110**R24** 108, 109, 110**R25** 117, 119, 120**R26** 117, 119, 120, 126, 127**R27** 118, 119, 120**R28** 118, 119, 120**R29** 126, 127, 128**R30** 126, 127, 128**R31** 126, 127, 128**R32** 97, 123, 124, 125, 126, 127**R33** 101, 102, 103, 141**RAD_L** 36, 76, 129**RAD_R** 36, 76, 129

Radio to ADC — tube radio's output transformer to the ADC 76

RADIO_TO_ADC 36, 76, 137**RDA5807M** 10, 11, 21, 31, 47, 59, 60, 134, 145, 146**RDA_3V3** 21, 48, 59**RDA_I2C** 21, 48, 60**RTC_3V3** 25, 31, 71**RTC_I2C** 22, 25, 72**S****S1** 98**S1A** 96, 97, 101, 102, 104, 108, 109, 110, 129, 136, 138, 152**S1B** 97, 108, 109, 110, 111, 112, 113, 117, 119**S1C** 97, 114, 115, 116, 117**S1D** 97, 111, 112, 121, 123, 124, 125, 126, 127**S1E** 97, 121, 123, 124, 125**S1F** 97, 114, 115, 116, 117**S1G** 96, 97, 114, 115, 117, 136, 138, 152**S2** 17, 98, 120, 125, 127, 128, 129, 142**S3_3V3** 21, 24, 26, 47, 48, 57, 59, 62, 64, 65, 140, 153**S3_5VIN** 21, 26, 60**S3_DIGIT** 20, 21, 43, 63**S3_DISPLAY** 20, 43, 61**S3_FM_LED** 21, 66**S3_LIMITS** 20, 24, 25, 48, 64**S3_PWR_DET** 20, 62, 63**S3_RDA** 21, 26, 48, 60**S3_STEPPER** 21, 48, 62**S3_to_A32** 21, 23, 66, 67**S3_TUNER** 21, 24, 25, 47, 48, 65**S3_UART** 21, 23, 26, 67

Safety-rated parts 14

SEG_A_BUS 41, 43, 80, 85**SEG_B_BUS** 43, 80, 85**SEG_C_BUS** 43, 80, 85**SEG_D_BUS** 43, 80, 85**SEG_E_BUS** 43, 80, 85**SEG_F_BUS** 43, 80, 85**SEG_G_BUS** 41, 43, 80, 85

Segment buses — display driver board to the display bus board 80

Segment drive — main board to display driver board 61

Source switch — audio board and 3.3 V to the source-switch board 73

STEPPER_5VIN 46, 48, 77**STEPPER_IN** 21, 48, 62**T****TDA7265** 90

The 5 V and 3.3 V buses 30

The 455 kHz AM IF and AM detector 111

The ADC (PCM1802) 34
 The aerial 87
 The amplifier power sensor 54
 The amplifier supply and the box fan 31
 The angle sensor (AS5600) 47
 The audio amplifier (V6, 12AV6, and V7, 50C5), tone switch and output transformer 117
 The audio harness — audio board to ADC and DAC 69
 The AUX path and the amplifier's input panel 38
 The back-panel USB-C ports 24
 The Bluetooth button and LED 51
 The boards 19
 The cabinet and how to open it 16
 The colon 43
 The control side 12
 The DAC (PCM5102A) 37
 The digit drivers 42
 The digital audio bus (I2S) 34
 The display at a glance 40
 The Faraday cages 17
 The feedline and its earthing 87
 The FM calibration receiver (RDA5807M) 47
 The FM IF path (V4, V5, 12BA6) 104
 The FM panel lamps 50
 The FM ratio detector and AFC control 108
 The four grounds 28
 The front panel 16
 The front panel at a glance 12

The front power switch does not make it safe 13
 The front volume knob 53
 The index sensor (A3144) and its magnet 46
 The input panel 91
 The low-voltage side 15
 The machine at a glance 10
 The mains side 28
 The mechanics 45
 The original transformerless AC supply and B+ 125
 The path at a glance 33
 The power at a glance 12
 The power at a glance 27
 The radio chassis is on mains neutral 14
 The real-time clock module 25
 The segment drivers 43
 The segment map 42
 The series heater string, the mode supply and the indicator lamps 121
 The sound path 11
 The source switch 52
 The stepper motor and its driver 46
 The three volume controls 38
 The tube radio into the ADC 36
 The volume control 90
 TL072 90
 TL074 90, 137
 Tuning angle sensor — main board to the angle sensor 65
 Tuning, needle and index 17

U

U17 85

U18 85
 U19 85
 U32 85
 U35 84, 85
 U71 85
 USB — back panel to each board 82

V

V5 93, 95, 96, 97, 104, 105, 106, 107, 122, 124, 129, 154
 VBUS 24, 25, 82, 137, 155
 VOLUME 23, 53, 67, 68

W

What is not measured 89
 What it does not cover 7
 What the amplifier and speakers are 90
 What this book covers 7
 What was measured at the panel and on the cable 91
 When something is wrong 25
 When something is wrong 32
 When something is wrong 39
 When something is wrong 44
 When something is wrong 48
 When something is wrong 55
 When something is wrong 86
 When something is wrong 89
 When something is wrong 92
 When something is wrong 129
 Where the mains is, and when 13
 Where the tube radio meets Ambersong 128